



DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING
COURSE STRUCTURE
B.TECH, I SEMESTER

S.No	Course Category	Course Code	Name of the Course	Course Type	Hours/week (L – T – P)	Max Marks			Credits
						Int	Ext	Total	
1	BS&H	23AM1T01	Communicative English	T	2 – 0 – 0	30	70	100	2
2	BS&H	23AM1T02	Linear Algebra & Calculus	T	3 – 0 – 0	30	70	100	3
3	BS&H	23AM1T03	Chemistry	T	3 – 0 – 0	30	70	100	3
4	ES	23AM1T04	Introduction to Programming	T	3 – 0 – 0	30	70	100	3
5	ES	23AM1T05	Basic Civil & Mechanical Engineering	T	1 – 0 – 4	30	70	100	3
6	BS&H	23AM1L06	Communicative English Lab	L	0 – 0 – 2	30	70	100	1
7	BS&H	23AM1L07	Chemistry Lab	L	0 – 0 – 2	30	70	100	1
8	ES	23AM1L08	Computer Programming Lab	L	0 – 0 – 3	30	70	100	1.5
9	ES	23AM1L09	Engineering Workshop	L	0 – 0 – 2	30	70	100	1.5
10	BS&H	23AM1L10	Health and wellness, Yoga and Sports	L	0 – 0 – 1	100	-	100	0.5
Total Number of Credits									19.5

B.TECH, II SEMESTER

S.No	Course Category	Course Code	Name of the Course	Course Type	Hours/week (L – T – P)	Max Marks			Credits
						Int	Ext	Total	
1	BS&H	23AM2T01	Differential Equations & Vector Calculus	T	3 – 0 – 0	30	70	100	3
2	BS&H	23AM2T02	Engineering Physics	T	3 – 0 – 0	30	70	100	3
3	ES	23AM2T03	Basic Electrical & Electronics Engineering	T	3 – 0 – 0	30	70	100	3
4	ES	23AM2T04	Data Structures	T	3 – 0 – 0	30	70	100	3
5	PC	23AM2T05	Engineering Graphics	T	3 – 0 – 0	30	70	100	3
6	BS&H	23AM2L06	Engineering Physics Lab	L	0 – 0 – 2	30	70	100	1
7	ES	23AM2L07	Electrical & Electronics Engineering workshop	L	0 – 0 – 3	30	70	100	1.5
8	ES	23AM2L08	IT Workshop	L	0 – 0 – 3	30	70	100	1
9	PC	23AM2L09	Data Structures Lab	L	0 – 0 – 3	30	70	100	1.5
10	BS&H	23AM2L10	NSS/NCC/Scouts & Guides/ Community Service	L	0 – 0 – 1	100	-	100	0.5
Total Number of Credits									20.5



B.TECH, III SEMESTER

S.No	Course Category	Course Code	Name of the Course	Course Type	Hours/week (L – T – P)	Max Marks			Credits
						Int	Ext	Total	
1	BS&H	23AM3T01	Discrete Mathematics & Graph Theory	T	3 – 0 – 0	30	70	100	3
2	BS&H	23AM3T02	Universal Human Values 2 Understanding Harmony	T	3 – 0 – 0	30	70	100	3
3	ES	23AM3T03	Artificial Intelligence	T	3 – 0 – 0	30	70	100	3
4	PC	23AM3T04	Advanced Data Structures Algorithms Analysis	T	3 – 0 – 0	30	70	100	3
5	PC	23AM3T05	Object Oriented Programming Through Java	T	3 – 0 – 0	30	70	100	3
6	PC	23AM3L06	Advanced Data Structures and Algorithms Analysis Lab	L	0 – 0 – 3	30	70	100	1.5
7	PC	23AM3L07	Object Oriented Programming Through Java Lab	L	0 – 0 – 3	30	70	100	1.5
8	SC	23AM3S08	Python Programming Lab	L	0 – 0 – 4	30	70	100	2
9	AC	23AM3M09	Environmental Science	T	2 – 0 – 0	30	-	30	-
Total Number of Credits									20

B.TECH, IV SEMESTER

S.No	Course Category	Course Code	Name of the Course	Course Type	Hours/week (L – T – P)	Max Marks			Credits
						Int	Ext	Total	
1	MC	23AM4T01	Optimization Techniques	T	3 – 0 – 0	30	70	100	3
2	ES	23AM4T02	Probability & Statistics	T	3 – 0 – 0	30	70	100	3
3	PC	23AM4T03	Machine Learning	T	3 – 0 – 0	30	70	100	3
4	PC	23AM4T04	Database Management Systems	T	3 – 0 – 0	30	70	100	3
5	PC	23AM4T05	Digital Logic & Computer Organization	T	3 – 0 – 0	30	70	100	3
6	PC	23AM4L06	Machine Learning Lab	L	0 – 0 – 3	30	70	100	1.5
7	PC	23AM4L07	Database Management Systems Lab	L	0 – 0 – 3	30	70	100	1.5
8	SC	23AM4S08	Full Stack Development-1	L	0 – 0 – 4	30	70	100	2
9	BS&H	23AM4L09	Design Thinking & Innovation	L	0 – 0 – 4	30	70	100	2
Total Number of Credits									20.5
Mandatory Community Service Project Internship for 08 Weeks duration during summer vacation									



B.TECH, V SEMESTER

S.No	Course Category	Course Code	Name of the Course	Course Type	Hours/week (L – T – P)	Max Marks			Credits
						Int	Ext	Total	
1	PC	23AM5T01	Deep Learning	T	3 – 0 – 0	30	70	100	3
2	PC	23AM5T02	Computer Networks	T	3 – 0 – 0	30	70	100	3
3	PC	23AM5T03	Operating Systems	T	3 – 0 – 0	30	70	100	3
Professional Elective-I									
4	PE-I	23AM5T04	Automata Theory & Compiler Design	T	3 – 0 – 0	30	70	100	3
		23AM5T05	Software Architecture and Design Pattern						
		23AM5T06	Internet of Things(IoT)						
		23AM5T07	Exploratory Data Analysis with Python						
		23AM5T08	12-Week SWAYAM /NPTEL Course suggested by the BoS						
5	Open Elective -I			T	3 – 0 – 0	30	70	100	3
6	PC	23AM5L10	Deep Learning Lab	L	0 – 0 – 3	30	70	100	1.5
7	PC	23AM5L11	Operating Systems & Computer Networks LAB	L	0 – 0 – 3	30	70	100	1.5
8	SC	23AM5S12	Full Stack Development -2 / SWAYAM Plus – Data Engineer / AI Engineer	L	0 – 0 – 4	30	70	100	2
9	ES	23AM5L13	User Interface Design using Flutter / SWAYAM Plus - Android Application Development (with Flutter)	L	0 – 0 – 2	30	70	100	1
10	PR	23AM5I14	Community Service Internship	P	-	-	50	50	2
Total Number of Credits									23

B.TECH, VI SEMESTER

S.No	Course Category	Course Code	Name of the Course	Course Type	Hours/week (L – T – P)	Max Marks			Credits
						Int	Ext	Total	
1	PC	23AM6T01	Reinforcement Learning	T	3 – 0 – 0	30	70	100	3
2	PC	23AM6T02	Software Engineering	T	3 – 0 – 0	30	70	100	3
3	PC	23AM6T03	Data Visualization	T	3 – 0 – 0	30	70	100	3
Professional Elective-II				T	3 – 0 – 0	30	70	100	3
4	PE-II	23AM6T04	Cryptography & Network Security						
		23AM6T05	Recommender Systems						
		23AM6T06	MEAN Stack Technologies						
		23AM6T07	Social Network Analysis						
		23AM6T08	Object Oriented Analysis and Design						
Professional Elective -III				T	3 – 0 – 0	30	70	100	3
5	PE-III	23AM6T09	Computer Vision						
		23AM6T10	Cloud Computing						
		23AM6T11	DevOps						
		23AM6T12	Soft Computing						
		23AM6T13	12-Week SWAYAM /NPTEL Course suggested by the BoS						
6	Open Elective -II			T	3 – 0 – 0	30	70	100	3
7	PC	23AM6L15	Software Engineering Lab	L	0 – 0 – 3	30	70	100	1.5
8	PC	23AM6L16	Data Visualization Lab	L	0 – 0 – 3	30	70	100	1.5
9	SC	23AM6S17	Soft skills / SWAYAM Plus - 21st Century Employability Skills	L	0 – 0 – 4	30	70	100	2
10	AC	23AM4M18	Technical Paper Writing & IPR	T	2 – 0 – 0	30	-	30	0
Total Number of Credits									23
Mandatory Industry Internship of 08 weeks duration during summer vacation									



OPEN ELECTIVE –I:

S. No.	Course Code	Course Name	L	T	P	C	Offered by
1	23CE5T08	Green Buildings	3	0	0	3	CE
2	23CE5T09	Construction technology and management	3	0	0	3	CE
3	23EE5T08	Renewable Energy Sources	3	0	0	3	EEE
4	23EE5T09	Concepts of Energy Auditing & Management	3	0	0	3	EEE
5	23ME5T09	Sustainable Energy Technologies	3	0	0	3	ME
6	23ME5T10	Design and Analysis of Experiments	3	0	0	3	ME
7	23ME5T11	Thermal Management of Electronic systems	3	0	0	3	ME
8	23EC5T09	Electronic Devices and Circuits	3	0	0	3	ECE
9	23EC5T10	Signals and Systems	3	0	0	3	ECE
10	23EC5T11	Digital Logic Design	3	0	0	3	ECE
11	23CS5T09	Operating Systems	3	0	0	3	CSE
12	23CS5T10	Computer organization and Architecture	3	0	0	3	CSE
13	23IT5T10	Database Management Systems	3	0	0	3	IT
14	23IT5T11	Computer Graphics	3	0	0	3	IT
15	23MB5T01	Entrepreneurship Development and Venture Creation	3	0	0	3	DMS
16	23MB5T03	Innovation and Technology Management	3	0	0	3	DMS
17	23AD5T09	Introduction to Data Science	3	0	0	3	AIDS
18	23AM5T09	Machine Learning	3	0	0	3	AIML

OPEN ELECTIVE –II:

S. No.	Course Code	Course Name	L	T	P	C	Offered by
1	23CE6T12	Disaster management	3	0	0	3	CE
2	23CE6T13	Sustainability in Engineering practices	3	0	0	3	CE
3	23EE6T10	Fundamentals of Electric Vehicles	3	0	0	3	EEE
4	23EE6T11	Electrical Wiring Estimation and Costing	3	0	0	3	EEE
5	23ME6T12	Introduction to Industrial Robotics	3	0	0	3	ME
6	23ME6T13	Introduction to Additive Manufacturing	3	0	0	3	ME
7	23ME6T14	Vehicle Technology	3	0	0	3	ME
8	23EC6T14	Linear & Digital IC Applications	3	0	0	3	ECE
9	23EC6T15	Principles Of Communications	3	0	0	3	ECE
10	23EC6T16	Microprocessors & Microcontrollers	3	0	0	3	ECE
11	23CS6T15	Database Management Systems	3	0	0	3	CSE
12	23CS6T16	Software Engineering	3	0	0	3	CSE
13	23IT6T14	Fundamentals of Machine Learning	3	0	0	3	IT
14	23IT6T15	Computer Networks	3	0	0	3	IT
15	23MB6T01	Industry Specific Employability Skills	3	0	0	3	DMS
16	23MB6T03	Supply Chain Management	3	0	0	3	DMS
17	23AD6T14	Artificial Intelligence	3	0	0	3	AIDS
18	23AM6T14	Social Media Analytics	3	0	0	3	AIML

HONORS/ MINOR COURSES OFFERED BY THE DEPARTMENT**Honors/ Minor Course Requirements:**

As per the curriculum structure, students opting for Minor or Honors programs are required to complete 18 additional credits beyond the regular degree requirements. The distribution and eligibility criteria for earning these credits are as follows:

1. Minor Degree Program:

- Students shall earn 15 credits by successfully completing the courses offered through the conventional mode, each carrying 3 credits, as per the curriculum.
- The remaining 3 credits shall be acquired through the completion of one domain-specific MOOC (SWAYAM/NPTEL), with a minimum duration of 12 weeks, as recommended by the Board of Studies.
- Minor courses offered by the AIML Department shall be available to students of other branches (excluding AIML students).

2. Honors Degree Program:

- Students shall earn 12 credits by successfully completing the courses offered through the conventional mode each carrying 3 credits, as per the curriculum.
 - The remaining 6 credits shall be acquired through the completion of two domain-specific MOOCs (SWAYAM/NPTEL), each carrying 3 credits and having a minimum duration of 12 weeks, as recommended by the Board of Studies.
 - The Honors program is offered to students within the same discipline.
3. At the beginning of each academic year, the Head of the Department shall publish a list of approved MOOC electives, aligned with the relevant domains of the Minor and Honors programs, based on the available courses on SWAYAM/NPTEL and subject to the recommendations of the Board of Studies.
 4. During Minor/Honors Course selection, there should not be any overlapping with Regular/Major/OPEN Electives

MINOR COURSES

S.No.	Course Code	Course Name	L	T	P	C	Offered By
1	23AMMN01	Introduction to Machine Learning	3	0	0	3	AIML
2	23AMMN02	Social Network Analysis	3	0	0	3	AIML
3	23AMMN03	Data Visualization	3	0	0	3	AIML
4	23AMMN04	Expert Systems	3	0	0	3	AIML
5	23AMMN05	Natural Language Processing	3	0	0	3	AIML
6	23AMMN06	Exploratory Data Analysis with Python	3	0	0	3	AIML

HONORS COURSES

S.No.	Course Code	Course Name	L	T	P	C	Offered By
1	20AMHN01	Optimization Techniques In Machine Learning	3	0	0	3	AIML
2	20AMHN02	Machine Learning For Security	3	0	0	3	AIML
3	20AMHN03	Sentiment Analysis	3	0	0	3	AIML
4	20AMHN04	Cyber Security	3	0	0	3	AIML
5	20AMHN05	Responsible & Ethical AI	3	0	0	3	AIML
6	20AMMN06	Design Of Artificial Intelligence Products	3	0	0	3	AIML
7	20AMHN07	Artificial Neural Networks	3	0	0	3	AIML
8	20AMHN08	Text Mining And Time Series Analysis	3	0	0	3	AIML
9	20AMMN09	MOOC1	-	-	-	3	AIML
10	20AMMN10	MOOC2	-	-	-	3	AIML



B.TECH I SEMESTER

BS & H	L	T	P	C
	3	0	0	3

23AM1T01 COMMUNICATIVE ENGLISH

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Facilitate effective listening, Reading, Speaking and Writing skills among the students
2. Enhance the same in their comprehending abilities, oral presentations ,reporting useful information
3. Provide knowledge of grammatical structures and vocabulary.
4. Help the students to make them effective in speaking and writing skills
5. Make the students industry ready

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the context, topic, and pieces of specific information from social or Transactional dialogues (Build or participate in social conversations in a meaningful manner that result in positive relationship development)
- CO2** Apply grammatical structures to formulate sentences and correct word forms (Enhance oral skills through various activities both in theory and lab activities and give input regarding grammar in remedial manner)
- CO3** Analyze different genres of texts and practice writing skills in the required formats; to speak clearly on a specific topic in informal discussions
- CO4** Evaluate reading /listening texts and to write summaries and essays based on global comprehension of these texts
- CO5** Create a coherent paragraph, essay, and resume (Use appropriate vocabulary and expressions based on the text/context)

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	3	-	2	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	2	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	2	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	2	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	2	-	-	-

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put '-'. 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

SYLLABUS

UNIT-I: Lesson: HUMAN VALUES: Gift of Magi (Short Story)

Listening: Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.

Speaking: Asking and answering general questions on familiar topics such as home, family, work, studies and interests.

Reading: Skimming to get the main idea of a text; scanning to look for specific pieces of information.

Writing: Mechanics of Writing-Capitalization, Spellings, Punctuation-Parts of Sentences.

Grammar: Parts of Speech, Basic Sentence Structures-forming questions

Vocabulary: Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words.

UNIT-II: Lesson: NATURE: The Brook by Alfred Tennyson (Poem)

Listening: Answering a series of questions about main ideas and supporting ideas after listening to audio texts.

Speaking: Discussion in pairs/small groups on specific topics followed by short structure talks.

Reading: Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.

Writing: Structure of a paragraph - Paragraph writing (specific topics)

Grammar: Cohesive devices - linkers, use of articles and zero article; prepositions.

Vocabulary: Homonyms, Homophones, Homographs.

UNIT-III: Lesson: BIOGRAPHY: Elon Musk

Listening: Listening for global comprehension and summarizing what is listened to.

Speaking: Discussing specific topics in pairs or small groups and reporting what is discussed

Reading: Reading a text in detail by making basic inferences -recognizing and interpreting specific context clues; strategies to use text clues for comprehension.

Writing: Summarizing, Note-making, paraphrasing

Grammar: Verbs - tenses; subject-verb agreement; Compound words, Collocations

Vocabulary: Compound words, Collocations

UNIT-IV: Lesson: INSPIRATION: The Toys of Peace by Saki

Listening: Making predictions while listening to conversations/ transactional dialogues without video; listening with video.

Speaking: Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions.

Reading: Studying the use of graphic elements in texts to convey information, reveal trends/patterns/relationships, communicate processes or display complicated data.

Writing: Letter Writing: Official Letters, Resumes, Email Writing SOP

Grammar: Reporting verbs, Direct & Indirect speech, Active & Passive Voice



Vocabulary: Words often confused, Jargons

UNIT-V: Lesson: MOTIVATION: The Power of Intrapersonal Communication (An Essay)

Listening: Identifying key terms, understanding concepts and answering a series of relevant questions that test comprehension.

Speaking: Formal oral presentations on topics from academic contexts

Reading: Reading comprehension.

Writing: Writing structured essays on specific topics.

Grammar: Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)

Vocabulary: Technical Jargons

Text Books

1. Pathfinder: Communicative English for Undergraduate Students, 1st Edition, Orient Black Swan, 2023 (Units 1,2 & 3)
2. Empowering with Language by Cengage Publications, 2023 (Units 4 & 5)

Reference Books

1. Dubey, Sham Ji & Co. English for Engineers, Vikas Publishers, 2020.
2. Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge, 2014.
3. Murphy, Raymond. English Grammar in Use, Fourth Edition, Cambridge University Press, 2019



B.TECH I SEMESTER

	L	T	P	C
BS & H	3	0	0	3

23AM1T02 LINEAR ALGEBRA AND CALCULUS

Prerequisites: Basic Knowledge of Matrices, Differentiation & Integration

Course Objectives:

- To equip the students with standard concepts and tools at an intermediate to advanced level mathematics to develop the confidence and ability among the students to handle various real-world problems and their applications.

Course Outcomes: At the end of the course, student will be able to

- CO1:** Develop and use of matrix algebra techniques that are needed by engineers for practical applications
- CO2:** Apply Eigen values and Eigen vectors Concepts in various Engineering fields
- CO3:** Utilize Mean value theorems to real life problems, Familiarize with functions of several Variables which is useful in optimization
- CO4:** Learn important tools of calculus in higher dimensions
- CO5:** Familiarize with double and triple integrals of functions of several variables in two Dimensions using Cartesian and polar coordinates and in three dimensions using Cylindrical and spherical coordinates

COURSE ARTICULATION MATRIX (Correlation between Cos, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	-	2	1	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	2	1	-	-	-	-	-	-	-	-	-	-
CO3	3	3	-	1	1	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	1	1	-	-	-	-	-	-	-	-	-	-
CO5	3	3	-	1	1		-	-	-	-	-	-	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I: Matrices

Linear Transformation (Introduction), Rank of a matrix by Echelon form, Normal form. Diagonalization of a matrix, Inverse of Non-singular matrices by Gauss-Jordan method, System of linear equations: Consistency and Inconsistency of system of Homogeneous, Non-Homogeneous

equations, Gauss-elimination method, Jacobi and Gauss Seidel Iteration Methods. Application: Finding Currents in the Electric circuits.

UNIT-II: Eigen values, Eigenvectors and Orthogonal Transformation

Eigen values, Eigen vectors and their properties, Cayley-Hamilton Theorem(without proof), finding inverse and power of a matrix by Cayley-Hamilton Theorem, Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT-III: Calculus

Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem with their geometrical interpretation, Cauchy's mean value theorem, Taylor's and Maclaurin's theorems with remainders (without proof), Problems and applications on the above theorems. Application: Maxima&Minima of functions of Single Variable.

UNIT-IV: Partial differentiation and Applications (Multivariable calculus)

Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Homogeneous functions & Euler's theorem, Taylor's and Maclaurin's series expansion of functions of two variables. Jacobians, Functional dependence,

Application: Maxima and minima of functions of two variables, Method of Lagrange multipliers.

UNIT-V: Multiple Integrals (Multi variable Calculus)

Double integrals, triple integrals, change of order of integration, change of variables to polar, cylindrical and spherical coordinates .Finding areas(by double integrals) and volumes(by double integrals and triple integrals)

Text Books

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition

Reference Books

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, Michael Greenberg, Pearson publishers, 9th edition
5. Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications, 2014, Third Edition (Reprint 2021).



B.TECH I SEMESTER

	L	T	P	C
BS & H	3	0	0	3

23AM1T03 CHEMISTRY

Prerequisites:

Course Objectives:

The main objectives of introducing this course are

- To familiarize engineering chemistry and its applications
- To train the students on the principles and applications of electrochemistry and polymers
- To introduce instrumental methods.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the fundamentals of Quantum mechanics and apply the MOT to homo and hetero diatomic molecular species.
- CO2** Apply the principle of band diagrams in the applications of conductor, semiconductor and applications of superconductors and nanomaterials.
- CO3** Understand the fundamentals of batteries and fuel cells.
- CO4** Preparation, properties and applications of thermoplastics and thermosetting plastics elastomers and conducting polymers.
- CO5** Understand the principles of spectroscopy and chromatographic techniques.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	-	-	1	-	-	-	-	-	-	1	-	-	-
CO2	3	2	2	2	2	3	2	-	-	-	-	1	1	1	-
CO3	3	3	2	2	2	1	1	-	-	-	-	1	1	1	-
CO4	3	-	2	1	1	2	-	-	-	-	-	1	-	-	-
CO5	3	2	2	2	2	3	1	-	-	-	-	1	-	1	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I Structure and Bonding Models

Fundamentals of Quantum mechanics, Schrodinger Wave equation(time independent) significance of Ψ and Ψ^2 , particle in one dimensional box, molecular orbital theory – bonding in homo- and hetero nuclear diatomic molecules – energy level diagrams of O_2 and CO , etc. π -molecular orbitals of

butadiene and benzene, calculation of bond order.

UNIT-II Modern Engineering materials

Semiconductors – Introduction, basic concept (elemental), applications.

Super conductors- Introduction basic concept, applications.

Super capacitors- Introduction, basic concept, classification, applications.

Nano materials: Introduction, classification, properties and applications of Fullerenes, carbon nano tubes and Graphines nano particles. Introduction and applications of HPC nano materials, Cores hell nano particles.

UNIT-III Electrochemistry and Applications

Electrochemical cell, Nernst equation(without derivation), cell potential calculations and numerical problems, potentiometry- potentiometric titrations (redox titrations), conductometric titrations (acid-base titrations).

Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples.

Primary cells – Zinc-air battery, Secondary cells –lithium-ion batteries- working of the batteries including cell reactions; Fuel cells, hydrogen-oxygen fuel cell– working of the cells. Polymer Electrolyte Membrane Fuel Cells (PEMFC).

UNIT-IV Polymer Chemistry

Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation.

Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of Poly ethene, Polystyrene, Poly acrylonitrile, Poly vinyl chloride, Teflon, Bakelite, Nylon-6,6, carbon fibres.

Elastomers–Buna-S, Buna-N–preparation, properties and applications.

Conducting polymers – poly acetylene, poly aniline, – mechanism of conduction and applications. Bio-Degradable polymers - Poly Glycolic Acid, Poly Lactic Acid, Poly hydroxy alkanoates.

UNIT-V Instrumental Methods and Applications

Electromagnetic spectrum. Absorption of radiation: UV-Visible Spectroscopy, electronic transition and instrumentation. IR spectroscopy- fundamental modes, selection rules, and applications. Chromatography-Introduction, principle, classification and applications.

Text Book

- 1.Jain and Jain, Engineering Chemistry, 16/e, Dhanpat Rai, 2013.
- 2.Peter Atkins, Julio de Paula and James Keeler, Atkins' Physical Chemistry, 10/e, Oxford University Press, 2010.
- 3.Shikha Agarwal, Engineering Chemistry, 2/e, Cambridge University Press.
- 4.G.V. Subba reddy, Engineering Chemistry, 1/e, McGraw Hill education (india) Pvt.ltd.2015.



Reference Books

1. Skoog and West, Principles of Instrumental Analysis, 6/e, Thomson, 2007.
2. J.D. Lee, Concise Inorganic Chemistry, 5th Edition, Wiley Publications, Feb.2008
3. Textbook of Polymer Science, Fred W. Billmayer Jr, 3rd Edition



B.TECH I SEMESTER

ES

L	T	P	C
3	0	0	3

23AM1T04 INTRODUCTION TO PROGRAMMING

Prerequisites:

Course Objectives:

To introduce students to the fundamentals of computer programming.

- To provide hands-on experience with coding and debugging.
- To foster logical thinking and problem-solving skills using programming.
- To familiarize students with programming concepts such as data types, control structures, functions, and arrays.
- To encourage collaborative learning and team work in coding projects.

Course Outcomes: A student after completion of the course will be able to

CO1: Understand basics of computers, the concept of algorithm and algorithmic thinking.

CO2: Analyze a problem and develop a program using selection and decision making statement.

CO3: Implement various algorithms using on arrays and string

CO4: Understand the concepts of functions and pointers

CO5: Develop programs using user defined data types and files

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	-	-	-	-	3	-
CO2	3	3	3	-	-	-	-	-	-	-	-	-	3	-
CO3	3	3	3	-	-	-	-	-	-	-	-	-	3	-
CO4	-	3	3	-	-	-	-	-	-	-	-	-	3	-
CO5	-	3	3	-	-	-	-	-	-	-	-	-	3	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I: Lesson: Introduction to Programming and Problem Solving

History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter. Introduction to Programming Languages, Basics of a Computer Program- , Generations of Languages, Application Software and System Software, Algorithms, flowcharts (Using Dia Tool), pseudo code..

Introduction to C Programming: Structure of C Program, Compilation and Execution of C Program, Primitive Data Types, Variable and Constants, Basic Input and Output statements, Operators, operator's precedence, Type Conversion and Casting.

Problem solving techniques: Algorithmic Approach, characteristics of algorithm, Problem solving strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms.

UNIT-II: Lesson: Control Structures

Simple sequential programs Conditional Statements (if, if-else, nested if, else if ladder, switch).

Loops (for, while, do- while), Break and Continue.

UNIT-III: Lesson: Arrays and Strings

Arrays: indexing, memory model, one dimensional Array, programs with array of integers, two dimensional arrays.

Strings: Introduction to Strings, string manipulation operations without using built in functions, String handling functions.

UNIT-IV: Lesson: Functions and Pointers

Functions: Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters, Recursive functions, Scope and Lifetime of Variables.

Pointers: Pointers dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers, Call by Value and Call by reference.

UNIT-V: Lesson: User Defined Data types & File handling

User-defined data types-Structures: Defining, Accessing and Initialization of Structure variable, Array of Structures, Nested Structures, passing structures to the functions, Unions.

File Handling: Types of files, defining and opening a file, closing a file, Input and output operations on files, Random Access files, Command line arguments.

Text Books

3. "The C Programming Language", Brian W .Kernighan and Dennis M.Ritchie , Prentice-Hall, 1988



4. Schaum's Outline of Programming with C ,Byron S Gottfried, McGraw-Hill Education, 1996

Reference Books

1. Computing fundamentals and C Programming, Balagurusamy , E., McGraw-Hill Education, 2008.
2. Programming in C, Rema Theraja, Oxford, 2016, 2nd edition
3. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad.



B.TECH I SEMESTER

	L	T	P	C
ES	3	0	0	3

23AM1T05 BASIC CIVIL & MECHANICAL ENGINEERING

Prerequisites:

Course Objectives:

1. Get familiarized with the scope and importance of Civil Engineering sub-divisions.
2. Introduce the preliminary concepts of surveying.
3. Acquire preliminary knowledge on Transportation and its importance in nation's economy.
4. Get familiarized with the importance of quality, conveyance and storage of water.
5. Introduction to basic civil engineering materials and construction techniques.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand various sub-divisions of Civil Engineering and to appreciate their role in ensuring better society & Understand the basic characteristics of Civil Engineering Materials and attain knowledge on prefabricated technology.
- CO2** Know the concepts of surveying and to understand the measurement of distances, angles and levels through surveying.
- CO3** Realize the importance of Transportation in nation's economy and the engineering measures related to Transportation & Understand the importance of Water Storage and Conveyance Structures so that the social responsibilities of water conservation will be appreciated.
- CO4** Understand the different manufacturing processes.
- CO5** Explain the basics of thermal engineering and its applications.
- CO6** Describe the working of different mechanical power transmission systems and power plants.
Describe the basics of robotics and its applications.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	3	-	3	-	3	-	-	1	1
CO2	3	-	-	-	-	3	-	3	-	3	-	-	1	1
CO3	3	-	-	-	-	3	-	3	-	3	-	-	1	1
CO4	3	3	2	2	2	3	2	2	2	2	2	3	3	1
CO5	3	3	2	2	-	3	-	-	-	-	-	1	3	2
CO6	3	3	3	-	-	2	-	-	-	-	-	1	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put '-'. 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

PART-A
SYLLABUS

UNIT-I: Basics of Civil Engineering

Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering - Hydraulics and Water Resources Engineering - Environmental Engineering-Scope of each discipline - Building Construction and Planning- Construction Materials-Cement - Aggregate - Bricks- Cement concrete- Steel. Introduction to Prefabricated construction Techniques.

UNIT-II: Surveying

Objectives of Surveying- Horizontal Measurements- Angular Measurements- Introduction to Bearings Levelling instruments used for levelling -Simple problems on levelling and bearings-Contour mapping

UNIT-III: Transportation Engineering , Water Resources and Environmental Engineering

Transportation Engineering Importance of Transportation in Nation's economic development- Types of Highway Pavements- Flexible Pavements and Rigid Pavements - Simple Differences. Basics of Harbour, Tunnel, Airport, and Railway Engineering.

Water Resources and Environmental Engineering: Introduction, Sources of water- Quality of water- Specifications- Introduction to Hydrology–Rainwater Harvesting-Water Storage and Conveyance Structures (Simple introduction to Dams and Reservoirs).

Textbooks:

1. Basic Civil Engineering, M.S.Palanisamy, , Tata McGraw Hill publications (India) Pvt. Ltd. Fourth Edition.
2. Introduction to Civil Engineering, S.S. Bhavikatti, New Age International Publishers. 2022. First Edition.
3. Basic Civil Engineering, Satheesh Gopi, Pearson Publications, 2009, First Edition.

Reference Books:

1. Surveying, Vol- I and Vol-II, S.K. Duggal, Tata McGraw Hill Publishers 2019. Fifth Edition.
2. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi. 2016
3. Irrigation Engineering and Hydraulic Structures - Santosh Kumar Garg, Khanna Publishers, Delhi 2023. 38th Edition.
4. Highway Engineering, S.K.Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications 2019. 10th Edition.
5. Indian Standard DRINKING WATER — SPECIFICATION IS 10500-2012.

PART-B

SYLLABUS

UNIT-I: Introduction to Mechanical Engineering & Engineering Materials

Introduction to Mechanical Engineering: Role of Mechanical Engineering in Industries and Society – Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

Engineering Materials- Metals- Ferrous and Non-ferrous, Ceramics, Composites, Smart Materials.

UNIT-II: Manufacturing Processes & Thermal Engineering

Manufacturing Processes: Principles of Casting, Forming, joining processes, Machining, Introduction to CNC machines, 3Dprinting, and Smart manufacturing.

Thermal Engineering – Working principle of Boilers, Otto cycle, Diesel cycle, Refrigeration and air – conditioning cycles, IC engines, 2-Stroke and 4-Stroke engines, SI/CI Engines, Components of Electric and Hybrid Vehicles.

UNIT-III: Power plants, Mechanical Power Transmission & Introduction to Robotics

Power plants – Working principle of Steam, Diesel, Hydro, Nuclear power plants.

Mechanical Power Transmission-Belt Drives, Chain, Rope drives, Gear Drives and their applications.

Introduction to Robotics - Joints &links, configurations, and applications of robotics

Textbooks:

1. Internal Combustion Engines by V. Ganesan, By Tata Mc Graw Hill publications (India) Pvt. Ltd.
2. A textbook of Theory of Machines by S. S. Rattan, Tata Mc Graw Hill Publications, (India)Pvt. Ltd.
3. An introduction to Mechanical Engg by Jonathan Wicker and Kemper Lewis, Cengage learning India Pvt. Ltd.

Reference Books:

1. G. Shanmugam and M.S. Palanisamy, Basic Civil and the Mechanical Engineering, Tata Mc Graw Hill publications (India)Pvt. Ltd.
2. Thermal Engineering by Mahesh M RathoreTata Mc Graw Hill publications (India) Pvt. Ltd.
3. 3D printing & Additive Manufacturing Technology - L.Jyothish Kumar, Pulak M Pandey, Springer publications
4. Appuu Kuttan K K, Robotics, I. K .International Publishing House Pvt. Ltd. Volume-I



B.TECH I SEMESTER

	L	T	P	C
BS & H	0	0	2	1

23AM1L06 COMMUNICATIVE ENGLISH LAB

Prerequisites:

Course Objectives: Objective of the course is to

- Expose the students to a variety of self-instructional, learner friendly modes of language learning.
- Train the students in basic communication skills
- Make them ready to face job interviews

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the different aspects of the English language proficiency with emphasis on LSRW skills.
- CO2** Analyze the English speech sounds, stress, rhythm, intonation and syllable division for better listening and speaking comprehension.
- CO3** Apply communication skills and enhance argumentative skills through various language learning activities like JAM/Role Plays and Debates
- CO4** Evaluate and exhibit professionalism in making PPT Presentations and Poster Presentations
- CO5** Nurture various nuances related to the interview process like participating in Group Discussions

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	1	3	-	2	-	-	-
CO2	-	-	-	-	-	-	-	-	1	3	-	2	-	-	-
CO3	-	-	-	-	-	-	-	-	1	3	-	2	-	-	-
CO4	-	-	-	-	-	-	-	-	1	3	-	2	-	-	-
CO5	-	-	-	-	-	-	-	-	1	3	-	2	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

1. Neutralization/Accent Rules
2. Narrating events, incidents and procedures
3. JAM and small talks



4. Self Introduction and Introduction of Others
5. Describing people, places and things
6. Role Play or Conversational Practice
7. Debates, Arguments to present ideas in a logical manner and persuade others
8. Group Discussions- polite turn taking strategies will be focused upon
9. PPT Presentations and Poster Presentations
10. Creative language use activities (stories, poems)

Suggested Software:

- Walden Infotech
- Young India Films

Reference Books:

1. Raman Meenakshi,Sangeeta-Sharma.*Technical Communication*. Oxford Press.2018.
2. Taylor Grant: *English Conversation Practice*, Tata McGraw-Hill Education India, 2016
3. Hewing's, Martin. *Cambridge Academic English(B2)*.CUP,2012.
4. J. Sethi & P.V. Dhamija. *A Course in Phonetics and Spoken English*, (2nd Ed), Kindle, 2013



B.TECH I SEMESTER

BS & H	L	T	P	C
	0	0	2	1

23AM1L07 CHEMISTRY LAB

Course Objectives:

- Verify the fundamental concepts with experiments.

Course Outcomes: At the end of the course, the students will be able to

CO1: Determine the cell constant and conductance of solutions.

CO2: Prepare advanced polymer Bakelite.

CO3: Measure the strength of an acid present in secondary batteries.

CO4: Analyze the IR spectra of simple organic compounds.

CO5: Analyze water quality parameters.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	-	-	1	-	-	1	2	1	-	1	-	-	-
CO2	3					3	-	1	2	1	-	1	1	1	-
CO3	3	3		2	2	1	1	1	-	-	-	1	1	1	-
CO4	3	2		2	1	2	-	-	-	-	1	1	-	-	-
CO5	3	2		2		1	1	1	1	-	-	1	-	1	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

NAME OF THE EXPERIMENT

- 1 Measurement of 10Dq by spectrophotometric method.
- 2 Conductometric titration of strong acid vs. strong base.
- 3 Conductometric titration of weak acid vs. strong base.
- 4 Determination of cell constant and conductance of solutions.
- 5 Potentiometry - determination of redox potentials and emfs.
- 6 Determination of Strength of an acid in Lead-Acid battery.
- 7 Preparation of a Bakelite.



-
- 8 Verification of Lambert-Beer's law.
 - 9 Wavelength measurement of sample through UV-Visible spectroscopy.
 - 10 Identification of simple organic compounds by IR.
 - 11 Preparation of nanomaterials by precipitation method.
 - 12 Estimation of Ferrous Iron by dichrometry.
 - 13 Estimate the total hardness of water using standard EDTA solution.
 - 14 Determine the pH of the given sample.
 - 15 Estimation of KMnO_4 by using standard oxalic acid.
 - 16 Determine the alkalinity of sample containing Na_2CO_3 and NaOH .

Reference:

- "Vogel's Quantitative Chemical Analysis 6th Edition 6th Edition" Pearson Publications by J. Mendham, R.C.Denney, J.D.Barnes and B. Sivasankar.



B.TECH I SEMESTER

ES	L	T	P	C
	0	0	3	1.5

23AM1L08 COMPUTER PROGRAMMING LAB

Prerequisites:

Course Objectives:

The course aims to give students hands –on experience and train the month e concepts of the C- programming language.

Course Outcomes: A student after completion of the course will be able to

CO1:

Read, understand, and trace the execution of programs written in C language.

CO2:

Select the right control structure for solving the problem.

CO3:

Develop C programs which utilize memory efficiently using programming constructs like pointers.

CO4:

Develop, Debug and Execute programs to demonstrate the applications of arrays, functions, basic concepts of pointers in C.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-	2	-
CO3	-	3	3	-	-	-	-	-	-	-	-	-	3	-
CO4	-	3	3	-	-	-	-	-	-	-	-	-	3	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put ‘-’**1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

LIST OF EXPERIMENTS

UNIT-I: Introduction to Programming and Problem Solving

WEEK 1:

Objective: Getting familiar with the programming environment on the computer and writing the first program.

Suggested Experiments/Activities:

Tutorial 1: Problem-solving using Computers.

Lab 1: Familiarization with programming environment

- Basic Linux environment and its editors like Vi, Vim&Emacsetc.

- ii) Exposure to Turbo C, gcc
- iii) Writing simple programs using printf(), scanf()

WEEK 2:

Objective: Getting familiar with how to formally describe a solution to a problem in a series of finite steps both using textual notation and graphic notation.

Suggested Experiments/Activities:

Tutorial 2: Problem-solving using Algorithms and Flow charts.

Lab 2: Converting algorithms/flow charts into C Source code. Developing the algorithms/flowcharts for the following sample programs

- i) Sum and average of 3 numbers
- ii) Conversion of Fahrenheit to Celsius and vice versa
- iii) Simple interest calculation

WEEK 3:

Objective: Learn how to define variables with the desired data-type, initialize them with appropriate values and how arithmetic operators can be used with variables and constants.

Suggested Experiments/Activities:

Tutorial 3: Variable types and type conversions:

Lab 3: Simple computational problems using arithmetic expressions.

- i) Finding the square root of a given number
- ii) Finding compound interest
- iii) Area of a triangle using heron's formulae
- iv) Distance travelled by an object

WEEK 4:

Objective: Explore the full scope of expressions, type-compatibility of variables & constants and operators used in the expression and how operator precedence works.

Suggested Experiments/Activities:

Tutorial 4: Operators and the precedence and as associativity:

Lab 4: Simple computational problems using the operator precedence and associativity

- i) Evaluate the following expressions. a. $A+B*C+(D*E)+F*G$ b. $A/B*C-B+A*D/3$ c. $A+++B---A$ d. $J=(i++)+(++i)$
- ii) Find the maximum of three numbers using conditional operator
- iii) Take marks of 5 subjects in integers, and find the total, average in float

UNIT-II: Lesson: Control Structures

WEEK 5: Objective: Explore the full scope of different variants of “if construct” namely if-else, null- else, ifelse if*-else, switch and nested-if including in what scenario each one of them can be used and how to use them. Explore all relational and logical operators while writing conditionals for “if construct”.

Suggested Experiments/Activities:

Tutorial 5: Branching and logical expressions:

Lab 5: Problems involving if-then-else structures.

- i) Write a C program to find the max and min of four numbers using if-else.
- ii) Write a C program to generate electricity bill.
- iii) Find the roots of the quadratic equation.
- iv) Write a C program to simulate a calculator using switch case.
- v) Write a C program to find the given year is a leap year or not.

WEEK 6 :

Objective : Explore the full scope of iterative on structs namely while loop, do-while loop and for loop in addition to structured jump constructs like break and continue including when each of these statements is more appropriate to use.

Suggested Experiments/Activities:

Tutorial 6: Loops, while and for loops

Lab 6: Iterative problems e.g., the sum of series

- i) Find the factorial of given number using any loop.
- ii) Find the given number is a prime or not.
- iii) Computes in e and cos series
- iv) Checking a number palindrome
- v) Construct a pyramid of numbers.

UNIT-III: Lesson: Arrays and Strings

WEEK 7:

Objective: Explore the full scope of Arrays construct namely defining and initializing 1-D and 2-D and more generically n-D array and referencing individual array elements from the defined array. Using integer 1-D arrays, explore each solution linear search.

Suggested Experiments/Activities:

Tutorial 7: 1 D Arrays: searching.

Lab 7: 1D Array manipulation, linear search

- i) Find the min and max of a 1-D integer array.
- ii) Perform linear search on 1D array.
- iii) The reverse of a 1D integer array
- iv) Find 2's complement of the given binary number.
- v) Eliminate duplicate elements in an array.

WEEK 8:

Objective: Explore the difference between other arrays and character arrays that can be used as Strings by using null character and get comfortable with string by doing experiments that will reverse a string and concatenate

two strings .Explore sorting solution bubble sort using integer arrays.

Suggested Experiments / Activities:

Tutorial 8: 2 D arrays, sorting and Strings.

Lab 8: Matrix problems, String operations, Bubble sort

- i) Addition of two matrices
- ii) Multiplication two matrices
- iii) Sort array elements using bubble sort
- iv) Concatenate two strings without built-in functions
- v) Reverse a string using built-in and without built-in string functions

UNIT-IV: Lesson: Functions and Pointers

WEEK 9:

Objective: Explore pointers to manage a dynamic array of integers ,including memory allocation & ;value initialization, resizing changing and re ordering the contents of an array And memory de-allocation using malloc() ,calloc(),realloc() and free() functions. Gain experience processing command-line arguments received by C

Suggested Experiments/Activities:

Tutorial 9: Pointers, structures and dynamic memory allocation

Lab 9: Pointers and structures, memory dereference.

- i) Write a C program to find the sum of a1D array using malloc()
- ii) Write a C program to find the total, average of n students using structures
- iii) Enter n students data using calloc()and display failed students list
- iv) Read student name and marks from the command line and display the student details along with the total.
- v) Write a C program to implement realloc()

WEEK 10:

Objective: Explore the Functions ,sub- routines ,scope and exten to f variables ,doing some experiments by parameter passing using call by value .Basic methods of numerical integration

Suggested Experiments/Activities:

Tutorial 10: Functions, call by value, scope and extent,

Lab 10: Simple functions using call by value, solving differential equations using Eulers theorem.

- i) Write a C function to calculate NCR value.
- ii) Write a C function to find the length of a string.
- iii) Write a C function to transpose of a matrix.
- iv) Write a C function to demonstrate numerical integration of differential equations using Euler's method

WEEK 11: Objective: Explore how recursive solution scan be programmed by writing recursive functions that

can be invoked from the main by programming at-least five distinct problems that have naturally recursive solutions. **Suggested Experiments/Activities:**

Tutorial 11: Recursion, the structure of recursive calls

Lab 11: Recursive functions

- i) Write a recursive function to generate Fibonacci series.
- ii) Write a recursive function to find the lcm of two numbers.
- iii) Write a recursive function to find the factorial of a number
- iv) Write a C Program to implement Ackermann function using recursion.
- v) Write a recursive function to find the sum of series.

WEEK 12:

Objective: Explore the basic difference between normal and pointer variables, Arithmetic operations using pointers and passing variables to functions using pointers

Suggested Experiments/Activities:

Tutorial 12: Call by reference, dangling pointers

Lab 12: Simple functions using Call by reference, Dangling pointers.

- i) Write a C program to swap two numbers using call by reference.
- ii) Demonstrate Dangling pointer problem using a C program.
- iii) Write a C program to copy one string into another using pointer.
- iv) Write a C program to find no of lower case ,uppercase ,digits and other characters using pointers.

UNIT-V: Lesson: User Defined Data types & File handling

WEEK 13:

Objective : Experiment with C Structures ,Unions ,bit fields and self-referential structures (Singly linked lists) and nested structures

Suggested Experiments/Activities:

Tutorial 13: Bit fields, Self-Referential Structures, Linked lists

Lab 13: Bit fields, linked lists Read and print a date using dd/mm/yyyy format using bit-fields and differentiate the same without using bit-fields

- i) Create and display a singly linked list using self-referential structure.
- ii) Demonstrate the differences between structures and unions using a C program.
- iii) Write a C program to shift/rotate using bit fields.

Write a C program to copy one structure variable to another structure of the same type.

WEEK 14:

Objective: To understand data files and file handling with various file I/O functions .Explore the differences between text and binary files.

Suggested Experiments/Activities:

Tutorial 14: File handling



Lab 14: File operations

- i) Write a C program to write and read text into a file.
- ii) Write a C program to write and read text into a binary file using f read() and f write()
- iii) Copy the contents of one file to another file.
- iv) Write a C program to merge two files into the third file using command-line arguments.
- v) Find no. of lines, words and characters in a file
- vi) Write a C program to print last n characters of a given file.

Text Books

- 5. Ajay Mittal, Programming in C: A practical approach, Pearson.
- 6. Byron Gottfried, Schaum's Outline of Programming with C, Mc Graw Hill

Reference Books

- 4. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
- 5. 2. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE



B.TECH: I SEMESTER

ES	L	T	P	C
	0	0	3	1.5

23AM1L09 ENGINEERING WORKSHOP

Course Objective: To familiarize students with wood working, sheet metal operations, fitting, electrical house wiring skills, and basic repairs of two-wheeler vehicle

Course Outcomes:

- CO1 Identify work shop tools and their operational capabilities.
- CO2 Practice on manufacturing of components using work shop trades including fitting, carpentry, foundry and welding.
- CO3 Apply fitting operations in various applications.
- CO4 Apply basic electrical engineering knowledge for House Wiring Practice

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	3	2	3	-	-	2	-	-	-	-	2	1
CO2	3	2	3	3	-	2	-	3	-	-	-	-	3	2
CO3	3	1	3	3	1	3	-	2	-	-	-	-	3	1
CO4	2	-	3	3	-	2	-	3	-	-	-	-	3	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

1. **Demonstration:** Safety practices and precautions to be observed in workshop.
2. **Wood Working:** Familiarity with different types of woods and tools used in wood working and make following joints.
 - a) Half – Lap joint
 - b) Mortise and Tenon joint
 - c) Corner Dovetail joint or Bridle joint
3. **Sheet Metal Working:** Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets.
 - a) Tapered tray
 - b) Conical funnel
 - c) Elbow pipe
 - d) Brazing
4. **Fitting:** Familiarity with different types of tools used in fitting and do the following fitting exercises.

- a) V-fit
 - b) Dovetail fit
 - c) Semi-circular fit
 - d) Bicycle tire puncture and change of two-wheeler tyre
5. **Electrical Wiring:** Familiarity with different types of basic electrical circuits and make the following connections.
- a) Parallel and series
 - b) Two-way switch
 - c) Godown lighting
 - d) Tube light
 - e) Three phase motor
 - f) Soldering of wires
6. **Foundry Trade:** Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.
7. **Welding Shop:** Demonstration and practice on Arc Welding and Gas welding.
Preparation of Lap joint and Butt joint.
8. **Plumbing:** Demonstration and practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameters.
9. **Basic repairs of Two-wheeler vehicle** – Demonstration of working of two-wheeler vehicle and its repairs

Textbooks:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.
2. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co., 2015 & 2017.

Reference Books:

1. Elements of Workshop Technology, Vol. I by S. K. Hajra Choudhury & Others,
Media Promoters and Publishers, Mumbai. 2007, 14th edition
2. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004.
3. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.;
Atul Prakashan, 2021-22.



B.TECH I SEMESTER

ES	L	T	P	C
	0	0	1	0.5

23AM1L10 NSS/NCC/SCOUTS & GUIDES/ COMMUNITY SERVICE

Course Objectives:

The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service.

Course Outcomes: After completion of the course the students will be able to

CO1: Understand the importance of discipline, character and service motto.

CO2: Solve some societal issues by applying acquired knowledge, facts, and techniques.

CO3: Explore human relationships by analyzing social problems.

CO4: Determine to extend their help for the fellow beings and downtrodden people.

CO5: Develop leadership skills and civic responsibilities.

UNIT I Orientation

General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.

Activities:

- Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills
- Conducting orientations programs for the students –future plans-activities-releasing road map etc.
- Displaying success stories-motivational biopics- award winning movies on societal issues etc.
- Conducting talent show in singing patriotic songs-paintings- any other contribution.

UNIT II Nature & Care

Activities:

- Best out of waste competition.
- Poster and signs making competition to spread environmental awareness.
- Recycling and environmental pollution article writing competition.
- Organising Zero-waste day.
- Digital Environmental awareness activity via various social media platforms.
- Virtual demonstration of different eco-friendly approaches for sustainable living.
- Write a summary on any book related to environmental issues.

UNIT III Community Service

Activities:

- Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media- authorities- experts-etc.



-
- ii) Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS,
 - iii) Conducting consumer Awareness. Explaining various legal provisions etc.
 - iv) Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education.
 - v) Any other programmes in collaboration with local charities, NGOs etc.

References Books

1. Nirmalya Kumar Sinha & Surajit Majumder, *A Text Book of National Service Scheme* Vol;.I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2. *Red Book - National Cadet Corps* – Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3. Davis M. L. and Cornwell D. A., “Introduction to Environmental Engineering”, McGraw Hill, New York 4/e 2008
4. Masters G. M., Joseph K. and Nagendran R. “Introduction to Environmental Engineering and Science”, Pearson Education, New Delhi. 2/e 2007
5. Ram Ahuja. *Social Problems in India*, Rawat Publications, New Delhi.



B.TECH I SEMESTER	BS & H	L	T	P	C
		3	0	0	3
23AM2T01 DIFFERENTIAL EQUATIONS & VECTOR CALCULUS					

Prerequisites: Basic Knowledge of Differentiation & Integration

Course Objectives:

- To enlighten the learners in the concept of Differential equations and multivariable calculus.
- To furnish the learners with basic concepts and techniques at plus two level to lead them into advanced level by handling various real-world applications.

Course Outcomes: At the end of the course, student will be able to

- CO1:** Analyze the differential Equations and then apply the methods to the real life situations
- CO2:** Solve the Differential equations related to various engineering fields
- CO3:** Identify solution methods for partial differential equations that model physical processes.
- CO4:** Interpret the physical meaning of different operators such as gradient, curl and Divergence
- CO5:** Estimate the work done against a field, circulation and flux using vector calculus

COURSE ARTICULATION MATRIX (Correlation between Cos, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	-	2	-	-	-	-	-	-	-	2	-	-	-
CO2	3	3	-	2	-	-	-	-	-	-	-	2	-	-	-
CO3	3	3	-	-	-	-	-	-	-	-	-	1	-	-	-
CO4	3	3	-	-	-	-	-	-	-	-	-	2	-	-	-
CO5	3	3	-	-	-	-	-	-	-	-	-	2	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I: Differential equations of first order and first degree

Linear differential equations–Bernoulli’s Equations–Exact equations and equations reducible to exact form. Applications: Newton’s Law of cooling, Law of natural growth and decay- Electrical circuits.

UNIT-II: Linear differential equations of higher order (Constant Coefficients)

Definitions, homogenous and non-homogenous, complimentary function, particular integral
General solution, Wronskian, Method of variation of parameters. Simultaneous linear equations,
Applications to L-C-R Circuit problems and Simple Harmonic motion.

UNIT-III: Partial Differential Equations

Introduction and formation of Partial Differential Equations by elimination of arbitrary constants
and arbitrary functions, solutions of first order linear equations using Lagrange's method.
Homogeneous & Non-Homogeneous Linear Partial differential equations with constant
coefficients.

UNIT-IV: Vector differentiation

Scalar and vector point functions, vector operator Del, Del applied to scalar point functions-
Gradient, Directional derivative, Del applied to vector point functions-Divergence and Curl,
Solenoidal & Irrotational vectors, Scalar Potential function, vector identities (Without proofs)

UNIT-V: Vector integration

Line integral-circulation-work done, surface integral-flux, Green's theorem in the plane (without
proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and
related problems.

Text Books

1. Higher Engineering Mathematics, B.S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition

Reference Books

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, R.K. Jain and S.R.K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5. Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017



B.TECH II SEMESTER

	L	T	P	C
BS & H	3	0	0	3

23AM2T02 ENGINEERING PHYSICS

Course Objectives:

- To bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

Course Outcomes: At the end of the course, student will be able to

- CO1:** Analyze the intensity variation of light due to polarization, interference and diffraction.
- CO2:** Summarize various types of polarization of dielectrics and classify the magnetic materials.
- CO3:** Familiarize with the basics of crystals and their structures.
- CO4:** Explanation of basic concepts of Quantum Mechanics and free electron theories.
- CO5:** Identify the types of solids and conductivity mechanisms in semiconductors.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3		2	-	-	-	-	-	-	-	-	-			-
CO2	3		2	-	-	-	-	-	-	-	-	-			-
CO3	3	2		2	-	-	-	-	-	-	-	-			-
CO4	3		2	-	-	-	-	-	-	-	-	-			-
CO5	3	2	2	2	-	-	-	-	-	-	-	-			-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I: Wave Optics

Interference: Introduction - Principle of superposition –Interference of light - Interference in thin films (Reflection Geometry) & applications - Colours in thin films- Newton’s Rings, Determination of wavelength and refractive index.

Diffraction: Introduction - Fresnel and Fraunhofer diffractions - Fraunhofer diffraction due to single slit, double slit & N-slits (Qualitative) – Diffraction Grating - Dispersive power and resolving power of Grating (Qualitative).

Polarization: Introduction -Types of polarization -Polarization by reflection, refraction and Double refraction - Nicol's Prism -Half wave and Quarter wave plates.

UNIT-II: Dielectric and Magnetic Materials

Dielectric Materials: Introduction-Dielectric polarization-Dielectric polarizability, Susceptibility, Dielectric constant and Displacement Vector – Relation between the electric vectors-Types of polarizations-Electronic(Quantitative) ,Ionic(Quantitative) and Orientation

Polarizations (Qualitative) - Lorentz internal field - Clausius- Mossotti equation – complex dielectric constant – Frequency dependence of polarization– dielectric loss

Magnetic Materials: Introduction-Magnetic dipole moment-Magnetization-Magnetic susceptibility and permeability – Atomic origin of magnetism - Classification of magnetic materials: Dia ,para ,Ferro ,anti-ferro & Ferri magnetic materials-Domain concept for Ferromagnetism &Domain walls(Qualitative)-Hysteresis-soft and hard magnetic materials- Applications.

UNIT-III: Crystallography and X-ray diffraction

Crystallography: Space lattice, Basis, Unit Cell and lattice parameters – Bravais Lattices –crystal systems (3D) – coordination number - packing fraction of SC, BCC & FCC – Miller indices– separation between successive(hkl)planes.

Ray diffraction: Bragg's law- X-ray Diffractometer–crystal structure determination by Laue's and powder methods

UNIT-IV: Quantum Mechanics and Free electron Theory

Quantum Mechanics :Dual nature of matter–Heisenberg's Uncertainty Principle–Significance and properties of wave function – Schrodinger's time independent and dependent wave equations– Particle in a one-dimensional infinite potential well.

Free Electron Theory: Classical free electron theory (Qualitative with discussion of merits and demerits) – Quantum free electron theory – electrical conductivity based on quantum free electron theory-Fermi-Dirac distribution –Density of states -Fermi energy

UNIT-V: Semiconductors

Semiconductors: Formation of energy bands – classification of crystalline solids – Intrinsic semiconductors: Density of charge carriers – Electrical conductivity – Fermi level – Extrinsic semiconductors: density of charge carriers–dependence of Fermi energy on carrier concentration and temperature-Drift and diffusion currents– Einstein's equation–Hall effect and its applications.

Text Books

1. A Textbook of Engineering Physics, M. N. Avadhanulu ,P.G. Kshirsagar & TVS Arun Murthy, S. Chand Publications, 11th Edition 2019.
2. Engineering Physics -D.K.Bhattacharya and Poonam Tandon, Oxford press(2015).
3. Engineering Physics –P.K.Palanisamy- Scitech Publications.



Reference Books

1. Engineering Physics -B.K.Pandey and S.Chaturvedi, Cengage Learning 2021.
2. Engineering Physics –Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018.
3. Engineering Physics”-SanjayD.Jain,D.SahasrabudheandGirish,UniversityPress.2010
4. Engineering Physics -M.R.Srinivasan, New Age international publishers (2009).

WebResources:<https://www.loc.gov/rr/scitech/selected-internet/physics.html>



B.TECH II SEMESTER

ES	L	T	P	C
	3	0	0	3

23AM2T03 BASIC ELECTRICAL & ELECTRONICS ENGINEERING

Prerequisites: Physics and Mathematics

Course Objectives:

- To expose to the field of electrical & electronics engineering, laws and principles of electrical/ electronic engineering and to acquire fundamental knowledge in the relevant field

Course Outcomes: At the end of the course, student will be able to

- CO1:** Describe fundamental laws, operating principles of motors/generators, MC/MI instruments
- CO2:** Demonstrate the working of electrical machines, measuring instruments and power generation stations.
- CO3:** Apply mathematical tools and fundamental concepts to derive various equations related to electrical circuits and machines & Calculate electrical load and electricity bill of residential and commercial buildings.
- CO4:** Demonstrate the working of different semiconductor devices and electronic circuits
- CO5:** Understand the basic functioning of Electronic circuits
- CO6:** Understand the basic functioning of digital electronics

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	-	-	-	-	3	3
CO2	3	3	-	-	-	-	-	-	-	-	-	-	3	3
CO3	3	3		-	-	-	-	-	-	-	-	-	3	3
CO4	3	3	3	-	-	-	-	-	-	-	-	-	3	3
CO5	3	3	3	-	1	-	-	-	-	-	-	-	3	3
CO6	3	3	3										3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put ‘-’ 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

SYLLABUS

UNIT-I: DC & AC Circuits

DC Circuits: Electrical circuit elements (R, L and C), Ohm’s Law and its limitations, KCL & KVL, series, parallel, series-parallel circuits, Super Position theorem, Simple numerical problems.

AC Circuits: A.C. Fundamentals: Equation of AC Voltage and current, waveform, time period, frequency, amplitude, phase, phase difference, average value, RMS value, form factor, peak factor,

Voltage and current relationship with phasor diagrams in R, L, and C circuits, Concept of Impedance, Active power, reactive power and apparent power, Concept of power factor (Simple Numerical problems).

UNIT-II: Machines and Measuring Instruments

Machines: Construction, principle and operation of (i) DC Motor, (ii) DC Generator, (iii) Single Phase Transformer, (iv) Three Phase Induction Motor and (v) Alternator, Applications of electrical machines.

Measuring Instruments: Construction and working principle of Permanent Magnet Moving Coil (PMMC), Moving Iron (MI) Instruments and Wheat Stone bridge.

UNIT-III: Energy Resources, Electricity Bill & Safety Measures

Energy Resources: Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Nuclear, Solar & Wind power generation.

Electricity bill: Power rating of household appliances including air conditioners, PCs, Laptops, Printers, etc. Definition of “unit” used for consumption of electrical energy, two-part electricity tariff, calculation of electricity bill for domestic consumers.

Equipment Safety Measures: Working principle of Fuse and Miniature circuit breaker (MCB), merits and demerits. Personal safety measures: Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

UNIT-IV: SEMICONDUCTOR DEVICES

Introduction - Evolution of electronics – Vacuum tubes to nano electronics - Characteristics of PN Junction Diode — Zener Effect — Zener Diode and its Characteristics. Bipolar Junction Transistor — CB, CE, CC Configurations and Characteristics — Elementary Treatment of Small Signal CE Amplifier.

UNIT-V: BASIC ELECTRONIC CIRCUITS AND INSTRUMENTATION

Rectifiers and power supplies: Block diagram description of a dc power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple zener voltage regulator. Amplifiers: Block diagram of Public Address system, Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response. Electronic Instrumentation: Block diagram of an electronic instrumentation system.

UNIT VI: DIGITAL ELECTRONICS

Overview of Number Systems, Logic gates including Universal Gates, BCD codes, Excess-3 code, Gray code, Hamming code. Boolean Algebra, Basic Theorems and properties of Boolean Algebra, Truth Tables and Functionality of Logic Gates – NOT, OR, AND, NOR, NAND, XOR and XNOR.

Simple combinational circuits–Half and Full Adders. Introduction to sequential circuits, Flip flops, Registers and counters (Elementary Treatment only)

Text Books

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition
4. R. L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.
5. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009

Reference Books

1. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, Mc Graw Hill, 2019, Fourth Edition
2. Principles of Power Systems, V.K. Mehtha, S.Chand Technical Publishers, 2020
3. Basic Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2017
4. Basic Electrical and Electronics Engineering, S. K. Bhattacharya, Person Publications, 2018, Second Edition.
5. R. S. Sedha, A Textbook of Electronic Devices and Circuits, S. Chand & Co, 2010.
6. Santiram Kal, Basic Electronics- Devices, Circuits and IT Fundamentals, Prentice Hall, India, 2002.
7. R. T. Paynter, Introductory Electronic Devices & Circuits – Conventional Flow Version, Pearson Education, 2009.



B.TECH II SEMESTER

ES	L	T	P	C
	3	0	0	3

23AM2T04 DATA STRUCTURES

Prerequisites: Basic Programming Knowledge

Course Objectives:

- To provide the knowledge of basic data structures and their implementations
- To understand importance of data structures in context of writing efficient programs.
- To develop skills to apply appropriate data structures in problem solving.

Course Outcomes: At the end of the course, Student will be able to

- CO1:** Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.
- CO2:** Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.
- CO3:** Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.
- CO4:** Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in Graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges.
- CO5:** Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO2	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO3	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO4	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO5	2	3	3	-	-	-	-	-	-	-	-	-	3	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I: Lesson: Introduction to Linear Data Structures

Introduction: Definition and importance of linear data structures, Abstract data types (ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures.

Searching Techniques: Linear & Binary Search,

Sorting Techniques: Bubble sort, Selection sort, Insertion Sort, Quick sort, Merge Sort.

UNIT-II: Lesson: Linked Lists

Singly linked lists: Representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.

UNIT-III: Lesson: Stacks

Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation and conversions: Infix, Prefix and Postfix expression backtracking, reversing list etc.

UNIT-IV: Lesson: Queues

Queues: Introduction to queues: properties and operations, implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.

Deque: Introduction to deque (double-ended queues), Operations on deque and their applications.

UNIT-V: Lesson: Trees and Hashing

Trees: Introduction to Trees, Representation of Trees in memory, Binary Tree Traversal Algorithms, Binary Search Tree– Insertion, Deletion & Traversal.

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.

Text Books

7. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
8. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan AndersonFreed, Silicon Press, 2008

Reference Books

6. Algorithms and Data Structures: The Basic Tool box by Kurt Mehlhorn and Peter Sanders
7. C Data Structures and Algorithms by Alfred V.Aho, Jeffrey D.Ullman, and John E. Hopcroft
8. Problem Solving with Algorithms and Data Structures by Brad Miller and David Ranum



- 9.** Introduction to Algorithms by Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest, and Clifford Stein
- 10.** Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick



B.TECH II SEMESTER

ES	L	T	P	C
	1	0	4	3

23AM2T05 ENGINEERING GRAPHICS

Course Objectives:

- To enable the students with various concepts like dimensioning, conventions and standards related to Engineering Drawing
- To impart knowledge on the projection of points, lines and plane surfaces
- To improve the visualization skills for better understanding of projection of solids
- To develop the imaginative skills of the students required to understand Section of solids and Developments of surfaces.
- To make the students understand the viewing perception of a solid object in Isometric and Perspective projections.

Course Outcomes: At the end of the course, student will be able to

- CO1:** Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections.
- CO2:** Draw and interpret orthographic projections of points, lines, planes and solids in front, top and side views
- CO3:** Understand and draw projection of solids in various positions in first quadrant
- CO4:** Explain principles behind development of surfaces.
- CO5:** Prepare isometric and perspective sections of simple solids.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	1	-	-	-	-	-	-	-	1	3	3
CO2	2	2	1	1	-	-	-	-	-	-	-	1	3	3
CO3	3	2	2	2	-	-	-	-	-	-	-	1	3	3
CO4	3	2	2	2	-	-	-	-	-	-	-	1	3	3
CO5	3	2	2	2	-	-	-	-	-	-	-	1	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I:

Introduction: Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.

Curves: construction of ellipse, parabola and hyperbola by general, Cycloids, Involute, Normal and tangent to Curves.

Scales: Plain scales, diagonal scales and vernier scales.

UNIT-II:

Orthographic Projections: Reference plane, importance of reference lines or Plane, Projections of a point situated in any one of the four quadrants.

Projections of Straight Lines: Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of Straight Line Inclined to both the reference planes

Projections of Planes: regular planes Perpendicular to both reference planes, parallel to one reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

UNIT-III:

Projections of Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to another plane.

UNIT-IV:

Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only.

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

UNIT-V:

Conversion of Views: Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Computer graphics: Creating 2D&3D drawings of objects including PCB and Transformations using Auto CAD (*Not for end examination*).

Vocabulary: Technical Jargons

Text Books

1. N. D. Bhatt, Engineering Drawing, Charotar Publishing House, 2016



Reference Books

1. Engineering Drawing, K.L. Narayana and P. Kannaiah, Tata McGraw Hill, 2013.
2. Engineering Drawing, M.B.Shah and B.C. Rana, Pearson Education Inc,2009.
3. Engineering Drawing with an Introduction to AutoCAD, Dhananjay Jolhe, Tata McGraw Hill, 2017.



B.TECH II SEMESTER

BS & H	L	T	P	C
	0	0	2	1

23AM2L06 ENGINEERING PHYSICS LAB

Course Objectives:

- To study the concepts of optical phenomenon like interference, diffraction etc., recognize the importance of energy gap in the study of conductivity and Hall effect in semiconductors and study the parameters and applications of dielectric and magnetic materials by conducting experiments.

Course Outcomes: At the end of the course, student will be able to

CO1: Implement the basic principles of Optics through various phenomenon of light

CO2: Implement the basic principles of Mechanics to measure various physical parameters.

CO3: Enhance the knowledge of usage of electronic devices in various applications

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	-	-	-	-	-	-	3	-	3	-			-
CO2	3	3	-	-	-	-	-	-	3	-	3	-			-
CO3	3	3		-	-	-	-	-	3	-	3	-			-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

LIST OF EXPERIMENTS

- Determination of radius of curvature of a given Plano-convex lens by Newton's rings.
- Determination of wave lengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration.
- Verification of Brewster's law
- Determination of dielectric constant using charging and discharging method.
- Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
- Determination of wave length of Laser light using diffraction grating.
- Estimation of Planck's constant using photo electric effect.
- Determination of the resistivity of semiconductors by four probe methods.
- Determination of energy gap of a semiconductor using p-n junction diode.
- Magnetic field along the axis of a current carrying circular coil by Stewart Gee's

Method.

11. Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall effect.
12. Determination of temperature coefficients of a thermistor.
13. Determination of acceleration due to gravity and radius of Gyration by using a compound pendulum.
14. Determination of magnetic susceptibility by Kundt's tube method.
15. Determination of rigidity modulus of the material of the given wire using Torsional pendulum.
16. Sonometer: Verification of laws of stretched string.
17. Determination of young's modulus for the given material of wooden scale by non-uniform bending (or double cantilever) method.
18. Determination of Frequency of electrically maintained tuning fork by Melde's experiment.

Note: Any TEN of the listed experiments are to be conducted. Out of which any TWO experiments may be conducted in virtual mode.

References:

- A Textbook of Practical Physics- S. Balasubramanian, M.N.Srinivasan, S.Chand Publishers, 2017.

Web Resources

- www.vlab.co.in
- <https://phet.colorado.edu/en/simulations/filter?subjects=physics&type=html.prototype>



B.TECH II SEMESTER

ES L T P C
0 0 3 1.5

23AM2L07 ELECTRICAL AND ELECTRONICS ENGINEERING WORKSHOP

Prerequisites: Basic Electrical and electronics engineering

Course Objectives:

- To expose to the field of electrical & electronics engineering, laws and principles of electrical/electronic engineering and to acquire fundamental knowledge in the relevant field **Course**.
- To impart knowledge on the principles of digital electronics and fundamentals of electron devices & its applications.

Course Outcomes: At the end of the course, student will be able to

- CO1:** Measure voltage, current and power and resistance in an electrical circuit.
- CO2:** Discover critical field resistance and critical speed of DC shunt generators
- CO3:** Investigate the effect of reactive power and power factor in electrical loads.
- CO4:** Identify & testing of various electronic components and understand measurement using electronic instruments
- CO5:** Understand the usage of electronic measuring instruments and explain the operation of a digital circuit.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3		-	-	3	-	-	-	-	-	-	-	3	3
CO2	3		-	-	3	-	-	-	-	-	-	-	3	3
CO3	3			-	3	-	-	-	-	-	-	-	3	3
CO4	3		3	-	3	-	-	-	-	-	-	-	3	3
CO5	3		3	-	3	3	-	-	-	-	-	-	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

Activities: 1. Familiarization of commonly used Electrical & Electronic Workshop Tools: Bread board, Solder, cables, relays, switches, connectors, fuses, Cutter, plier, screwdriver set, wire stripper, flux, knife/blade, soldering iron, de-soldering pump etc.

- Provide some exercises so that hardware tools and instruments are learned to be used by the students.
- 2. Familiarization of Measuring Instruments like Voltmeters, Ammeters, multimeter, LCR-Q meter, Power Supplies, CRO, DSO, Function Generator, Frequency counter.
 - Provide some exercises so that measuring instruments are learned to be used by the students.
- 3. Components:
 - Familiarization/Identification of components (Resistors, Capacitors, Inductors, Diodes, transistors, IC's etc.) – Functionality, type, size, colour coding package, symbol, cost etc.
 - Testing of components like Resistor, Capacitor, Diode, Transistor, ICs etc. - Compare values of components like resistors, inductors, capacitors etc with the measured values by using instruments

LIST OF EXPERIMENTS:

PART -A

1. Verification of KCL and KVL
2. Verification of Superposition theorem
3. Measurement of Resistance using Wheat stone bridge
4. Magnetization Characteristics of DC shunt Generator
5. Measurement of Power and Power factor using Single-phase wattmeter
6. Measurement of Earth Resistance using Megger
7. Calculation of Electrical Energy for Domestic Premises

PART -B

1. Plot V-I characteristics of PN Junction diode A) Forward bias B) Reverse bias.
2. Plot V – I characteristics of Zener Diode and its application as voltage Regulator.
3. Implementation of half wave and full wave rectifiers
4. Plot Input & Output characteristics of BJT in CE and CB configurations
5. Frequency response of CE amplifier.
6. Simulation of RC coupled amplifier with the design supplied
7. Verification of Truth Table of AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR gates using ICs.
8. Verification of Truth Tables of S-R, J-K& D flip flops using respective ICs

Reference Books:

PART –A

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Editio

PART -B

1. R. L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.
2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009
3. R. T. Paynter, Introductory Electronic Devices & Circuits – Conventional Flow Version, Pearson Education, 2009.



B.TECH II SEMESTER

ES	L	T	P	C
	0	0	2	1

23AM2L08 IT WORKSHOP

Prerequisites:

Course Objectives:

To introduce the internal parts of a computer, peripherals, I/O ports, connecting cables

- To demonstrate configuring the system as Dual boot both Windows and other Operating Systems Viz. Linux, BOSS
- To teach basic command line interface commands on Linux.
- To teach the usage of Internet for productivity and self-paced life-long learning
- To introduce Compression, Multimedia and Antivirus tools and Office Tools such as Word processors, Spreadsheets and Presentation tools.

Course Outcomes: A student after completion of the course will be able to

CO1: Perform Hardware troubleshooting.

CO2: Understand Hardware components and inter dependencies.

CO3: Safeguard computer systems from viruses/worms.

CO4: Document/ Presentation preparation.

CO5: Perform calculations using spreadsheets.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	2	-	-	-	1	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO3	3	2	-	-	-	-	-	-	2	-	-	-	1	-
CO4	3	-	-	-	-	-	-	-	-	2	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	2	-	-	1	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

PC Hardware & Software Installation

Task1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task2: Every student should disassemble and assemble the PC back to working condition .Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC.A video would be given as part of the course content.

Task3: Every student should individually install MS windows on the personal computer .Lab instructor should verify the installation and follow it up with a Viva.

Task4: Every student should install Linux on the computer .This computer should have windows installed. The system should be configured as dual boot (VMWare) with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva.

Task5: Every student should install BOSS on the computer .The system should be configured as dual boot (VMWare) with both Windows and BOSS. Lab instructors should verify the installation and follow it up with a Viva.

Internet &World Wide Web

Task1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet .In the process they configure the TCP/IP setting .Finally students should demonstrate, to the instructor, how to access the websites and email. If there is No internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings , bookmarks, search tool bars and popup blockers. Also, plug-ins like Macro media Flash and JRE for applets should be configured.

Task3: Search Engines & Netiquette: Students should know what search engines are and how to use the search engines .A few to pictures would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

LaTeX and WORD

Task1–WordOrientation: The mentor needs to give an overview of LaTeX and Microsoft (MS)office or equivalent (FOSS)tool word :Importance of LaTeX and MSoffice or equivalent (FOSS) tool Word as word Processors ,Details of the four tasks and features that would b covered in each ,Using LaTeX and word– Accessing ,overview of toolbars ,saving files ,Using help and resources ,rulers, format painter in word.

Task2:Using LaTeX and Word to create a project certificate .Features to be covered:- Formatting Fonts in word ,Drop Capin word ,Applying Text effects ,Using Character Spacing, Borders and Colors ,Inserting Header and Footer ,Using Date and Time option in both LaTeX and Word.

Task3: Creating project abstract Features to be covered:-Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task4: Creating a Newsletter :Features to be covered:- Table of Content ,News paper columns, Images from

files and clipart ,Drawing toolbar and Word Art ,Formatting Images ,Textboxes, Paragraphs and Mail Merge in word.

EXCEL

Excel Orientation: The mentor needs to tell the importance of MS office or equivalent (FOSS) tool Excel as a Spread sheet tool ,give the details of the four tasks and features that would be covered in each. Using Excel– Accessing, overview of toolbars, saving excels files, Using help and resources.

Task1: Creating a Scheduler- Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text.

Task2: Calculating GPA-.Features to be covered:-Cell Referencing ,Formulae in excel– average ,std. deviation, Charts ,Renaming and Inserting worksheets ,Hyper linking ,Count function.

LOOKUP/VLOOKUP

Task3: Split cells , freeze panes ,group and outline, Sorting, Boolean and logical operators, Conditional formatting.

POWER POINT

Task1: Students will be working on basic power point utilities and tools which help them create basic power point presentations. PPT Orientation, Slide Layouts ,Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task2: Interactive presentations-Hyperlinks, Inserting–Images, ClipArt, Audio, Video, Objects, Tables and Charts.

Task3: Master Layouts (slide, template, and notes), Types of views (Basic, presentation, slide slotter, notes etc), and Inserting– Background, textures, Design Templates, Hidden slides.

AI TOOLS– Chat GPT

Task1: Prompt Engineering: Experiment with different types of prompts to see how the model responds .Try asking questions, starting conversations, or even providing in complete sentences to see how the model completes them.

- Ex: Prompt:"You are knowledge able AI. Please answer the following question: What is the capital of France?"

Task2: Creative Writing: Use the model as a writing assistant .Provide the beginning of a story or a description of ascene , and let the model generate the rest of the content. This can be a fun way to brainstorm creative ideas

- Ex: Prompt:"In a world where gravity suddenly stopped working, people started floating upwards. Write a story about how society adapted to this new reality."

Task3: Language Translation: Experiment with translation tasks by providing a sentence in one language and asking the model to translate it into another language .Compare the out put to see how accurate and fluent the translations are.



- Ex: Prompt: "Translate the following English sentence to French : 'Hello ,how are you doing today?'"

Reference Books

11. Comdex Information Technology course tool kit, Vikas Gupta, WILEY Dream tech, 2003
12. The Complete Computer upgrade and repair book, Chery la Schmidt, WILEY Dream tech, 2013, 3rd edition
13. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education, 2012, 2nd edition
14. PC Hardware -A Handbook, Kate J. Chase, PHI(Microsoft)
15. LaTeX Companion, Leslie Lamport, PHI/Pearson.
16. IT Essentials PC Hardware and Software Companion Guide ,David An finson and Ken Quamme. – CISCO Press, Pearson Education, 3rd edition
17. IT Essentials PC Hardware and Software Labs and Study Guide, Patrick Regan–CISCO Press, Pearson Education, 3rd edition



B.TECH I SEMESTER

ES	L	T	P	C
	0	0	3	1.5

23AM2L09 Data Structures Lab

Prerequisites: Basic Programming Knowledge

Course Objectives:

- The course aims to strengthen the ability of the students to identify and apply the suitable data structure for the given real-world problem. It enables them to gain knowledge in practical applications of data structures.

Course Outcomes: At the end of the course, Student will be able to

CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.

CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.

CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.

CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in Graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges.

CO5: Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO2	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO3	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO4	2	3	3	-	-	-	-	-	-	-	-	-	3	-
CO5	2	3	3	-	-	-	-	-	-	-	-	-	3	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

LIST OF EXPERIMENTS:

Exercise 1: Array Manipulation

- Write a program to reverse an array.
- C Programs to implement the Searching Techniques– Linear & Binary Search

- iii) C Programs to implement Sorting Techniques– Bubble, Selection and Insertion Sort

Exercise 2: Linked List Implementation

- i) Implement a singly linked list and perform insertion and deletion operations.
- ii) Develop a program to reverse a linked list iteratively and recursively.
- iii) Solve problems involving linked list traversal and manipulation.

Exercise 3: Linked List Applications

- i) Create a program to detect and remove duplicates from a linked list.
- ii) Implement a linked list to represent polynomials and perform addition.
- iii) Implement a double-ended queue(deque) with essential operations.

Exercise 4: Double Linked List Implementation

- i) Implement a doubly linked list and perform various operations to understand its properties and applications.
- ii) Implement a circular linked list and perform insertion, deletion, and traversal.

Exercise 5: Stack Operations

- i) Implement a stack using arrays and linked lists.
- ii) Write a program to evaluate a postfix expression using a stack.
- iii) Implement a program to check for balanced parentheses using a stack.

Exercise 6: Queue Operations

- i) Implement a queue using arrays and linked lists.
- ii) Develop a program to simulate a simple printer queue system.
- iii) Solve problems involving circular queues.

Exercise 7: Stack and Queue Applications

- i) Use a stack to evaluate an infix expression and convert it to postfix.
- ii) Create a program to determine whether a given string is a palindrome or not.
- iii) Implement a stack or queue to perform comparison and check for symmetry.

Exercise 8: Binary Search Tree

- i) Implementing a BST using Linked List.
- ii) Traversing of BST.

Exercise 9: Hashing

- i) Implement a hash table with collision resolution techniques.
- ii) Write a program to implement a simple cache using hashing.

Text Books

- 9. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
- 10. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, Silicon Press, 2008

Reference Books

- 1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders



2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures "by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
- 5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms by Robert Sedgewick.**

B.TECH: II SEMESTER

	L	T	P	C
PC	0	0	1	0.5

23EC2L10 HEALTH AND WELLNESS, YOGA AND SPORTS**Course Objectives:**

The main objective of introducing this course is to make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.

Course Outcomes: After completion of the course the student will be able to

CO1: Understand the importance of yoga and sports for Physical fitness and sound health.

CO2: Demonstrate an understanding of health-related fitness components.

CO3: Compare and contrast various activities that help enhance their health.

CO4: Assess current personal fitness levels.

CO5: Develop Positive Personality

SYLLABUS**UNIT I**

Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity
Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index (BMI) of all age groups.

Activities:

- Organizing health awareness programmes in community
- Preparation of health profile
- Preparation of chart for balance diet for all age groups

UNIT II

Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context, classification of yoga, Physiological effects of Asanas- Pranayama and meditation, stress management and yoga, Mental health and yoga practice.

Activities:

Yoga practices – Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar

UNIT III

Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games

Activities:

- i Participation in one major game and one individual sport viz., Athletics, Volleyball, Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc.
Practicing general and specific warm up, aerobics
- ii Practicing cardiorespiratory fitness, treadmill, run test, 9 min walk, skipping and running.

Reference Books:

1. Gordon Edlin, Eric Golanty. Health and Wellness, 14th Edn. Jones & Bartlett Learning, 2022
2. T.K.V.Desikachar. The Heart of Yoga: Developing a Personal Practice
3. Archie J.Bahm. Yoga Sutras of Patanjali, Jain Publishing Company, 1993
4. Wiseman, John Lofty, SAS Survival Handbook: The Ultimate Guide to Surviving Anywhere
Third Edition, William Morrow Paperbacks, 2014
5. The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc.2014

General Guidelines:

1. Institutes must assign slots in the Timetable for the activities of Health/Sports/Yoga.
2. Institutes must provide field/facility and offer the minimum of five choices of as many as Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit.
Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva voce on the subject.



B.TECH III SEMESTER

	L	T	P	C
BS & H	3	0	0	3

23AM3T01 DISCRETE MATHEMATICS & GRAPH THEORY

Course Objectives:

- To introduce the students to the topics and techniques of discrete methods and combinatorial reasoning.
- To introduce a wide variety of applications. The algorithmic approach to the solution of problems is fundamental in discrete mathematics, and this approach reinforces the close ties between this discipline and the area of computer science.

Course Outcomes:

CO1: Build skills in solving mathematical problems

CO2: Understand the concepts and perform the operations related to sets, relations and Functions

CO3: Demonstrate knowledge of mathematical modeling and proficiency in using mathematical software

CO4: Manipulate and analyze data numerically and/or graphically using appropriate Software

CO5: How to communicate effectively mathematical ideas/results verbally or in writing

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	1	-	1	-	-	-	-	-	-	-	-	-	-
CO4	3	2	1	1	-	-	-	-	-	-	-	-	-	-	-
CO5	3	2	1	1	-	-	-	-	-	-	-	-	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I: Mathematical Logic:

Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT-II: Set Theory:

Sets: Operations on Sets, Principle of Inclusion-Exclusion, Relations: Properties, Operations, Partition and Covering Transitive Closure, Equivalence Relation, Compatibility Relation and Partial Ordering relation, Hasse Diagrams Functions: Bijective, Composition, Inverse, Permutation and Recursive Functions, Lattice and its Properties..

UNIT-III: Combinatorics and Recurrence Relations:

Basics of Counting, Permutations, Permutations with Repetitions, Circular and Restricted Permutations, Combinations, Restricted Combinations, Binomial and Multinomial Coefficients and theorems. Recurrence Relations: Generating Functions, Functions of Sequences, Calculating Partial Fractions, Coefficients of Generating Functions, Recurrence relations, Formulation as Recurrence Relations, Solving Recurrence Relations by substitution and Generating functions, The Method of Characteristic roots, Solving Inhomogeneous Recurrence Relations.

UNIT-IV: Graph Theory:

Basic Concepts, Graph Theory and its Applications, Sub graphs, Graph Representations, Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs.

Unit-V: Multi Graphs:

Multigraphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Colouring and covering, Chromatic Number, Spanning Trees, Minimal Spanning tree, Prim's and Kruskal's Algorithms, BFS and DFS Algorithms for Spanning Trees (Problems Only and Theorems without Proofs).

Textbooks:

1. Discrete Mathematical Structures with Applications to Computer Science, J.P. Tremblay and P. Manohar, Tata McGraw Hill.
2. Elements of Discrete Mathematics-A Compute Oriented Approach, C.L. Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
3. Theory and Problems of Discrete Mathematics, Schaum's Outline Series, Seymour Lipschutz and Marc Lars Lipson, 3rd Edition, McGraw Hill.

Reference Books:

1. Discrete Mathematics for Computer Scientists and Mathematicians, J.L. Mott, A. Kandel and T. P. Baker, 2nd Edition, Prentice Hall of India.
2. Discrete Mathematical Structures, Bernard Kolman, Robert C. Busby and Sharon Cutler Ross, PHI.
3. Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.
4. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K.H. Rosen, 7th Edition, Tata McGraw Hill.



B.TECH III SEMESTER

	L	T	P	C
BS&H	3	0	0	3

**23AD3T02 UNIVERSAL HUMAN VALUES 2 UNDERSTANDING
HARMONY**

Course Objectives:

- To help the students appreciate the essential complementary between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
- To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such-holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way.
- To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with Nature.

Course Outcomes:

- Define the terms like Natural Acceptance, Happiness and Prosperity(L1,L2)
- Identify one's self, and one's surroundings(family, society nature)(L1,L2)
- Apply what they have learnt to their own self in different day-to-day settings in real life (L3)
- Relate human values with human relationship and human society.(L4)
- Justify the need for universal human values and harmonious existence(L5)
- Develop as socially and ecologically responsible engineers(L3,L6)

SYLLABUS

The course has 28 lectures and 14 tutorials in 5 modules. The lectures and tutorials are of 1- hour duration. Tutorial sessions are to be used to explore and practice what has been proposed during the lecture sessions.

The Teacher's Manual provides the outline for lectures as well as practice sessions. The teacher is expected to present the issues to be discussed as propositions and encourage the students to have a dialogue.

UNIT I: Introduction to Value Education (6 lectures and 3 tutorials for practice session)

Lecture1: Right Understanding, Relationship and Physical Facility (Holistic development and the Role of Education)

Lecture2: Understanding Value Education

Tutorial 1: Practice Session PS1 Sharing about One self

Lecture 3: self-exploration as the Process for Value Education

Lecture4: Continuous Happiness and Prosperity–the Basic Human Aspirations

Tutorial 2: Practice Session PS2 Exploring Human Consciousness

Lecture 5: Happiness and Prosperity – Current Scenario

Lecture 6: Method to Fulfil the Basic Human Aspirations

Tutorial 3: Practice Session PS3 Exploring Natural Acceptance

UNIT II: Harmony in the Human Being (6 lectures and 3 tutorials for practice session)

Lecture 7: Understanding Human being as the Co-existence of the self and the body.

Lecture8: Distinguishing between the Needs of the self and the body

Tutorial 4: Practice Session PS4 Exploring the difference of Needs of self and body.

Lecture 9: The body as an Instrument of the self

Lecture 10: Understanding Harmony in the self

Tutorial 5: Practice Session PS5 Exploring Sources of Imagination in the self

Lecture 11: Harmony of the self with the body

Lecture12: Programme to ensure self-regulation and Health

Tutorial6: Practice Session PS6 Exploring Harmony of self with the body

UNIT III Harmony in the Family and Society (6 lectures and 3tutorials for practice session)

Lecture 13: Harmony in the Family – the Basic Unit of Human Interaction

Lecture 14: 'Trust' – the Foundational Value in Relationship

Tutorial 7: Practice Session PS7 Exploring the Feeling of Trust

Lecture 15: 'Respect' – as the Right Evaluation

Tutorial 8: Practice Session PS8 Exploring the Feeling of Respect

Lecture 16: Other Feelings, Justice in Human-to-Human Relationship

Lecture 17: Understanding Harmony in the Society

Lecture18: Vision for the Universal Human Order

Tutorial 9: Practice Session PS9 Exploring Systems to fulfil Human Goal

UNIT IV: Harmony in the Nature/Existence (4 lectures and 2 tutorials for practice session)

Lecture19: Understanding Harmony in the Nature

Lecture 20: Interconnectedness, self-regulation and Mutual Fulfilment among
the Four Orders of Nature

Tutorial 10: Practice Session PS10 Exploring the Four Orders of Nature



Lecture 21: Realizing Existence as Co-existence at All Levels

Lecture22: The Holistic Perception of Harmony in Existence

Tutorial11: Practice Session PS11 Exploring Co-existence in Existence.

UNIT V: Implications of the Holistic Understanding – a Look at Professional Ethics

(6 lectures and 3 tutorials for practice session)

Lecture 23: Natural Acceptance of Human Values

Lecture 24: Definitiveness of (Ethical) Human Conduct

Tutorial12: Practice Session PS12 Exploring Ethical Human Conduct

Lecture25: A Basis for Humanistic Education, Humanistic Constitution and
Universal Human Order

Lecture26: Competence in Professional Ethics

Tutorial13: Practice Session PS13 Exploring Humanistic Models in Education

Lecture 27: Holistic Technologies, Production Systems and Management Models-
Typical Case Studies

Lecture28: Strategies for Transition towards Value-based Life and Profession

Tutorial 14: Practice Session PS14 Exploring Steps of Transition towards Universal
Human Order

Practice Sessions for UNIT I - Introduction to Value Education

PS1 Sharing about Oneself

PS2 Exploring Human Consciousness

PS3 Exploring Natural Acceptance

Practice Sessions for UNIT II - Harmony in the Human Being

PS4 Exploring the difference of Needs of self and body

PS5 Exploring Sources of Imagination in the self

PS6 Exploring Harmony of self with the body

Practice Sessions for UNIT III - Harmony in the Family and Society

PS7 Exploring the Feeling of Trust

PS8 Exploring the Feeling of Respect

PS9 Exploring Systems to fulfil Human Goal

Practice Sessions for UNIT IV - Harmony in the Nature (Existence)

PS10 Exploring the Four Orders of Nature

PS11 Exploring Co-existence in Existence



Practice Sessions for UNIT V - Implications of the Holistic Understanding - a Look at Professional Ethics

PS12 Exploring Ethical Human Conduct

PS13 Exploring Humanistic Models in Education

PS14 Exploring Steps of Transition towards Universal Human Order

READINGS:

Textbook and Teachers Manual

a. The Textbook

R R. Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1

b. The Teacher's Manual

R R Gaur, R Asthana, G P Bagaria, Teachers' Manual for A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books

1. Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. The Story of Stuff (Book).
4. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5. Small is Beautiful - E. F Schumacher.
6. Slow is Beautiful - Cecile Andrews
7. Economy of Permanence - J C Kumarappa
8. Bharat Mein Angreji Raj - PanditSunderlal
9. Rediscovering India - by Dharampal
10. Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11. India Wins Freedom - Maulana Abdul Kalam Azad
12. Vivekananda - Romain Rolland (English)
13. Gandhi - Romain Rolland (English)

MODE OF CONDUCT:

Lecture hours are to be used for interactive discussion, placing the proposals about the topics at hand and motivating students to reflect, explore and verify them.

Tutorial hours are to be used for practice sessions.

While analyzing and discussing the topic, the faculty mentor's role is in pointing to essential elements to help in sorting them out from the surface elements. In other words, help the students explore the important or critical elements.

In the discussions, particularly during practice sessions (tutorials), the mentor encourages the student to connect with one's own self and do self-observation, self-reflection and self-exploration. Scenarios may be used to initiate discussion. The student is encouraged to take up "ordinary" situations rather than "extra-ordinary" situations. Such observations and their analyses are shared and discussed with other students and faculty mentor, in a group sitting.

Tutorials (experiments or practical) are important for the course. The difference is that the laboratory is everyday life, and practical are how you behave and work in real life. Depending on the nature of topics, worksheets, home assignment and/or activity are included. The practice sessions (tutorials) would also provide support to a student in performing actions commensurate to his/her beliefs. It is intended that this would lead to development of commitment, namely behaving and working based on basic human values.

It is recommended that this content be placed before the student as it is, in the form of a basic foundation course, without including anything else or excluding any part of this content. Additional content may be offered in separate, higher courses. This course is to be taught by faculty from every teaching department, not exclusively by any one department.

Teacher preparation with a minimum exposure to at least one 8-day Faculty Development Program on Universal Human Values is deemed essential.

Online Resources:

1. <https://fdp-si.aicte-india.org/UHVII%20Class%20Notes%20&%20Handouts/UHV%20Handout%201-Introduction%20to%20Value%20Education.pdf>
2. <https://fdp-si.aicte-india.org/UHVII%20Class%20Notes%20&%20Handouts/UHV%20Handout%202-Harmony%20in%20the%20Human%20Being.pdf>
3. <https://fdp-si.aicte-india.org/UHVII%20Class%20Notes%20&%20Handouts/UHV%20Handout%203-Harmony%20in%20the%20Family.pdf>
4. <https://fdp-si.aicte-india.org/UHV%201%20Teaching%20Material/D3-S2%20Respect%20July%202023.pdf>



-
5. <https://fdp-si.aicte-india.org/UHVII%20Class%20Notes%20&%20Handouts/UHV%20Handout%205-Harmony%20in%20the%20Nature%20and%20Existence.pdf>
 6. <https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDPSI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf>
 7. <https://fdp-si.aicteindia.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf>
 8. <https://www.studocu.com/in/document/kiet-group-of-institutions/universal-humanvalues/chapter-5-holistic-understanding-of-harmony-on-professional-ethics/62490385>
 9. https://onlinecourses.swayam2.ac.in/aic22_ge23/preview



B.TECH III SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM3T03 ARTIFICIAL INTELLIGENCE

Pre-requisite:

- Knowledge in Computer Programming.
- A course on “Mathematical Foundations of Computer Science”.
- Background in linear algebra, data structures and algorithms, and probability.

Course Objectives:

- The student should be made to study the concepts of Artificial Intelligence.
- The student should be made to learn the methods of solving problems using Artificial Intelligence.
- The student should be made to introduce the concepts of Expert Systems.
- To understand the applications of AI, namely game playing, theorem proving, and machine learning.
- To learn different knowledge representation techniques

Course Outcomes:

CO1: Explain the history and foundations of Artificial Intelligence.

CO2: Apply the basic principles of AI in problem solving.

CO3: Choose the appropriate representation of Knowledge.

CO4: Explain the perspectives in Machine Learning.

CO5: Discuss the concepts in Machine Learning.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	2	3	1	2	-	-	-	-	-	-	-	1
CO3	3	2	2	1	-	-	-	-	-	-	-	-
CO4	2	2	2	-	-	-	-	-	-	-	-	-
CO5	2	2	-	-	-	-	-	-	-	-	-	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

SYLLABUS

UNIT – I: Introduction: foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents.

UNIT – II: Searching-Types of Search Strategies, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning

UNIT – III: Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems.

UNIT – IV: Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Reinforcement Learning.

UNIT – V: Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Textbooks:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, Second Edition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel, “Computational Intelligence: a logical approach”, Oxford University Press.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problemsolving”, Fourth Edition, Pearson Education.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.

Online Learning Resources:

1. <https://ai.google/>
2. https://swayam.gov.in/nd1_noc19_me71/preview

B.TECH III SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM3T04 ADVANCED DATA STRUCTURES ALGORITHMS ANALYSIS**Course Objectives:**

The main objectives of the course is to

- provide knowledge on advance data structures frequently used in Computer Science domain
- Develop skills in algorithm design techniques popularly used
- Understand the use of various data structures in the algorithm design

Course Outcomes:

CO1: Understand the concepts of Algorithm Analysis and the concepts of Balanced Trees

CO2: Understand and Apply various operations on Graphs and Divide and Conquer Strategy

CO3: Synthesize Efficient Algorithms for common engineering problems using Greedy Method

CO4: Ability to solve complex problems using Back Tracking and Branch & Bound

CO5: Apply and analyze the complexity of dynamic programming strategy.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	3	-	2
CO2	3	2	1	-	-	-	-	-	-	3	-	2
CO3	3	2	-	1	-	-	-	-	-	3	-	2
CO4	-	3	2	1	-	-	-	-	-	3	-	2
CO5	-	3	2	1	-	-	-	-	-	3	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS**UNIT – I:**

Heap Trees (Priority Queues)- Min and Max Heaps, Operations and Applications

AVL Trees – Creation, Insertion, Deletion operations and Applications

B-Trees – Creation, Insertion, Deletion operations and Applications

UNIT – II:

Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations

Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications
Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull

UNIT – III:

Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths
Dynamic Programming: General Method, All pairs shortest paths, 0/1 Knapsack, Travelling Salesperson problem

UNIT – IV:

Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring
Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem

UNIT – V:

NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem
String Matching: Introduction, String Matching Meaning and Application
NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling.

Textbooks:

1. Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2nd Edition Universities Press
2. Computer Algorithms in C++, Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, 2nd Edition University Press

Reference Books:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995
5. Algorithms + Data Structures & Programs:, N.Wirth, PHI
6. Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgottia Pub.
7. Data structures in Java:, Thomas Standish, Pearson Education Asia



Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. <http://peterindia.net/Algorithms.html>
3. Abdul Bari, [Introduction to Algorithms \(youtube.com\)](#)



B.TECH III SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM3T05 OBJECT ORIENTED PROGRAMMING THROUGH JAVA

Course Objectives:

The learning objectives of this course are to:

- identify Java language components and how they work together in applications
- Learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.
- learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications
- understand how to design applications with threads in Java
- understand how to use Java APIs for program development

Course Outcomes:

CO1: Understand the concepts of Object-Oriented Programming and Java programming constructs.

CO2: Demonstrate the concepts – Strings, Inheritance and Interfaces.

CO3: Build efficient and error-free codes using exception handling and demonstrate Multi-threading.

CO4: Design GUI applications using Event Handling and Abstract Window Toolkit.

CO5: Develop real-time applications using Applets and Swings.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	-	-	1	1	-	2
CO2	2	2	2	-	-	-	-	-	1	1	-	2
CO3	2	2	3	-	-	-	-	-	1	1	-	2
CO4	2	2	3	-	-	-	-	-	1	1	-	2
CO5	2	2	2	-	-	-	-	-	1	1	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT I

Object Oriented Programming: Basic concepts, Principles,

Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in

Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators : Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, **Introduction to Operators**, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if-else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do-while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

UNIT II

Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

UNIT III

Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays.

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface.

UNIT IV

Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes

into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes,

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java(Text Book 2)

UNIT V

String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded programming for Multi-core Processor, Thread Class, Main Thread-Creation of NewThreads, Thread States, Thread Priority-Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.

Java Database Connectivity: Introduction, JDBC Architecture, Installing MySQL and MySQL Connector/J, JDBC Environment Setup, Establishing JDBC Database Connections, Result Set Interface

Text Books:

- 1) JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
- 2) Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
- 3) JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

References Books:

- 1) The complete Reference Java, 11thedition, Herbert Schildt,TMH
- 2) Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Resources:

- 1) <https://nptel.ac.in/courses/106/105/106105191/>
- 2) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview



B.TECH III SEMESTER

	L	T	P	C
PC	0	0	3	1.5

**23AM3L06 ADVANCED DATA STRUCTURES AND ALGORITHMS
ANALYSIS LAB**

Course Objectives:

The objectives of the course is to

- acquire practical skills in constructing and managing Data structures
- apply the popular algorithm design methods in problem-solving scenarios

Course Outcomes:

CO1: Students will be able to write programs for performing various operations on AVL Trees

CO2: Students will be able to write programs for the problems using Divide and Conquer.

CO3: Students will be able to write programs for the problems using Greedy Method.

CO4: Students will be able to write programs for the problems using Dynamic programming.

CO5: Students will be able to write programs for the problems using Backtracking.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	2	2	-	-	-	-	1	1	-	-
CO2	2	1	2	2	-	-	-	-	1	1	-	1
CO3	2	1	2	2	-	-	-	-	1	1	-	1
CO4	2	1	2	2	-	-	-	-	1	1	-	1
CO5	2	1	2	2	-	-	-	-	1	1	-	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

EXPERIMENTS COVERING THE TOPICS:

- Operations on AVL trees, B-Trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

SAMPLE PROGRAMS:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.
4. Implement BFT and DFT for given graph, when graph is represented by
 - a) Adjacency Matrix
 - b) Adjacency Lists
5. Write a program for finding the biconnected components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).
7. Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.
8. Implement Job Sequencing with deadlines using Greedy strategy.
9. Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.
10. Implement N-Queens Problem Using Backtracking.
11. Use Backtracking strategy to solve 0/1 Knapsack problem.
12. Implement Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures in C++, Horowitz Ellis, Sahni Sartaj, Mehta, Dinesh, 2nd Edition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, 2nd Edition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill

Online Learning Resources:

1. <http://cse01-iiith.vlabs.ac.in/>
2. <http://peterindia.net/Algorithms.html>



B.TECH III SEMESTER

PC	L	T	P	C
	0	0	3	1.5

**23AM3L07 OBJECT ORIENTED PROGRAMMING THROUGH
JAVA LAB**

Course Objectives:

The aim of this course is to

- Practice object oriented programming in the Java programming language
- Implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism
- Illustrate inheritance, Exception handling mechanism, JDBC connectivity
- Construct Threads, Event Handling, implement packages, Java FX GUI

Course Outcomes:

CO1: Implement java applications using OOP principles and proper program structuring.

CO2: Develop java programs using packages, inheritance and interfaces.

CO3: Implement error and exception handling techniques.

CO4: Design event driven GUI and real-time web related applications.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	-	-	-	-	1	2	-	2
CO2	2	2	2	1	-	-	-	-	1	2	-	2
CO3	2	2	2	1	-	-	-	-	1	2	-	2
CO4	2	2	2	1	-	-	-	-	1	2	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

EXPERIMENTS COVERING THE TOPICS:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, Java FX GUI

SAMPLE EXPERIMENTS:

Exercise – 1:

- Write a JAVA program to display default value of all primitive data type of JAVA

- b) Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminant D and basing on value of D, describe the nature of root.

Exercise - 2

- a) Write a JAVA program to search for an element in a given list of elements using binarysearch mechanism.
- b) Write a JAVA program to sort for an element in a given list of elements using bubble sort
- c) Write a JAVA program using String Buffer to delete, remove character.

Exercise - 3

- a) Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method.
- b) Write a JAVA program implement method overloading.
- c) Write a JAVA program to implement constructor.
- d) Write a JAVA program to implement constructor overloading.

Exercise - 4

- a) Write a JAVA program to implement Single Inheritance
- b) Write a JAVA program to implement multi level Inheritance
- c) Write a JAVA program for abstract class to find areas of different shapes

Exercise - 5

- a) Write a JAVA program give example for “super” keyword.
- b) Write a JAVA program to implement Interface. What kind of Inheritance can be achieved?
- c) Write a JAVA program that implements Runtime polymorphism

Exercise - 6

- a) Write a JAVA program that describes exception handling mechanism
- b) Write a JAVA program Illustrating Multiple catch clauses
- c) Write a JAVA program for creation of Java Built-in Exceptions
- d) Write a JAVA program for creation of User Defined Exception

Exercise – 7

Write a JAVA program that import and use the user defined packages

Exercise – 8

- a) Write a java program that connects to a database using JDBC
- b) Write a java program to connect to a database using JDBC and insert values into it.
- c) Write a java program to connect to a database using JDBC and delete values from it



B.TECH III SEMESTER

SC	L	T	P	C
	0	0	4	2

23AM3S08 PYTHON PROGRAMMING LAB

Course Objectives:

The main objectives of the course are to

- Introduce core programming concepts of Python programming language.
- Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries
- Implement Functions, Modules and Regular Expressions in Python Programming and to create practical and contemporary applications using these.

Course Outcomes:

CO1: Develop programs using python fundamentals

CO2: Make use of control statements and strings for developing applications

CO3: Develop applications using data structures and functions

CO4: Apply OOPs concepts and files for developing programs

CO5: Make use of Exception Handling to handle runtime errors

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	1	2	-	-	-	1	1	-	2
CO2	2	2	3	1	2	-	-	-	1	1	-	2
CO3	2	2	2	1	2	-	-	-	1	1	-	2
CO4	2	2	2	1	2	-	-	-	1	1	-	2
CO5	2	1	2	1	2	-	-	-	1	1	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I:

History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook.

Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language.

Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement,

while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.

Sample Experiments:

1. Write a program to find the largest element among three Numbers.
2. Write a Program to display all prime numbers within an interval
3. Write a program to swap two numbers without using a temporary variable.
4. Demonstrate the following Operators in Python with suitable examples.
 - i) Arithmetic Operators ii) Relational Operators iii) Assignment Operatorsiv) Logical Operators v) Bit wise Operators vi) Ternary Operator vii) Membership Operatorsviii) Identity Operators
5. Write a program to add and multiply complex numbers
6. Write a program to print multiplication table of a given number.

UNIT-II:

Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments.

Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings.

Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement.

Sample Experiments:

1. Write a program to define a function with multiple return values.
2. Write a program to define a function using default arguments.
3. Write a program to find the length of the string without using any library functions.
4. Write a program to check if the substring is present in a given string or not.
5. Write a program to perform the given operations on a list:
 - i. Addition ii. Insertion iii. slicing
6. Write a program to perform any 5 built-in functions by taking any list.

UNIT-III:

Dictionaries: Creating Dictionary, Accessing and Modifying key:value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement.

Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and

Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.

Sample Experiments:

1. Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
2. Write a program to count the number of vowels in a string (No control flow allowed).
3. Write a program to check if a given key exists in a dictionary or not.
4. Write a program to add a new key-value pair to an existing dictionary.
5. Write a program to sum all the items in a given dictionary.

UNIT-IV:

Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules.

Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.

Sample Experiments:

1. Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
2. Python program to print each line of a file in reverse order.
3. Python program to compute the number of characters, words and lines in a file.
4. Write a program to create, display, append, insert and reverse the order of the items in the array.
5. Write a program to add, transpose and multiply two matrices.
6. Write a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.

UNIT-V:

Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.

Sample Experiments:

1. Python program to check whether a JSON string contains complex object or not.
2. Python Program to demonstrate NumPy arrays creation using array () function.
3. Python program to demonstrate use of ndim, shape, size, dtype.
4. Python program to demonstrate basic slicing, integer and Boolean indexing.
5. Python program to find min, max, sum, cumulative sum of array
6. Create a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows:
 - a) Apply head () function to the pandas data frame
 - b) Perform various data selection operations on Data Frame
7. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib

Reference Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2nd Edition, Pearson, 2024
3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Online Learning Resources/Virtual Labs:

1. <https://www.coursera.org/learn/python-for-applied-data-science-ai>
2. <https://www.coursera.org/learn/python?specialization=python#syllabus>



B.TECH III SEMESTER	MC	L	T	P	C
		2	0	0	-

23AM3M09

ENVIRONMENTAL SCIENCE
(Audit Course)

Course Objectives:

- To make the students to get awareness on environment
- To understand the importance of protecting natural resources, ecosystems for future generations and pollution causes due to the day-to-day activities of human life
- To save earth from the inventions by the engineers.

Course Outcomes:

CO1: Understand the multidisciplinary nature of environmental studies and various renewable and non-renewable resources.

CO2: Understand flow and bio-geo- chemical cycles and ecological pyramids.

CO3: Understand various causes of pollution and solid waste management and related preventive measures.

CO4: Understand the rainwater harvesting, watershed management, ozone layer depletion and waste land reclamation

CO5: Illustrate the causes of population explosion, value education and welfare programmes.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	-	-	3	-	1	1	-	-
CO2	2	-	-	-	-	-	3	-	1	1	-	1
CO3	2	-	-	-	-	-	3	-	1	1	-	1
CO4	2	-	-	-	-	-	3	-	1	1	-	1
CO5	2	-	-	-	-	-	3	-	1	1	-	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT – I

Multidisciplinary Nature of Environmental Studies: – Definition, Scope and Importance – Need for Public Awareness.

Natural Resources : Renewable and non-renewable resources – Natural resources and associated problems – Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people – Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies – Food resources:

World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. – Energy resources:

UNIT – II

Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem:

1. Forest ecosystem.
2. Grassland ecosystem
3. Desert ecosystem
4. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Biodiversity and Its Conservation : Introduction and Definition: genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT – III

Environmental Pollution: Definition, Cause, effects and control measures of:

- a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution
- f. Thermal pollution g. Nuclear hazards

Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides.

UNIT – IV

Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy – Water conservation, rain water harvesting, watershed management – Resettlement and rehabilitation of people; its problems and concerns. Case studies – Environmental ethics: Issues and possible solutions – Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies –

Wasteland reclamation. – Consumerism and waste products. – Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act – Issues involved in enforcement of environmental legislation – Public awareness.

UNIT – V

Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies.

Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.

Text books:

1. Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2. Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3. S.Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4. K.Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.

Reference Books:

1. Deeksha Dave and E.Sai Baba Reddy, Textbook of Environmental Science, 2/e, Cengage Publications, 2012.
2. M.Anji Reddy, “Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3. J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4. J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5. G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018.
6. Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.

Online Learning Resources:

- https://onlinecourses.nptel.ac.in/noc23_hs155/preview



-
- https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-d84c-4773-8388-1b2f8f6a75f2&webview=false&campaign=AP%C2%AE+Environmental+Science+Part+3%3A+Pollution+and+Resources&source=edX&product_category=course&placement_url=https%3A%2F%2Fwww.edx.org%2Flearn%2Fenvironmental-science
 - <http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf>
 - <https://www.youtube.com/watch?v=5QxxaVfgQ3k>



B.TECH IV SEMESTER

BS&H

L	T	P	C
3	0	0	3

23AM4T01 OPTIMIZATION TECHNIQUES

Pre-requisite:

Course Objectives:

1. Identify and develop operational research models from the verbal description of the real system.
2. Understand the mathematical tools that are needed to solve optimization problems. Model strategic behavior in different economic situations.
3. Use mathematical software to solve the proposed models.
4. Develop a report that describes the model and the solving technique, analyze the results
5. Explain scheduling and sequencing of production runs and develop proper replacement policies.

Course Outcomes: At the end of the course, student will be able to

CO1: Understand the methodology of Operations Research& concepts of linear programming

CO2: Formulate the solutions to transportation problems

CO3: Explain the solutions for various sequencing problems

CO4: Illustrate the solutions to different replacement policies

CO5: Apply game theory to solve real world problems

COURSE ARTICULATION MATRIX (Correlation between COs, POs &PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	2	-	-	-	-	-
CO2	3	2	-	-	1	-	-	-	-	-	-	-
CO3	3	-	-	1	-	-	2	-	-	-	-	1
CO4	3	-	-	-	-	-	-	-	-	-	-	1
CO5	3	-	-	-	1	-	1	-	-	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1-** Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT I

Introduction to Operations Research: Basics definition, scope, objectives, phases, models and limitations of Operations Research. Linear Programming Problem –Formulation of LPP, Graphical solution of LPP. Simplex Method, Artificial variables, big-M method, two-phase method, degeneracy

and unbound solutions.

UNIT II

Transportation Problem. Formulation, Solution, unbalanced Transportation problem. Finding basic feasible solutions – Northwest corner rule, least cost method and Vogel’s approximation method. Optimality test: MODI method. Assignment Problem – Formulation, Hungarian Method for Solving Assignment Problems, Traveling Salesman problem.

UNIT III

Assignment model. Formulation. Hungarian Method for optimal solution. Solving unbalanced problem. Sequencing Models. Solution of Sequencing Problem –Processing n Jobs through 2 Machines – Processing n Jobs through 3 Machines –Processing 2 Jobs through m machines Processing n Jobs through m Machines.

UNIT IV

Replacement Models. Replacement of Items that deteriorate whose maintenance costs increase with time without change in the money value. Replacement of items that fail suddenly: individual replacement policy, group replacement policy.

UNIT V

Game Theory. Competitive games, rectangular game, saddle point, minimax (maximin) method of optimal strategies, value of the game. Solution of games with saddle points, dominance principle. Rectangular games without saddle point – mixed strategy for 2X games. Inventory models. Inventory costs. Models with deterministic demand – model (a) demand rate uniform and production rate infinite, model (b) demand rate non-uniform and production rate infinite, model (c) demand rate uniform and production rate finite.

Text Books

1. P. Sankara Iyer, "Operations Research", Tata Mc-Graw-Hill, 2008.
2. A.M. Natarajan, P. Balasubramani, A. Tamilarasi, "Operations Research", Pearson Education, 2005.

References

1. Hiller F.S., and Liberman G.J., Introduction to Operations Research, 7th Edition, Tata McGraw Hill, 2010.
2. Sharma J.K., Operations Research: Theory and Applications, 4th Edition, Laxmi Publications, 2009.

B.TECH IV SEMESTER

BS&H	L	T	P	C
	3	0	0	3

23AM4T02 PROBABILITY&STATISTICS**Course Objectives:**

- To familiarize the students with the foundations of probability and statistical methods
- To impart probability concepts and statistical methods in various applications in Engineering

Course Outcomes:

CO1: Classify the concepts of data science and its importance

CO2: Interpret the association of characteristics and through correlation and regression tools

CO3: Apply discrete and continuous probability distributions

CO4: Design the components of a classical hypothesis test

CO5: Infer the statistical inferential methods based on small and large sampling tests

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-
CO3	3	2	1	-	-	-	-	-	-	-	-	-
CO4	3	2	1	-	-	-	-	-	-	-	-	-
CO5	3	2	1	-	-	-	-	-	-	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS**UNIT-I: Descriptive statistics and methods for data science**

Data science, Statistics Introduction ,Population vs. Sample, Collection of data ,primary and secondary data, Type of variable: dependent and independent Categorical and Continuous variables, Data visualization, Measures of Central tendency ,Measures of Variability – Skewness – Kurtosis.

UNIT-II: Correlation and Regression:

Correlation, Correlation coefficient, Rank correlation. Linear Regression: Straight line, Multiple Linear Regression, Regression coefficients and properties – Curvilinear Regression: Parabola, Exponential, Power curves.

UNIT-III: Probability and Distributions:

Probability, Conditional Probability, Baye’s Theorem Random variables, Discrete and Continuous random variables, Distribution functions, Probability mass function, Probability density function and Cumulative distribution functions, Mathematical Expectation and Variance, Binomial, Poisson

Uniform and Normal distributions.

UNIT-IV: Sampling Theory:

Introduction, Population and Samples, Sampling distribution of Means and Variance (definition only)–Point and Interval estimations, Maximum error of estimate, Central Limit theorem (without proof)–Estimation using t, χ^2 and F-distributions.

UNIT-V: Tests of Hypothesis:

Introduction – Hypothesis – Null and Alternative Hypothesis – Type I and Type II errors – Level of significance – One tail and two-tail tests – Test of significance for large samples and Small Samples: Single and difference means – Single and two proportions –Student's-test, F-test, χ^2 test

Text Books:

1. Millerand Freund's, Probability and Statistics for Engineers, 7/e, Pearson, 2008.
2. S. C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.

Reference Books:

1. Shron L. Myers, Keying Ye, Ronald E Walpole, Probability and Statistics Engineers and the Scientists, 8th Edition, Pearson 2007.
2. Jay I. Devore, Probability and Statistics for Engineering and the Sciences, 8th Edition, Cengage.
3. Sheldon M. Ross, Introduction to probability and statistics Engineers and the Scientists, 4th Edition, Academic Foundation, 2011.
4. Johannes Ledolter and Robert V. Hogg, Applied statistics for Engineers and Physical Scientists, 3rd Edition, Pearson, 2010

B.TECH IV SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM4T03 MACHINE LEARNING**Pre-requisite:** Probability and Statistics, Linear Algebra**Course Objectives:**

- To understand the need for machine learning for various problem solving
- To study the various supervised, semi-supervised and unsupervised learning algorithms in machine learning
- To understand the latest trends in machine learning
- To design appropriate machine learning algorithms for problem solving

Course Outcomes:

CO1: Basic understanding of Machine Learning algorithms

CO2: Apply various ANN algorithms with machine learning techniques.

CO3: Apply the Bayesian concepts to machine learning.

CO4: Implementation of SVM.

CO5: Exposure to types of reinforced learning.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	1	2	-	-	-	1	1	-	2
CO2	2	2	3	1	2	-	-	-	1	1	-	2
CO3	2	2	2	1	2	-	-	-	1	1	-	2
CO4	2	2	2	1	2	-	-	-	1	1	-	2
CO5	2	1	2	1	2	-	-	-	1	1	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).**SYLLABUS****UNIT-I: Introduction to Machine Learning:**

Evolution of Machine Learning, Paradigms for ML, Well posed Learning Problems, Perspectives and Issues, Designing a Learning system, Concept Learning, FIND-S Algorithm, Version Spaces and Candidate Eliminations, Inductive bias, Decision Tree learning, Decision Tree Representation – ID3 Algorithm, Heuristic Space search

UNIT-II: Neural Networks and Genetic Algorithms:

Neural Network Representation, Appropriate Problems for ANN, Perceptron Learning, Perceptron Models Types, Multi-layered perceptron model, Multilayer Networks and Back Propagation

Algorithms, Advanced Topics in ANN, Genetic Algorithm in machine Learning ,Genetic Programming, Models of Evaluation and Learning.

UNIT-III: Bayesian and Computational Learning

Bayes Theorem Representation, Brute force Concept Learning , Maximum Likelihood-Least Squared error Hypothesis, Maximum Likelihood hypothesis-Predicting Probabilities, Minimum Description Length Principle, Bayes Optimal Classifier, Naïve Bayes Classifier (NBC), Bayesian Belief Network, EM Algorithm.

UNIT-IV: Instant Based Learning

Introduction of Instant based learning, K- Nearest Neighbor Learning-Algorithm , Regression-Linear Regression, Locally weighted regression, Logistic Regression, Random Forests for Classification and Regression, Radial Basis Functions, Case Based Learning-CADET System, Support Vector Machines (SVM).

UNIT-V: Reinforcement Learning:

Introduction – Terms, Key features, Approaches, Elements, The Bellman Equation, Types of Reinforcement Learning, Reinforcement Learning Algorithms, The Learning Task, Q – Learning, non-deterministic rewards and actions, temporal difference learning, Algorithms in temporal difference learning, generalizing from examples, relationship to dynamic programming.

Text Books:

1. Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.

Reference books:

1. EthemAlpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
2. Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009



B.TECH IV SEMESTER

ES	L	T	P	C
	3	0	0	3

23AM4T04 DATABASE MANAGEMENT SYSTEMS

Course Objectives:

The main objectives of the course is to

- Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra
- Introduce the concepts of basic SQL as a universal Database language
- Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization
- Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques

Course Outcomes:

CO1: Understand the concept of database, database models and familiarize with Entity relationship models

CO2: Demonstrate the use of constraints, relational algebra operations.

CO3: Apply SQL queries to interact with database and understand the basics of NOSQL.

CO4: Apply normalization in database design to eliminate anomalies.

CO5: Understand the basic concepts of transaction processing and concurrency control.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	2	2	1	1	1	-	-	-	-	-	-	1
CO3	2	2	2	-	-	-	-	-	-	-	-	1
CO4	2	2	3	-	-	-	-	-	-	-	-	1
CO5	2	2	-	1	-	-	-	-	-	-	-	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT I:

Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture

for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT II:

Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT III:

SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT IV:

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT V:

Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

Text Books:

- 1) Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
- 2) Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

- 1) Introduction to Database Systems, 8th edition, C J Date, Pearson.



-
- 2) Database Management System, 6th edition, RamezElmasri, Shamkant B. Navathe, Pearson
 - 3) Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Web-Resources:

- 1) <https://nptel.ac.in/courses/106/105/106105175/>
- 2) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

B.TECH IV SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM4T05 DIGITAL LOGIC & COMPUTER ORGANIZATION

Course Objectives: The main objectives of the course is to

- provide students with a comprehensive understanding of digital logic design principles and computer organization fundamentals
- Describe memory hierarchy concepts
- Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices

Course Outcomes:

CO1: Understand the Data representation and the fundamentals of Digital logic circuits

CO2: Understand the architecture of a modern computer with its various processing units.

CO3: Understand the Computer Arithmetic and Processor organization

CO4: Understand the Memory hierarchy system and performance improvement by cache memory

CO5: Analyze the communication methods of I/O devices and standard I/O interfaces.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	-	1	-	-	-	-	-	-	1
CO2	2	2	2	-	1	-	-	-	-	-	-	-
CO3	2	2	2	1	1	-	-	-	-	-	-	-
CO4	2	2	2	2	1	-	-	-	-	-	-	2
CO5	2	2	2	2	2	-	-	-	-	-	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT – I: Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Number base conversions, Octal and Hexadecimal Numbers, components, Signed binary numbers, Binary codes

Digital Logic Circuits-I: Basic Logic Functions, Logic gates, universal logic gates, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers

UNIT – II: Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters

Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi computers, Computer Generations, Von- Neumann Architecture

UNIT – III: Computer Arithmetic : Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control

UNIT – IV: The Memory Organization: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage

UNIT – V: Input/Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

Textbooks:

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw Hill, 2023.
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education, 2018.
3. Computer Organization and Architecture, William Stallings, 11th Edition, Pearson, 2022.

Reference Books:

1. Computer Systems Architecture, M. Morris Mano, 3rd Edition, Pearson, 2017.
2. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier, 2004.
3. Fundamentals of Logic Design, Roth, 5th Edition, Thomson, 2003.

Online Learning Resources:

<https://nptel.ac.in/courses/106/103/106103068/>

B.TECH IV SEMESTER

	L	T	P	C
PC	0	0	3	1.5

23AM4L06 MACHINE LEARNING LAB**Course Objectives:**

- Make use of Data sets in implementing the machine learning algorithms
- Implement the machine learning concepts and algorithms in any suitable language of choice

Course Outcomes:

CO1: Understand the implementation procedures for the machine learning algorithms

CO2: Design Java/Python programs for various Learning algorithms

CO3: Apply appropriate data sets to the Machine Learning algorithms

CO4: Identify and apply Machine Learning algorithms to solve real world problems

CO5: Implement various Classification and Clustering algorithms using Java/Python

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1	-	-	-	-	-	-	-	-
CO2	2	2	2	2	-	-	-	-	-	-	-	-
CO3	2	2	2	3	-	-	-	-	-	-	-	-
CO4	2	2	2	2	-	-	-	-	-	-	-	-
CO5	2	2	2	1	-	-	-	-	-	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

LIST OF EXPERIMENTS

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file
2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples
3. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample
4. Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
5. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.

6. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.
7. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set. You can use Java/Python ML library classes/API.
8. Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.
9. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem.
10. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs

References:

1. Vijayvargia, Abhishek, Machine Learning with Python: An Approach to Applied Machine Learning, BPB Publications, 1st edition, 2018.
2. Aurelien Geron, Hands-On Machine Learning with Scikit-Learn and TensorFlow, Oreilly, March 2017.
3. Dr. M Gopal, Applied Machine Learning, 1st Edition, McGraw-Hill, 2018



B.TECH IV SEMESTER

	L	T	P	C
PC	0	0	3	1.5

23AM4L07 DATABASE MANAGEMENT SYSTEMS LAB

Course Objectives:

This Course will enable students to

- Populate and query a database using SQL DDL/DML Commands
- Declare and enforce integrity constraints on a database
- Writing Queries using advanced concepts of SQL
- Programming PL/SQL including procedures, functions, cursors and triggers

Course Outcomes:

CO1: Design database schema for a given application and apply normalization

CO2: Acquire skills in using SQL commands for data definition and data manipulation.

CO3: Develop solutions for database applications using procedures, cursors and triggers

CO4: Develop solutions using PL/SQL procedures

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	2	2	1	1	1	-	-	-	-	-	-	1
CO3	2	2	-	-	-	-	1	-	-	2	-	-
CO4	2	2	1	-	-	-	-	1	-	-	-	-

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put '-' 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

EXPERIMENTS COVERING THE TOPICS:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,
- Database connectivity- ODBC/JDBC

SAMPLE EXPERIMENTS:

1. Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.

3. Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.
 - i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)
 - ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
7. Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE-APPLICATION ERROR.
8. Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9. Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
12. Create a table and perform the search operation on table using indexing and non-indexing techniques

Text Books/Suggested Reading:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
3. Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007



		L	T	P	C
B.TECH IV SEMESTER	SC	0	0	4	2

23AM4S08 FULL STACK DEVELOPMENT – 1
(Skill Enhancement Course)

Course Objectives

The main objectives of the course are to

- Make use of HTML elements and their attributes for designing static web pages
- Build a web page by applying appropriate CSS styles to HTML elements
- Experiment with JavaScript to develop dynamic web pages and validate forms

Course Outcomes:

CO1: Use HTML elements and their attributes for developing static web pages

CO2: Use CSS for front end development

CO3: Use javascript for developing interactive applications

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	1	-	-	-	1	-	-	-	1
CO3	2	2	-	-	-	-	-	1	-	1	-	1
CO4	2	2	-	-	-	-	-	-	-	-	1	1
CO5	2	2	-	1	-	-	-	3	-	-	-	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

EXPERIMENTS COVERING THE TOPICS:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

SAMPLE EXPERIMENTS:

1. Lists, Links and Images

- a. Write a HTML program, to explain the working of lists.

Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.

- b. Write a HTML program, to explain the working of hyperlinks using `<a>` tag and href, target Attributes.
- c. Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.
- d. Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

2. HTML Tables, Forms and Frames

- a. Write a HTML program, to explain the working of tables. (use tags: `<table>`, `<tr>`, `<th>`, `<td>` and attributes: border, rowspan, colspan
- b. Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use `<caption>` tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
- c. Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using `<select>` & `<option>` tags, `<text area>` and two buttons ie: submit and reset. Use tables to provide a better view).
- d. Write a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame ☐ image, second frame ☐ paragraph, third frame ☐ hyperlink. And also make sure of using "no frame" attribute such that frames to be fixed).

3. HTML 5 and Cascading Style Sheets, Types of CSS

- a. Write a HTML program, that makes use of `<article>`, `<aside>`, `<figure>`, `<figcaption>`, `<footer>`, `<header>`, `<main>`, `<nav>`, `<section>`, `<div>`, `<span>` tags.
- b. Write a HTML program, to embed audio and video into HTML web page.
- c. Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value).

4. Selector forms

- a. Write a program to apply different types of selector forms
 - i. Simple selector (element, id, class, group, universal)
 - ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
 - iii. Pseudo-class selector
 - iv. Pseudo-element selector

v. Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

- a. Write a program to demonstrate the various ways you can reference a color in CSS.
- b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
- c. Write a program using the following terms related to CSS font and text:
 - i. font-size ii. font-weight iii. font-style
 - iv. text-decoration v. text-transformation vi. text-alignment
- d. Write a program, to explain the importance of CSS Box model using
 - i. Content ii. Border iii. Margin iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

- a. Write a program to embed internal and external JavaScript in a web page.
- b. Write a program to explain the different ways for displaying output.
- c. Write a program to explain the different ways for taking input.
- d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

7. Java Script Pre-defined and User-defined Objects

- a. Write a program using document object properties and methods.
- b. Write a program using window object properties and methods.
- c. Write a program using array object properties and methods.
- d. Write a program using math object properties and methods.
- e. Write a program using string object properties and methods.
- f. Write a program using regex object properties and methods.
- g. Write a program using date object properties and methods.
- h. Write a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. Java Script Conditional Statements and Loops

- a. Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words "LARGER NUMBER" in an information message dialog. If the numbers are equal, output HTML text as "EQUAL NUMBERS".
- b. Write a program to display week days using switch case.
- c. Write a program to print 1 to 10 numbers using for, while and do-while loops.
- d. Write a program to print data in object using for-in, for-each and for-of loops

- e. Develop a program to determine whether a given number is an 'ARMSTRONG NUMBER' or not. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $1^3 + 5^3 + 3^3 = 153$]
- f. Write a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1-100's, 1-50's, 1-10's, 1-2's & 1-1's)

9. Java Script Functions and Events

- a. Design a appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- c. Write a program to validate the following fields in a registration page Name (start with alphabet and followed by alphanumeric and the length shouldnot be less than 6 characters)
 - i. Mobile (only numbers and length 10 digits)
 - ii. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

Text Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.

Web Links:

1. <https://www.w3schools.com/html>
2. <https://www.w3schools.com/css>
3. <https://www.w3schools.com/js/>
<https://www.w3schools.com/nodejs>



B.TECH IV SEMESTER	ES	L	T	P	C
		0	0	4	2

23AM4L09 DESIGN THINKING & INNOVATION

Course Objectives: The objectives of the course are to

- Bring awareness on innovative design and new product development.
- Explain the basics of design thinking.
- Familiarize the role of reverse engineering in product development.
- Train how to identify the needs of society and convert into demand.
- Introduce product planning and product development process.

Course Outcomes:

CO1: Define the concepts related to Design Thinking

CO2: Explain the fundamentals of Design Thinking and Innovation

CO3: Apply the Design Thinking Techniques for problems in various sectors

CO4: Analyze working in Multidisciplinary Environment

CO4: Evaluate the values of Creativity

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	2	-	1	-	-	3	-	-
CO2	-	-	-	-	-	-	-	2	-	3	1	1
CO3	-	-	-	-	-	-	-	2	-	3	-	1
CO4	-	-	-	-	2	-	1	-	-	3	1	1
CO5	-	-	-	-	-	-	-	2	-	3	-	1

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put '-'. 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

SYLLABUS

UNIT-I: Introduction to Design Thinking

Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.

UNIT-II: Design Thinking Process

Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development

Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.

UNIT-III: Innovation

Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity.

Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.

UNIT-IV: Product Design

Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.

Activity: Importance of modeling, how to set specifications, Explaining their own product design.

UNIT-V: Design Thinking in Business Processes

Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes.

Activity: How to market our own product, about maintenance, Reliability and plan for startup.

Text books:

1. Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2. Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.

Reference Books:

1. David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2. Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3. William Lidwell, Kritina Holden, & Jill Butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4. Chesbrough J.H., The era of open innovation, 2003.

Online Learning Resources:

- <https://nptel.ac.in/courses/110/106/110106124/>
- <https://nptel.ac.in/courses/109/104/109104109/>
- https://swayam.gov.in/nd1_noc19_mg60/preview
- https://onlinecourses.nptel.ac.in/noc22_de16/preview



B.TECH V SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM5T01 DEEP LEARNING

Prerequisites: Basic knowledge in P&S, ML, Python programming.

Course Objectives: The main objectives of introducing this course are

1. Provide foundational knowledge in linear algebra and vectorization techniques essential for deep learning
2. Introduce and explain the architecture and training of neural networks.
3. Explore advanced optimization and regularization techniques to improve training.
4. Develop an understanding of recurrent and convolutional neural networks and generative models.
5. Introduce recent trends in generative AI and transformer-based architectures with real-world applications

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand and apply linear algebra concepts in the context of deep learning.
- CO2** Analyze and implement perceptron models and basic learning algorithms
- CO3** Design and train deep feedforward networks using various optimization techniques.
- CO4** Apply and evaluate advanced neural network architectures like CNNs, RNNs, and LSTMs.
- CO5** Identify applications of generative AI in vision, NLP, and speech, and understand agentic AI.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	1	-	-	-	-	-	-	-	3	2
CO2	3	3	3	3	2	-	-	-	-	-	-	-	-	3
CO3	3	3	3	2	3	-	-	-	-	-	-	-	2	2
CO4	3	2	3	3	3	-	-	-	-	-	-	-	2	3
CO5	2	2	2	2	1	-	-	-	-	-	-	-	2	3

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put '-'1-Slight(Low),2-Moderate(Medium),3-Substantial(High).

SYLLABUS

UNIT-I: Basics of linear algebra for Deep Learning

Basics-Basics of linear algebra for Deep Learning: Matrices,types,Inverse,Eigen values and vectors, Principal component analysis, Image and text vectorization, Biological Neuron, Idea of computational units, McCulloch–Pitts unit and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Linear separability, Convergence theorem for Perceptron Learning Algorithm.

UNIT-II: Feed forward Networks

Feed forward Networks-Multilayer Perceptron, Gradient Descent, Backpropagation, Empirical Risk Minimization, regularization. Deep Neural Networks: Difficulty of training deep neural networks, Greedy layer wise training.

UNIT-III: Better Training of Neural Networks

Better Training of Neural Networks-Newer optimization methods for neural networks (Adagrad, adadelta, rmsprop, adam, NAG), second order methods for training, Saddle point problem in neural networks, Regularization methods (dropout, drop connect, batch normalization).

UNIT-IV: Recurrent Neural Networks

Recurrent Neural Networks- Back propagation through time, Long Short Term Memory, Gated Recurrent Units, Bidirectional LSTMs, Bidirectional RNNs.

Convolutional Neural Networks: Convolution operation, types of convolutions, LeNet, AlexNet. Generative models: Restrictive Boltzmann Machines (RBMs), Introduction to MCMC and Gibbs Sampling, gradient computations in RBMs, Deep Boltzmann.

UNIT-V: Recent trends

Recent trends-Variational Auto encoders, Transformers, GPT Applications: Vision, NLP, Speech, Introduction to Generative AI-Types,Applications,Agentic AI-Applications.

Text Books:

1. Deep Learning, Ian Good fellow and YoshuaBengio and Aaron Courville, MIT Press, 2016.

Reference Books

1. Neural Networks: A Systematic Introduction, Raúl Rojas,1996
2. Pattern Recognition and Machine Learning, Christopher Bishop,2007
3. Deep Learning with Python, François Chollet, Manning Publications,2017

B.TECH V SEMESTER**PC**

L	T	P	C
3	0	0	3

23AM5T02 COMPUTER NETWORKS**Course Objectives:** The main objectives of introducing this course are

1. To provide insight about networks, topologies, and the key concepts
2. To gain comprehensive knowledge about the layered communication architectures (OSI and TCP/IP) and its functionalities
3. To understand the principles, key protocols, design issues, and significance of each layers in ISO and TCP/IP.
4. To know the basic concepts of network services and various network applications.

Course Outcomes: At the end of the course, student will be able to

Demonstrate different network models for networking links OSI, TCP/IP, B-ISDN, N-

CO1 BISDN and get knowledge about various communication techniques, methods and protocol standards.**CO2** Discuss different transmission media and different switching networks.**CO3** Analyze data link layer services, functions and protocols like HDLC and PPP.**CO4** Compare and Classify medium access control protocols like ALOHA, CSMA, CSMA/CD, CSMA/CA, Polling, Token passing, FDMA, TDMA, CDMA protocols**CO5** Determine application layer services and client server protocols working with the client server paradigms like WWW, HTTP, FTP, e-mail and SNMP etc.**COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):**

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	-	-	-	-	-	-	-	-	3	2
CO2	3	2	2	-	1	-	-	-	-	-	-	-	2	2
CO3	3	3	2	1	1	-	-	-	-	-	-	-	2	3
CO4	3	2	2	2	2	-	-	-	-	-	-	-	3	3
CO5	2	2	2	1	2	-	-	-	-	-	-	-	3	3

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put '-' 1-Slight(Low),2-Moderate(Medium),3-Substantial(High).**SYLLABUS****UNIT-I: Introduction****Introduction:** Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs TCP/IP.**Physical Layer** –Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber

optic cable and introduction about unguided media.

UNIT-II: Data link layer:

Data link layer: Design issues, **Framing:** fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer, **Elementary Data Link Layer protocols:** simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel.

Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol, Data link layer in HDLC, Point to point protocol (PPP) .

UNIT-III: Media Access Control: Random Access & The Network Layer Design Issues

Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, **Controlled Access:** Reservation, Polling, Token Passing, **Channelization:** frequency division multiple Access(FDMA), time division multiple access(TDMA), code division multiple access(CDMA).

The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service- Comparison of Virtual Circuit and Datagram Networks.

UNIT-IV: Routing Algorithms & Internet Working

Routing Algorithms-The Optimality principle- Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms-General principles of congestion control, Congestion prevention policies, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm-Leaky bucket & Token bucket.

Internet Working: How networks differ- How networks can be connected- Tunneling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6

UNIT-V: The Transport Layer & Application Layer

The Transport Layer: Transport layer protocols: Introduction-services- port number-User data gram protocol-User datagram-UDP services-UDP applications-Transmission control protocol: TCP services- TCP features- Segment- A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP.

Application Layer — World Wide Web: HTTP, Electronic mail-Architecture- web based mail-email security- TELNET-local versus remote Logging-Domain Name System.



Text Books:

1. Computer Networks, Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks, Behrouz A. Forouzan, Fifth Edition TMH

Reference Books

1. Data Communications and Networks- Achut S Godbole, AtulKahate
2. Computer Networks, Mayank Dave, CENGAGE



B.TECH V SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM5T03 OPERATING SYSTEMS

Course Objectives: The main objectives of introducing this course are

1. Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection
2. Make use of process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.
3. Illustrate different conditions for deadlock and their possible solutions.

Course Outcomes: At the end of the course, student will be able to

CO1 Understand the role and structure of an operating system.

CO2 Apply process and thread concepts.

CO3 Analyze CPU scheduling algorithms.

CO4 Manage memory effectively.

CO5 Understand and implement file system concepts.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	-	-	-	-	-	-	-	-	3	2
CO2	3	3	2	2	1	-	-	-	-	-	-	-	2	3
CO3	3	3	3	2	1	-	-	-	-	-	-	-	2	3
CO4	3	3	3	2	2	-	-	-	-	-	-	-	3	3
CO5	3	2	2	1	1	-	-	-	-	-	-	-	3	2

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put'-'1-Slight(Low),2-Moderate(Medium),3-Substantial(High).

SYLLABUS

UNIT-I: Operating Systems Overview

Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems System Structures: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure, Building and Booting an Operating System, Operating system debugging

UNIT-II: Processes:

Process Concept, Process scheduling, Operations on processes, Inter-process communication. Threads and Concurrency: Multithreading models, Thread libraries, Threading issues. CPU Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.

UNIT-III: Synchronization Tools:

The Critical Section Problem, Peterson's Solution, Mutex Locks, semaphores, Monitors, Classic problems of Synchronization. Deadlocks: system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock

UNIT-IV: Memory- Management Strategies:

Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping. Virtual Memory Management: Introduction, Demand paging, Copy-on-write, Page replacement Allocation of frames, Thrashing Storage Management: Overview of Mass Storage Structure, HDD Scheduling.

UNIT-V: File System:

File System Interface: File concept, Access methods, Directory Structure; File system Implementation: File-system structure, File-system Operations, Directory implementation, Allocation method, Free space management; File-System Internals: File-System Mounting, Partitions and Mounting, File Sharing. Protection: Goals of protection, Principles of protection, Protection Rings, Domain of protection, Access matrix.

Text Books:

- 1) Operating System Concepts, Silberschatz A, GalvinPB, GagneG, 10th Edition, Wiley, 2018.
- 2) Modern Operating Systems, Tanenbaum AS, 4th Edition, Pearson, 2016

Reference Books

1. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018
2. Operating Systems: A Concept Based Approach, D. M Dhamdhare, 3rd Edition, McGraw- Hill, 2013

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/106/106106144/>
2. <http://peterindia.net/OperatingSystems.html>

B.TECH V SEMESTER

PE	L	T	P	C
	3	0	0	3

23AM5T04 AUTOMATA THEORY & COMPILER DESIGN
(Professional Elective-I)

Course Objectives: The main objectives of introducing this course are

1. To introduce the fundamental concepts of formal languages, grammars, and automata.
2. To provide a deep understanding of deterministic and non-deterministic machines
3. To impart knowledge of compiler design principles.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the fundamental concepts of formal languages, grammars, and automata theory.
- CO2** Design finite automata, regular expressions, and context-free grammars for given language specifications.
- CO3** Analyze and construct deterministic and non-deterministic automata, pushdown automata, and Turing machines
- CO4** Understand the phases of a compiler and implement key components such as lexical analyzer, parser, and semantic analyzer
- CO5** Apply syntax-directed translation and intermediate code generation techniques to translate source programs into target code.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	-	-	-	-	-	-	-	-	2	2
CO2	3	3	2	2	2	-	-	-	-	-	-	-	2	3
CO3	3	3	2	2	-	-	-	-	-	-	-	-	2	2
CO4	3	2	2	3	2	-	-	-	-	-	-	-	1	2
CO5	3	2	3	2	2	-	-	-	-	-	-	-	3	2

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put '-'1-Slight(Low),2-Moderate(Medium),3 Substantial(High).

SYLLABUS**UNIT-I: Introduction to Finite Automata**

Structural Representations, Automata and Complexity, the Central Concepts of Automata Theory – Alphabets, Strings, Languages, Problems. Nondeterministic Finite Automata: Formal Definition, an application, Text Search, Finite Automata with Epsilon-Transitions.

Deterministic Finite Automata: Definition of DFA, How A DFA Process Strings, The language of DFA, Conversion of NFA with ϵ -transitions to NFA without ϵ -transitions. Conversion of NFA to DFA

UNIT-II: Regular Expressions:

Finite Automata and Regular Expressions, Applications of Regular Expressions, Algebraic Laws for Regular Expressions, Conversion of Finite Automata to Regular Expressions. Pumping Lemma for Regular Languages: Statement of the pumping lemma, Applications of the Pumping Lemma. Context-Free Grammars: Definition of Context-Free Grammars, Derivations Using a Grammar, Leftmost and Rightmost Derivations, the Language of a Grammar, Parse Trees, Ambiguity in Grammars and Languages.

UNIT-III: Push Down Automata:

Definition of the Pushdown Automaton, the Languages of a PDA, Equivalence of PDA's and CFG's, Acceptance by final state Turing Machines: Introduction to Turing Machine, Formal Description, Instantaneous description, The language of a Turing machine Undecidability: Undecidability, A Language that is Not Recursively Enumerable, An Undecidable Problem That is RE, Undecidable Problems about Turing Machines

UNIT-IV: Introduction:

The structure of a compiler, Lexical Analysis: The Role of the Lexical Analyzer, Input Buffering, Recognition of Tokens, The Lexical- Analyzer Generator Lex, Syntax Analysis: Introduction, Context-Free Grammars, Writing a Grammar, Top-Down Parsing, Bottom- Up Parsing, Introduction to LR Parsing: Simple LR, More Powerful LR Parsers

UNIT-V: Syntax-Directed Translation:

Syntax-Directed Definitions, Evaluation Orders for SDD's, SyntaxDirected Translation Schemes, Implementing L-Attributed SDD's. Intermediate-Code Generation: Variants of Syntax Trees, Three-Address Code Run-Time Environments: Stack Allocation of Space, Access to Nonlocal Data on the Stack, Heap Management

Text Books:

1. Introduction to Automata Theory, Languages, and Computation, 3rd Edition, John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, Pearson Education.
2. Compilers: Principles, Techniques and Tools, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, 2nd Edition, Pearson.
3. Theory of Computer Science – Automata languages and computation, Mishra and Chandrashekar, 2nd Edition, PHI.



Reference Books

1. Introduction to Formal languages Automata Theory and Computation, Kamala Krithivasan, Rama R, Pearson.
2. Introduction to Languages and The Theory of Computation, John C Martin, TMH.
3. lex&yacc – John R. Levine, Tony Mason, Doug Brown, O'reilly
4. Compiler Construction, Kenneth C. Loudon, Thomson. Course Technology.



B.TECH V SEMESTER

PE	L	T	P	C
	3	0	0	3

**23AM5T05 SOFTWARE ARCHITECTURE AND DESIGN PATTERN
(Professional Elective-I)**

Course Objectives: The main objectives of introducing this course are

1. Understand the creational and structural patterns.
2. Be capable of applying his knowledge to create an architecture for given application.
3. Be able to explain the role of analyzing architectures.
4. Be able to identify different structural patterns.

Course Outcomes: At the end of the course, student will be able to

CO1 Understand the architecture, creating it and moving from one to any, different structural patterns.

CO2 Analyze the architecture and build the system from the components

CO3 Design creational and structural patterns

CO4 Learn about behavioral patterns

CO5 Do a case study in utilizing architectural structures

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	1	1	1	1	2	2	2	2	2
CO2	2	3	3	2	2	1	1	1	2	2	2	2	2	3
CO3	2	2	3	2	2	1	1	1	1	2	2	2	3	2
CO4	1	1	2	1	2	1	1	1	1	2	2	2	1	2
CO5	2	2	3	3	3	1	1	2	2	3	2		3	2

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put '-'1-Slight(Low),2-Moderate(Medium),3 Substantial(High).

SYLLABUS

UNIT-I: Envisioning Architecture & Creating an Architecture

Envisioning Architecture: The Architecture Business Cycle, What is Software Architecture, Architectural patterns, reference models, reference architectures, architectural structures and views.

Creating an Architecture: Quality Attributes, Achieving qualities, Architectural styles and patterns, designing the Architecture, Documenting software architectures, Reconstructing Software Architecture.

UNIT-II: Analyzing Architectures & Moving from one system to many:

Analyzing Architectures: Architecture Evaluation, Architecture design decision making, ATAM, CBAM.

Moving from one system to many: Software Product Lines, Building systems from off the shelf components, Software architecture in future.

UNIT-III: Patterns, Creational and Structural patterns

Patterns: Pattern Description, Organizing catalogs, role in solving design problems, Selection and usage.

Creational and Structural patterns: Abstract factory, builder, factory method, prototype, singleton, adapter, bridge, composite, façade, flyweight, Proxy.

UNIT-IV: Behavioral patterns

Chain of responsibility, command, Interpreter, iterator, mediator, memento, observer state, strategy, template method, visitor.

UNIT-V: Case Studies

A-7E– A case study in utilizing architectural structures, The World Wide Web–a case study in inter operability, Air Traffic Control–a case study in designing for high availability, Celsius Tech– a case study in product line development.

Text Books:

1. Len Bass, Paul Clements & Rick Kazman, Software Architecture in Practice, 2nd Edition, Pearson Education, 2003.
2. Erich Gamma, Design Patterns, 1st Edition, Pearson Education, 1995

Reference Books

1. Luke Hohmann, Beyond Software architecture, Addison wesley, 2003.
2. David M. Dikel, David Kane and James R. Wilson, Software architecture, 1st Edition, Prentice hall, 2001.
3. F. Buschmann, Pattern Oriented Software Architecture, Wiley & Sons, 1st Edition, 2001.

B.TECH V SEMESTER

PE	L	T	P	C
	3	0	0	3

23AM5T06 INTERNET OF THINGS (IOT)
(Professional Elective-I)

Course Objectives: The main objectives of introducing this course are

1. Vision and Introduction to Internet of Things (IoT).
2. Understand IoT Market perspective.
3. Data and Knowledge Management and use of Devices in IoT Technology.
4. Understand State of the Art – IoT Architecture.
5. Understand Real World IoT Design Constraints, Industrial Automation and Commercial.

Course Outcomes: At the end of the course, student will be able to

- CO1** Explain in a concise manner how the general Internet as well as Internet of Things work.
- CO2** Understand constraints and opportunities of wireless and mobile networks for Internet of Things.
- CO3** Use basic sensing and measurement and tools to determine the real-time performance of network of devices.
- CO4** Develop prototype models for various applications using IoT technology.
- CO5** Explain in a concise manner how the general Internet as well as Internet of Things work.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	1	1	1	1	2	2	2	3	2
CO2	3	3	2	2	2	2	1	1	1	2	2	2	-	2
CO3	2	2	3	3	2	1	1	1	2	2	3	2	3	2
CO4	1	1	2	1	2	1	1	1	1	2	2	2	-	2
CO5	2	3	3	3	3	2	1	1	2	3	2	2	3	2

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-' 1-Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS**UNIT-I: The Internet of Things**

An Overview of Internet of things, Internet of Things Technology, behind IoTs Sources of the IoTs, M2M Communication, Examples of IoTs, Design Principles For Connected Devices Internet Connectivity Principles, Internet connectivity, Application Layer Protocols: HTTP, HTTPS, FTP, Telnet.

UNIT-II: Business Models

Business Models for Business Processes in the Internet of Things, IoT/M2M systems LAYERS AND designs standardizations, Modified OSI Stack for the IoT/M2M Systems ,ETSI M2M domains and High- level capabilities ,Communication Technologies, Data Enrichment and Consolidation and Device Management Gateway Ease of designing and affordability

UNIT-III: Design Principles for the Web Connectivity for Connected

Devices, Web Communication protocols for Connected Devices, Message Communication protocols for Connected Devices, Web Connectivity for Connected-Devices.

UNIT-IV:

Data Acquiring, Organizing and Analytics in IoT/M2M, Applications /Services /Business Processes, IOT/M2M Data Acquiring and Storage, Business Models for Business Processes in the Internet Of Things, Organizing Data, Transactions, Business Processes, Integration and Enterprise Systems.

UNIT-V:

Data Collection, Storage and Computing Using a Cloud Platform for IoT/M2M Applications/Services, Data Collection, Storage and Computing Using cloud platform Everything as a service and Cloud Service Models, IOT cloud-based services using the Xively (Pachube/COSM), Nimbits and other platforms Sensor, Participatory Sensing, Actuator, Radio Frequency Identification, and Wireless, Sensor Network Technology, Sensors Technology, Sensing the World.

Text Books:

1. Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill Higher Education
2. Internet of Things, A. Bahgya and V.Madisetti, Univesity Press, 2015

Reference Books

1. Designing the Internet of Things, Adrian McEwen and Hakim Cassimally, Wiley
2. Getting Started with the Internet of Things, CunoPfister , Oreilly



B.TECH V SEMESTER

PE **L T P C**
3 0 0 3

23AM5T07 EXPLORATORY DATA ANALYSIS WITH PYTHON
(Professional Elective-I)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Introduce the fundamentals of Exploratory Data Analysis
2. Cover essential exploratory techniques for understanding multivariate data by summarizing it through statistical methods and graphical methods.
3. Evaluate the Models and select the best model

Course Outcomes: At the end of the course, student will be able to

CO1 Understand the fundamental concepts of Exploratory Data Analysis

CO2 Apply data handling techniques using Python libraries

Demonstrate the use of various data visualization tools like matplotlib lib and sea born to

CO3 generate insightful visualizations such as bar charts, histograms, scatter plots, and polar charts.

Analyze datasets using statistical measures such as central tendency, dispersion,

CO4 correlation, and distributions, and perform uni variate, bi variate, multivariate, and time series analysis.

Develop and evaluate machine learning models by performing hypothesis testing, model

CO5 training, and applying evaluation metrics such as R2 score, MAE, and MSE to select the best performing models.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	1	-	-	2	3
CO2	3	3	2	2	2	-	-	-	-	1	-	-	2	2
CO3	2	2	3	2	3	-	-	-	1	2	-	-	2	2
CO4	3	2	3	3	2	-	-	-	-	2	-	-	2	2
CO5	3	2	3	3	3	-	-	-	-	2	2	1	2	2

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put“-”1-Slight(Low),2-Moderate(Medium),3 Substantial(High).

SYLLABUS

UNIT-I: Exploratory Data Analysis Fundamentals:

Understanding data science, The significance of EDA, Steps in EDA, Making sense of data, Numerical data, Categorical data, Measurement scales, Comparing EDA with classical and Bayesian analysis, Software tools available for EDA, Getting started with EDA.

Sample Experiments:

1. a) Download Dataset from Kaggle using the following link :

<https://www.kaggle.com/datasets/sukhmanibedi/cars4u>

b) Install python libraries required for Exploratory Data Analysis (numpy, pandas, matplotlib, seaborn)

2. Perform Numpy Array basic operations and Explore Numpy Built-in functions.

3. Loading Dataset into pandas dataframe

4. Selecting rows and columns in the dataframe

UNIT-II: Visual Aids for EDA:

Technical requirements, Line chart, Bar charts, Scatter plot using seaborn, Polar chart, Histogram, Choosing the best chart

Case Study: EDA with Personal Email, Technical requirements, Loading the dataset, Data transformation, Data cleansing, Applying descriptive statistics, Data refactoring, Data analysis.

Sample Experiments:

1. Apply different visualization techniques using sample dataset

a. Line Chart b. Bar Chart c. Scatter Plots d. Bubble Plot

2. Generate Scatter Plot using seaborn library for iris dataset

3. Apply following visualization Techniques for a sample dataset

a. Area Plot b. Stacked Plot c. Pie chart d. Table Chart

4. Generate the following charts for a dataset.

a. Polar Chart b. Histogram c. Lollipop chart

5. Case Study: Perform Exploratory Data Analysis with Personal Email Data

UNIT-III: Data Transformation:

Merging database-style dataframes, Concatenating along with an axis, Merging on index, Reshaping and pivoting, Transformation techniques, Handling missing data, Mathematical operations with NaN, Filling missing values, Discretization and binning, Outlier detection and filtering, Permutation and random sampling, Benefits of data transformation, Challenges.

Sample Experiments:

1. Perform the following operations

-
- a) Merging Dataframes b) Reshaping with Hierarchical Indexing
 - c) Data Deduplication d) Replacing Values
 2. Apply different Missing Data handling techniques
 - a) NaN values in mathematical Operations b) Filling in missing data
 - c) Forward and Backward filling of missing values d) Filling with index values
 - e) Interpolation of missing values
 3. Apply different data transformation techniques
 - a) Renaming axis indexes b) Discretization and Binning
 - c) Permutation and Random Sampling d) Dummy variables

UNIT-IV: Descriptive Statistics:

Distribution function, Measures of central tendency,

Measures of dispersion, Types of kurtosis, Calculating percentiles, Quartiles, Grouping Datasets, Correlation, Understanding univariate, bivariate, multivariate analysis, Time Series Analysis

Sample Experiments:

1. Study the following Distribution Techniques on a sample data
 - a) Uniform Distribution b) Normal Distribution
 - c) Gamma Distribution d) Exponential Distribution
 - e) Poisson Distribution f) Binomial Distribution
2. Perform Data Cleaning on a sample dataset.
3. Compute measure of Central Tendency on a sample dataset
 - a) Mean b) Median c) Mode
4. Explore Measures of Dispersion on a sample dataset
 - a) Variance b) Standard Deviation c) Skewness d) Kurtosis
5. a) Calculating percentiles on sample dataset
 - b) Calculate Inter Quartile Range(IQR) and Visualize using Box Plots
6. Perform the following analysis on automobile dataset.
 - a) Bivariate analysis b) Multivariate analysis
7. Perform Time Series Analysis on Open Power systems dataset

UNIT-V: Model Development and Evaluation:

Unified machine learning workflow, Data preprocessing,

Data preparation, Training sets and corpus creation, Model creation and training,

Model evaluation, Best model selection and evaluation, Model deployment

Case Study: EDA on Wine Quality Data Analysis

Sample Experiments:

1. Perform hypothesis testing using stats models library



a) Z-Test b) T-Test

2. Develop model and Perform Model Evaluation using different metrics such as prediction score, R2 Score, MAE Score, MSE Score.
3. Case Study: Perform Exploratory Data Analysis with Wine Quality Dataset

Text Books

1. Suresh Kumar Mukhiya, Usman Ahmed, Hands-On Exploratory Data Analysis with Python, Packt Publishing, 2020.

Reference Books

1. Ronald K. Pearson, Exploratory Data Analysis Using R, CRC Press, 2020
2. Radhika Datar, Harish Garg, Hands-On Exploratory Data Analysis with R: Become an expert in exploratory data analysis using R packages, 1st Edition, Packt Publishing, 2019

B.TECH V SEMESTER

PC	L	T	P	C
	0	0	3	1.5

23AM5L10 DEEP LEARNING LAB**Prerequisites:**

Course Objectives: The main objectives of introducing this course are

1. To provide hands-on experience in building and training neural networks for classification and prediction tasks.
2. To familiarize students with the implementation of convolution neural networks for image-based applications.
3. To introduce the concept of transfer learning using pre-trained models such as VGG16.
4. To enable students to apply various text encoding techniques including one-hot encoding and word embeddings for natural language processing tasks.
5. To develop the ability to design and evaluate recurrent neural networks for sequence modeling.
6. To expose students to real-time image and video processing using advanced object detection algorithms like YOLO.

Course Outcomes: At the end of the course, student will be able to

CO1 Implement deep neural networks to solve real world problems

CO2 Choose appropriate pre-trained model to solve real time problem

CO3 Interpret the results of two different deep learning models

CO4 Demonstrate the ability to use word representation techniques such as one-hot encoding and word embeddings to preprocess textual data for neural network models.

Utilize advanced deep learning architectures and tools such as pre-trained CNNs (e.g.,

CO5 VGG16), YOLO for video/image detection, and RNNs for sequence modeling and next-word prediction.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3	-	-	-	-	2	-	-	3	3
CO2	3	3	3	2	3	-	-	-	-	2	-	-	2	2
CO3	3	2	3	2	2	-	-	-	-	1	-	-	2	2
CO4	3	2	2	2	2	-	-	-	-	2	-	-	2	2
CO5	3	3	3	2	3	-	-	-	-	2	2	1	2	2

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put“-”1-Slight(Low),2-Moderate(Medium),3 Substantial(High).

LIST OF EXPERIMENTS

1. Implement multi-layer perceptron algorithm for MNISTH and written Digit Classification.
2. Design a neural network for classifying movie reviews (Binary Classification) using IMDB dataset.
3. Design a neural Network for classifying news wires (Multi class classification) using Reuters dataset.
4. Design a neural network for predicting house prices using Boston Housing Price dataset.
5. Build a Convolution Neural Network for MNISTH and written Digit Classification.
6. Build a Convolution Neural Network for simple image (dogs and Cats) Classification
7. Use a pre-trained convolution neural network (VGG16) for image classification.
8. Implement one hot encoding of words or characters.
9. Implement word embeddings for IMDB dataset.
10. Implement video, image processing using YOLO algorithms
11. Implementation of Recurrent Neural Network for next word prediction in a sentence.

Text Books

1. Reza Zadeh and Bharath Ramsundar, "Tensor flow for Deep Learning", O'Reilly publishers, 2018

Reference Books

1. <https://github.com/fchollet/deep-learning-with-python-notebooks>

Software Packages required:

- Keras
- Tensorflow
- PyTorch



B.TECH V SEMESTER

	L	T	P	C
PC	0	0	3	1.5

**23AM5L11 OPERATING SYSTEMS & COMPUTER NETWORKS
LAB**

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. To develop the concepts of process and memory management techniques.
2. To develop the concepts of process and memory management techniques.
3. To impart knowledge on developing shell scripts.

Course Outcomes: At the end of the course, student will be able to

CO1 Implement CPU and disk scheduling algorithms.

CO2 Demonstrate memory management techniques.

CO3 Demonstrate algorithms for Deadlock Detection and prevention.

CO4 Develop shell scripts in order to perform shell programming.

Design and implement network layer protocols and routing algorithms like Sliding

CO5 Window, Dijkstra's algorithm, and Distance Vector Routing to model efficient communication.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2	-	-	-	-	1	-	2	2	2
CO2	3	2	3	2	2	-	-	-	-	1	-	2	2	2
CO3	2	3	2	2	3	-	-	-	-	2	-	2	2	2
CO4	3	2	3	3	2	-	-	-	-	2	-	-	2	2
CO5	3	2	3	3	3	-	-	-	-	2	-	1	2	2

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1-Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

LIST OF EXPERIMENTS

1. Simulate the following CPU scheduling algorithms
 - a) FCFS b) SJF c) Priority d) Round Robin
2. Simulate the following page replacement algorithms
3. Implement LRU page replacement algorithm.
4. a) Illustrate Dead Lock Avoidance Algorithm
 - b) Illustrate Dead Lock Detection Algorithm
5. Illustrate UNIX commands and Vi editor



-
- 6.a) Write a Shell program to check the given number is even or odd
 - b) Write a shell script to print the factorial of first n natural numbers.
 - c) Write a shell script that counts the number of lines and words present in a given file.
 7. Write a Program to implement the data link layer framing methods such as
 - i) Character stuffing ii) bit stuffing.
 8. Write a program for Hamming Code generation for error detection and correction.
 9. Write a Program to implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
 10. Write a Program to implement Sliding window protocol for Selective repeat.
 11. Write a Program to implement Dijkstra's algorithm to compute the Shortest path through a graph.
 12. Write a Program to implement Distance vector routing algorithm by obtaining routing Table at each node (Take an example subnet graph with weights indicating delay between nodes).



B.TECH V SEMESTER

SC	L	T	P	C
	0	0	4	2

**23AM5S12 FULL STACK DEVELOPMENT-2
(Skill Enhanced Course)**

Course Objectives: The main objectives of introducing this course are

1. To provide comprehensive knowledge of front-end and back-end technologies used in modern web development.
2. To enable students to design, develop, and deploy scalable and responsive web applications using advanced tools, frameworks, and programming languages..
3. To impart hands-on experience with RESTful APIs, database integration, and authentication mechanisms
4. To cultivate problem-solving skills and coding practices through project-based learning and realtime case studies.

Course Outcomes: At the end of the course, student will be able to

- CO1** Design and implement RESTful APIs using ExpressJS, incorporating routing, HTTP methods (GET, POST, PUT, DELETE), and middleware for request handling and validation.
- CO2** Utilize templating engines like EJS to render dynamic HTML pages and handle form data submission and processing in ExpressJS applications.
- CO3** Implement session management using cookies and sessions in ExpressJS, and develop user authentication mechanisms to secure web applications.
- CO4** Connect ExpressJS applications to MongoDB using Mongoose, and perform CRUD operations to manage data effectively.
- CO5** Develop ReactJS applications by rendering HTML elements, writing JSX syntax, and creating both functional and class-based components for dynamic user interfaces.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2-	3	-	-	-	-	2	-	2	3	3
CO2	2	2	2	2	3	-	-	-	-	2	-	2	3	3
CO3	3	2	3	2	3	-	-	-	-	2	-	3	2	2
CO4	3	2	3	2	3	-	-	-	-	2	-	3	2	2
CO5	3	2	3	2	3	-	-	-	-	2	-	3	2	2

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put '-'.1-Slight(Low),2-Moderate(Medium),3 Substantial(High).

LIST OF EXPERIMENTS

Experiment 1: Express JS – Routing, HTTP Methods, Middleware.

- a. Write a program to define a route, Handling Routes, Route Parameters, Query

Parameters and URL building.

- b. Write a program to accept data, retrieve data and delete a specified resource using http methods.
- c. Write a program to show the working of middleware.

Experiment 2: Express JS – Templating, Form Data

- a. Write a program using templating engine.
- b. Write a program to work with form data.

Experiment 3: Express JS – Cookies, Sessions, Authentication

- a. Write a program for session management using cookies and sessions.
- b. Write a program for user authentication.

Experiment 4: ExpressJS – Database, RESTful APIs

- a. Write a program to connect MongoDB database using Mongoose and perform CRUD operations.
- b. Write a program to develop a single page application using RESTful APIs.

Experiment 5: ReactJS – Render HTML, JSX, Components – function & Class

- a. Write a program to render HTML to a web page.
- b. Write a program for writing markup with JSX.
- c. Write a program for creating and nesting components (function and class).

Experiment 6: ReactJS – Props and States, Styles, Respond to Events

- a. Write a program to work with props and states.
- b. Write a program to add styles (CSS & Sass Styling) and display data.
- c. Write a program for responding to events.

Experiment 7: ReactJS – Conditional Rendering, Rendering Lists, React Forms

- a. Write a program for conditional rendering.
- b. Write a program for rendering lists.
- c. Write a program for working with different form fields using react forms.

Experiment 8: ReactJS – React Router, Updating the Screen

- a. Write a program for routing to different pages using react router.
- b. Write a program for updating the screen.

Experiment 9: ReactJS – Hooks, Sharing data between Components

- a. Write a program to understand the importance of using hooks.
- b. Write a program for sharing data between components.

Experiment 10: ReactJS Applications – To-do list and Quiz

- a. Design to-do list application.

Experiment 11: MongoDB – Installation, Configuration, CRUD operations

- a. Install MongoDB and configure ATLAS
- b. Write MongoDB queries to perform CRUD operations on document using insert(), find(), update(), remove()

Experiment 12: MongoDB – Databases, Collections and Records

- a. Write MongoDB queries to Create and drop databases and collections.
- b. Write MongoDB queries to work with records using find(), limit(), sort(), createIndex(), aggregate().

Experiment 13-15: Augmented Programs: (Any 2 must be completed)

13. Design a to-do list application using NodeJS and ExpressJS.
14. Design a Quiz app using ReactJS.
15. Complete the MongoDB certification from MongoDB University website.

Text Books

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.

Web Links

1. ExpressJS - <https://www.tutorialspoint.com/expressjs>
2. ReactJS - <https://www.w3schools.com/REACT> (and) <https://react.dev/learn#>
3. MongoDB - <https://learn.mongodb.com/learning-paths/introduction-to-mongodb>



B.TECH V SEMESTER

ES	L	T	P	C
	0	0	2	1

23AM5L13 USER INTERFACE DESIGN USING FLUTTER

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Learns to Implement Flutter Widgets and Layouts
2. Understands Responsive UI Design and with Navigation in Flutter
3. Knowledge on Widges and customize widgets for specific UI elements, Themes
4. Understand to include animation apart from fetching data

Course Outcomes: At the end of the course, student will be able to

- CO1** **Develop** responsive user interfaces in Flutter using core widgets, layouts,media-query breakpoints, and multi-screen navigation.
- CO2** **Apply** state management, custom widgets, theming, forms, and validation techniques to build interactive and maintainable mobile app UIs.
- CO3** **Integrate** RESTful data, animations, testing, and debugging to design polished, cross-platform Flutter applications
- CO4** Create custom widgets, apply themes and styles, and implement forms with validation to enhance user interaction and experience.
- CO5** Integrate APIs for data handling, apply animations, and use testing and debugging tools to build robust and dynamic Flutter applications.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	3	-	-	-	-	2	-	2	3	3
CO2	2	2	3	2	3	-	-	-	-	2	-	2	3	3
CO3	2	2	3	2	3	-	-	-	-	2	-	2	3	2
CO4	2	2	3	2	3	-	-	-	-	2	-	2	2	2
CO5	3	2	3	3	3	-	-	-	-	2	2	3	2	2

Note: Enter Correlation Levels **1** or **2** or **3**.If there is no correlation,Put **-** **1**-Slight(Low),**2**-Moderate(Medium),**3** Substantial(High).

SYLLABUS

LIST OF EXPERIMENTS

1. a) Install Flutter and Dart SDK.
b) Write a simple Dart program to understand the language basics.
2. a) Explore various Flutter widgets (Text, Image, Container, etc.).
b) Implement different layout structures using Row, Column, and Stack widgets.
3. a) Design a responsive UI that adapts to different screen sizes.



-
- b) Implement media queries and breakpoints for responsiveness.
 - 4. a) Set up navigation between different screens using Navigator.
 - b) Implement navigation with named routes.
 - 5. a) Learn about stateful and stateless widgets.
 - b) Implement state management using set State and Provider.
 - 6. a) Create custom widgets for specific UI elements.
 - b) Apply styling using themes and custom styles.
 - 7. a) Design a form with various input fields.
 - b) Implement form validation and error handling.
 - 8. a) Add animations to UI elements using Flutter's animation framework.
 - b) Experiment with different types of animations (fade, slide, etc.).
 - 9. a) Fetch data from a REST API.
 - b) Display the fetched data in a meaningful way in the UI.
 - 10. a) Write unit tests for UI components.
 - b) Use Flutter's debugging tools to identify and fix issues.

Text Books

1. Marco L. Napoli, Beginning Flutter: A Hands-on Guide to App Development.
2. Rap Payne, Beginning App Development with Flutter: Create Cross-Platform Mobile Apps 1st Edition, Apres.



B.TECH VI SEMESTER

	L	T	P	C
PC	3	0	0	3

23AM6T01 REINFORCEMENT LEARNING

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. To familiarize students with core RL algorithms, such as Dynamic Programming, Monte Carlo methods, and Temporal-Difference Learning.
2. To develop the ability to design and implement RL algorithms for solving sequential decision-making problems.
3. **To enable students to evaluate and apply RL techniques**

Course Outcomes: At the end of the course, student will be able to

- CO1** Enumerate the elements of Reinforcement Learning
- CO2** Solve then-armed Bandit problem
- CO3** Compare different Finite Markov Decision Process
- CO4** Discuss about Monte Carlo Methods in solving real world problems
- CO5** List the Applications and Case Studies of Reinforcement Learning

COURSE ARTICULATION MATRIX(Correlation between COs,POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	2	3	-
CO2	3	3	2	-	2	-	-	-	-	-	-	2	3	2
CO3	3	3	3	2	2	-	-	-	-	-	-	2	3	2
CO4	2	3	3	3	3	-	-	-	2	2	-	2	3	3
CO5	3	2	-	-	-	-	-	-	-	-	-	2	3	-

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put -1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: The Reinforcement Learning Problem

Reinforcement Learning, Examples, Elements of Reinforcement Learning, Limitations and Scope, An Extended Example: Tic-Tac-Toe, Summary, History of Reinforcement Learning.

UNIT-II: Multi-arm Bandits

An n-Armed Bandit Problem, Action-Value Methods, Incremental Implementation, tracking a Non stationary Problem, Optimistic Initial Values, Upper-Confidence-Bound Action Selection, Gradient Bandits, Associative Search (Contextual Bandits)

UNIT-III: Finite Markov Decision Processes

The Agent–Environment Interface, Goals and Rewards, Returns, Unified Notation for Episodic and Continuing Tasks, The Markov Property, Markov Decision Processes, Value Functions, Optimal Value Functions, Optimality and Approximation.

UNIT-IV: Monte Carlo Methods

Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control without Exploring Starts, Off- policy Prediction via Importance Sampling, Incremental Implementation, Off-Policy Monte Carlo Control, Importance Sampling on Truncated Returns.

UNIT-V: Applications and Case Studies

TD-Gammon, Samuel’s Checkers Player, The Acrobot, Elevator Dispatching, Dynamic Channel Allocation, Job-Shop Scheduling, Robot path planning.

Text Books

1. Richard S. Sutton and Andrew G. Barto, “Reinforcement Learning-An Introduction”, 2ndEdition, The MIT Press,2018
2. Marco Wiering, Martijn van Otterlo Reinforcement Learning: State-of-the-Art (Adaptation, Learning, and Optimization(12)) 2012thEdition

Reference Books

1. Vincent François-Lavet, Peter Henderson, Riashat Islam, An Introduction to Deep Reinforcement Learning (Foundations and Trends(r) in Machine Learning) , 2019.



B.TECH VI SEMESTER

PC	L	T	P	C
	3	0	0	3

23AM6T02 SOFTWARE ENGINEERING

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Gain knowledge about software process models.
2. Familiarize the basic software engineering methods, practices and its applications.
3. Facilitate students in software design.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the software life cycle models.
- CO2** Understand the scrum approach to agile project management.
- CO3** Analyze the software requirements and generate SRS document.
- CO4** Understand some of the different models that may be used to design.
- CO5** Understand various software testing approaches and quality control to ensure good quality software.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	-	-	-	-	1	-	1	-	3
CO2	3	2	2	1	-	-	-	-	2	1	2	1	-	3
CO3	3	3	3	2	2	-	-	-	-	2	-	2	3	-
CO4	3	3	3	2	2	-	-	-	-	1	-	1	3	-
CO5	3	2	2	2	2	-	-	-	-	1	-	2	-	-

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put - '1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction to software engineering & Process models:

Introduction to software engineering: Nature of software, Software engineering, The Software Processes, Software Myths.

Process models: A Generic Process Model, Process Assessment and Improvement, Prescriptive Process Models, Specialised Process models, The Unified Process, Personal and Team Process Models.

UNIT-II: Requirements engineering & Requirements modelling

Requirements engineering: Functional and Non-Functional Requirements, The Software

Requirements Document, Requirements Specification, requirementsEngineering, Requirements Elicitation and Analysis, Requirement Validation, Requirement Management.

Requirements modelling: Requirement Analysis, Scenario-Based Modelling, Data Modelling Concepts, Class-Based Modelling.

UNIT-III: Design concepts:

The Design Process, Design Concepts, The Design Models, Architectural Design: Software Architecture, Architectural Genres, Architectural Styles. User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Design Evaluation.

UNIT-IV: Understanding of UML diagrams:

Structural diagrams - class diagram, object diagram, component diagram, deployment diagram, Behavioural diagrams - Use-casediagram, activity diagram, sequence diagram, collaboration diagram, statechart diagram.

UNIT-V: Implementation & Software testing strategies:

Implementation: Structured coding Techniques, Coding Styles-Standards and Guidelines, Implementation Issues.

Software testing strategies: A Strategic approach to Software Testing, Strategic Issues and Test Strategies for Conventional Software, Validation Testing, System Testing, The Art of Debugging, White-Box Testing, Black Box Testing ,Software Quality concepts.

Text Books

- 1.Roger S. Pressman (2010), Software Engineering, A Practitioner's Approach,7th Edition, McGraw-Hill International Edition, India.
- 2.Ian Sommerville (2011), Software Engineering, 9th Edition, Pearson education, India.
- 3.Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Bobbi J. Young, Ph.D. Jim ConallenKelli A. Houston," Object-Oriented Analysis and Design with Applications", 3rd edition

Reference Books

1. PankajJalote (2010), Software Engineering, A Precise Approach, Wiley India.
2. Waman S. Jawadekar (2008), Software Engineering: A Primer, McGraw-Hill, India.
3. Hans Van Vilet (2008), Software Engineering Principles and Practice, 3rd Edition, John Wiley & Sons Ltd.
4. Rajib Mall (2005), Fundamental of Software Engineering, PHI.
5. Deepak Jain, Software Engineering, Principles and Practices, Oxford,University Press, India

B.TECH VI SEMESTER

	L	T	P	C
PC	3	0	0	3

23AM6T03 DATA VISUALIZATION**Prerequisites:**

Computer Graphics, Image Processing

Course Objectives: The main objectives of introducing this course are

1. familiarize students with the basic and advanced techniques of information visualization and scientific visualization
2. learn key techniques of the visualization process
3. a detailed view of visual perception, the visualized data and the actual visualization, interaction and distorting techniques

Course Outcomes: At the end of the course, student will be able to

CO1 Explain the fundamental concepts of visualization, including its history, the visualization process, and principles of visual perception such as Gestalt laws to effectively represent data.

CO2 Apply various visual mapping and design techniques to create meaningful visual representations and develop visualization applications based on the visualization reference model.

CO3 Classify different types of visualization systems and techniques, including interaction methods, and identify common pitfalls such as misleading visualizations.

CO4 Demonstrate visualization methods for complex data structures such as groups, trees, graphs, clusters, and networks, including metaphorical visualization approaches.

CO5 Evaluate advanced visualization techniques for volumetric data, vector fields, geographic information systems (GIS), and collaborative visualizations, incorporating recent trends and perception techniques.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	-	-	-	-	-	2	-	3	3	-
CO2	2	3	2	2	3	-	-	-	1	2	1	3	3	2
CO3	2	3	-	3	2	-	-	-	-	2	-	3	3	2
CO4	2	3	2	2	3	-	-	-	1	2	1	3	3	3
CO5	2	2	3	3	3	1	2	1	1	2	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).



SYLLABUS

UNIT-I:

What Is Visualization?, History of Visualization, Relationship between Visualization and Other Fields. The Visualization Process, Introduction of visual perception, visual representation of data, Gestalt principles, and information overloads.

UNIT-II:

Creating visual representations, visualization reference model, visual mapping, visual analytics, Design of visualization applications

UNIT-III:

Classification of visualization systems, Interaction and visualization techniques misleading, Visualization of one, two and multi-dimensional data, text and text documents.

UNIT-IV:

Visualization of groups, trees, graphs, clusters, networks, software, Metaphorical visualization

UNIT-V:

Visualization of volumetric data, vector fields, processes and simulations, Visualization of maps, GIS systems, collaborative visualizations, Evaluating visualizations

Recent trends in various perception techniques, various visualization techniques, data structures used in data visualization.

Text Books

WARD, GRINSTEIN, KEIM. Interactive Data Visualization: Foundations, Techniques, and Applications. Natick : A K Peters, Ltd.

E. Tufte, The Visual Display of Quantitative Information, Graphics Press.

Reference Books

https://kdd.cs.ksu.edu/Courses/CIS536/Lectures/Slides/Lecture-34-Main_6up.pdf

B.TECH VI SEMESTER

	L	T	P	C
PE-II	3	0	0	3

23AM6T04 CRYPTOGRAPHY & NETWORK SECURITY
(Professional Elective-II)**Prerequisites:**

Course Objectives: The main objectives of introducing this course are

1. To provide a comprehensive understanding of fundamental security concepts, classical cryptographic techniques, and the need for network security.
2. To enable students to explore mathematical foundations essential for cryptographic algorithms, including symmetric and asymmetric key cryptography.
3. To study and analyze encryption techniques like DES, AES, RSA, and ECC and their role in ensuring secure communications.
4. To understand the working of cryptographic hash functions, message authentication codes (MACs), and digital signatures to maintain data integrity and authenticity.
5. To introduce network and internet security protocols such as SSL/TLS, IPsec, and email security standards like S/MIME and PGP.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the need of information security and its importance.
- CO2** Apply symmetric security mechanisms for confidentiality
- CO3** Apply asymmetric security mechanisms for confidentiality
- CO4** Apply Hash Function and digital signature techniques for authentication
- CO5** Understand network security designs using available secure solutions (such as PGP, SSL, IPsec)

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	2	1	1	1	1	-	-	-	2	3	-
CO2	3	3	2	2	2	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	1	-	1	-	1	1	2	3	3
CO4	3	3	2	3	2	1	-	2	-	1	1	2	3	3
CO5	3	2	2	3	3	2	1	2	2	2	1	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put - 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS**UNIT-I: Security Concepts**

Introduction, The need for security, Security approaches, Principles of security, Types of

Security attacks, Security services, Security Mechanisms, A model for Network Security Cryptography.

Classical Encryption Techniques-symmetric cipher model, Substitution techniques, Transposition techniques.

UNIT-II: Introduction to Symmetric Cryptography

Algebraic Structures-Groups, Rings, Fields, $GF(2^n)$ fields, Polynomials.

Mathematics of Asymmetric cryptography: Primes, Checking For Primness, Eulers phi-functions, Fermat's Little Theorem, Euler's Theorem, Generating Primes, Primality Testing, Factorization, Chinese Remainder Theorem.

UNIT-III: Symmetric key Ciphers

Block Cipher principles, DES, AES, Block cipher operation, Stream ciphers: RC4, RC5.

Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, Elgamal Cryptographic system, Elliptic Curve Arithmetic, Elliptic Curve Cryptography.

UNIT-IV: Cryptographic Hash Functions

Applications of Cryptographic Hash Functions, Two Simple Hash Functions, Requirements and Security, Hash Functions Based on Cipher Block Chaining, Secure Hash Algorithms (SHA).

Message Authentication Codes: Message Authentication Requirements, Message Authentication Functions, Requirements for Message Authentication Codes, Security of MAC'S, MAC'S Based On Hash Functions: HMAC, MAC'S Based On Block Ciphers: DAA And CMAC

Digital Signatures: Digital Signatures, Elgamal Digital Signature Scheme, Elliptic Curve Digital Signature Algorithm.

UNIT-V: Network and Internet Security:

Transport-Level Security: Web Security Considerations, Transport Level Security, HTTPS, SSH.

IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Authentication Header Protocol.

Electronic-Mail Security: Internet-mail Security, Email Format, Email Threats and Comprehensive Email Security, S/MIME, PGP.

Text Books

1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 7th Edition, 2017
2. Cryptography and Network Security: Behrouz A. Forouzan Debdeep, McGraw Hill, 3rd Edition, 2015



Reference Books

1. Cryptography and Network Security: AtulKahate, McGraw Hill, 3rd Edition
2. Introduction to Cryptography with Coding Theory: Wade Trappe, Lawrence C. Washington, Pearson.
3. Modern Cryptography: Theory and Practice ByWenbo Mao. Pearson

B.TECH VI SEMESTER

PE-II	L	T	P	C
	3	0	0	3

23AM6T05 RECOMMENDER SYSTEMS
(Professional Elective-II)**Prerequisites:**

Course Objectives: The main objectives of introducing this course are

1. This course covers the basic concepts of recommender systems, including personalization algorithms, evaluation tools, and user experiences
2. Understanding collaborative filtering techniques and address challenges like scalability and attacks.
3. This course covers different types of recommendation systems
4. Describe system evaluation methods from both algorithmic and users' perspectives
5. Describe applications of recommender systems in various domains

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the fundamental concepts, applications, and challenges of recommender systems, including linear algebra notations and rating mechanisms.
- CO2** Apply user-based and item-based collaborative filtering techniques and analyze their vulnerabilities.
- CO3** Design content-based recommender systems using various profiling, similarity, and classification methods; explore knowledge-based recommender approaches.
- CO4** Analyze and implement hybrid recommender systems using monolithic, parallelized, and pipelined strategies.
- CO5** Evaluate recommender systems using historical datasets, various error and decision-support metrics, and understand the role of communities and social tagging in personalized recommendations.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	–	–	–	–	1	–	1	3	2
CO2	2	3	3	2	2	–	–	–	1	1	1	2	3	3
CO3	2	3	3	2	3	–	–	–	2	2	2	2	3	3
CO4	2	2	3	3	3	–	–	–	1	2	2	2	3	3
CO5	2	3	2	3	2	2	1	1	2	2	1	3	3	2

Note: Enter Correlation Levels 1 or 2 or 3. If there is nocorrelation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction

Recommender system functions, Linear Algebra notation: Matrix addition, Multiplication, transposition, and inverses, covariance matrices, Understanding ratings, Applications of recommendation systems, Issues with recommender system.

UNIT-II: Collaborative Filtering

User-based nearest neighbor recommendation, Item-based nearest neighbor recommendation, Model based and pre-processing based approaches, Attacks on collaborative recommender systems.

UNIT-III: Content-based recommendation

High level architecture of content-based systems, Advantages and drawbacks of content based filtering, Item profiles, Discovering features of documents, Obtaining item features from tags, Representing item profiles, Methods for learning user profiles, Similarity based retrieval, Classification algorithms.

Knowledge based recommendation: Knowledge representation and reasoning, Constraint based recommenders, Case based recommenders.

UNIT-IV: Hybrid approaches

Opportunities for hybridization, Monolithic hybridization design: Feature combination, Feature augmentation, Parallelized hybridization design: Weighted, Switching, Mixed, Pipelined hybridization design: Cascade Meta-level, Limitations of hybridization strategies.

UNIT-V: Evaluating Recommender System

Introduction, General properties of evaluation research, Evaluation designs, Evaluation on historical datasets, Error metrics, Decision-Support metrics, User-Centered metrics.

Recommender Systems and communities: Communities, collaboration and recommender systems in personalized web search, Social tagging recommender systems, Trust and recommendations

Text Books

1. Jannach D., Zanker M. and FelFering A., Recommender Systems: An Introduction, Cambridge University Press(2011), 1st ed.
2. Ricci F., Rokach L., Shapira D., Kantor B.P., Recommender Systems Handbook, Springer(2011), 1st ed.

Reference Books

1. Manouselis N., Drachsler H., Verbert K., Duval E., Recommender Systems For Learning, Springer (2013), 1st ed.



B.TECH VI SEMESTER

PE-II	L	T	P	C
	3	0	0	3

23AM6T06 MEAN STACK TECHNOLOGIES

(Professional Elective-II)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client.
2. Writing optimized front end code HTML and JavaScript.
3. Monitor the performance of web applications & infrastructure and Troubleshooting web application with a fast and accurate are solution.
4. Design and implementation of Robust and Scalable Front End Applications

Course Outcomes: At the end of the course, student will be able to

- CO1** Enumerate the Basic Concepts of Web & Markup Languages.
CO2 Develop web Applications using Scripting Languages & Frameworks
CO3 Make use of Express JS and Node JS frameworks.
CO4 Illustrate the uses of web services concepts like restful, Angular.
CO5 Apply Deployment Techniques & Working with cloud platform

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	—	—	—	—	1	—	1	3	2
CO2	2	3	3	2	3	—	—	—	1	2	1	2	3	3
CO3	2	3	3	2	3	—	—	—	1	2	1	2	3	3
CO4	2	2	3	2	3	—	—	—	1	2	1	2	3	3
CO5	2	2	3	3	3	—	—	—	2	2	1	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '—' 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction to Web

Internet and World Wide Web, Domain name service, Protocols: HTTP, FTP, SMTP. Html5 concepts, CSS3, Anatomy of a webpage. XML: Document type Definition, XML schemas, Document object model, XSLT, DOM and SAX Approaches.

UNIT-II: Java Script

The Basic of Java Script: Objects, Primitives Operations and Expressions, Control Statements, Arrays, Functions, Constructors, Pattern Matching using Regular Expressions. Angular Java Script Angular JS Expressions: ARRAY, Objects, oval, Strings, Angular JS Form Validation & Form Submission, Single Page Application development using Angular JS.

UNIT-III: Node js

Introduction, Advantages, Node.js Process Model, Node JS Modules. Express.js: Introduction to Express Frame work, Introduction to Node js, What is Node s, Getting Started with Express, Your first Express App, Express Routing, Implementing MVC in Express, Middleware, Using Template Engines, Error Handling, API Handling, Debugging, Developing Template Engines, Using Process Managers, Security & Deployment

UNIT-IV: REST full Web Services

Using the Uniform Interface, Designing URIs, Web Linking, Conditional Requests. React Js: Welcome to React, Obstacles and Road blocks, React's Future, Keeping Up with the Changes, Working with the Files, Pure React, Page Setup, The Virtual DOM, React Elements, React DOM, Children, Constructing Elements with Data, React Components, DOM Rendering, Factories.

UNIT-V: Mongo DB

Introduction, Architecture, Features, Examples, Database Creation & Collection in Mongo DB. Deploying Applications: Web hosting & Domains, Deployment Using Cloud Platforms.

Text Books

1. Programming the World Wide Web, Robert W Sebesta, 7ed, Pearson.
2. Web Technologies, Uttam K Roy, Oxford.
3. Pro Mean Stack Development, Elad Elrom, Apress.
4. Restful Web Services Cookbook, Subbu Allamraju, O'Reilly.
5. JavaScript & jQuery the missing manual, David Sawyer McFarland, O'Reilly.
6. Web Hosting for Dummies, Peter Pollock, John Wiley Brand

Reference Books

1. Ruby on Rails up and Running, Lightning fast Web development, Bruce Tate, Curt Hibbs, Oreilly (2006).
2. Programming Perl, 4ed, Tom Christiansen, Jonathan Orwant, Oreilly (2012).
3. Web Technologies, HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.
4. An Introduction to Web Design, Programming, Paul Swang, Sanda S Katila, Engage Learning.
5. Express. JS Guide, The Comprehensive Book on Express.js, Azat Mardan, Lean Publishing.

B.TECH VI SEMESTER

PE-II	L	T	P	C
	3	0	0	3

23AM6T07 SOCIAL NETWORK ANALYSIS
(Professional Elective-II)**Prerequisites:**

Course Objectives: The main objectives of introducing this course are

1. To understand the concept of semantic web and related applications.
2. To understand functionality of various communities in web social networks.
3. To understand human behaviour and privacy issues in social web and related communities.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand and Develop semantic web related applications
CO2 Represent knowledge using ontology
CO3 Understand Web Mining
CO4 Predict human behavior in social web and related communities
CO5 Interpret data from Social Networks using visualization techniques

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	—	—	—	1	2	—	1	3	2
CO2	3	2	3	2	3	—	—	—	—	2	—	1	3	2
CO3	3	3	3	3	3	—	—	—	—	2	—	2	3	3
CO4	2	3	3	3	3	2	2	2	2	2	—	2	3	3
CO5	2	2	2	3	3	—	—	—	1	2	—	2	3	-

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS**UNIT-I: INTRODUCTION**

Introduction to Semantic Web: Limitations of current Web – Development of Semantic Web – Emergence of the Social Web – Social Network analysis: Development of Social Network Analysis – Key concepts and measures in network analysis – Electronic sources for network analysis: Electronic discussion networks, Blogs and online communities – Web-based networks – Applications of Social Network Analysis.

UNIT-II: MODELLING, AGGREGATING AND KNOWLEDGE REPRESENTATION

Ontology and their role in the Semantic Web: Ontology-based knowledge Representation – Ontology languages for the Semantic Web: Resource Description Framework – Web

Ontology Language – Modelling and aggregating social network data: State-of-the-art in network data representation – Ontological representation of social individuals – Ontological representation of social relationships – Aggregating and reasoning with social network data – Advanced representations.

UNIT-III: EXTRACTION AND MINING COMMUNITIES IN WEB SOCIAL NETWORKS

Extracting evolution of Web Community from a Series of Web Archive – Detecting communities in social networks – Definition of community – Evaluating communities – Methods for community detection and mining – Applications of community mining algorithms – Tools for detecting communities social network infrastructures and communities – Decentralized online social networks – Multi-Relational characterization of dynamic social network communities.

UNIT-IV: PREDICTING HUMAN BEHAVIOR AND PRIVACY ISSUES

Understanding and predicting human behavior for social communities – User data management – Inference and Distribution – Enabling new human experiences – Reality mining – Context – Awareness – Privacy in online social networks – Trust in online environment – Trust models based on subjective logic – Trust network analysis – Trust transitivity analysis – Combining trust and reputation – Trust derivation based on trust comparisons – Attack spectrum and counter measures.

UNIT-V: VISUALIZATION AND APPLICATIONS OF SOCIAL NETWORKS

Graph theory – Centrality – Clustering – Node-Edge Diagrams – Matrix representation – Visualizing online social networks, visualizing social networks with matrix-based representations – Matrix and Node-Link Diagrams – Hybrid representations – Applications – Cover networks – Community welfare – Collaboration networks – Co-Citation networks.

Text Books

1. Peter Mika, —Social Networks and the Semantic Web, First Edition, Springer 2007.
2. Borko Furht, —Handbook of Social Network Technologies and Applications, 1st Edition, Springer, 2010.

Reference Books

1. Guandong Xu, Yanchun Zhang and Lin Li, -Web Mining and Social Networking – Techniques and applications, First Edition, Springer, 2011.
2. Dion Goh and Schubert Foo, -Social information Retrieval Systems: Emerging Technologies and Applications for Searching the Web Effectively, IGI Global Snippet, 2008.



B.TECH VI SEMESTER

PE-II	L	T	P	C
	3	0	0	3

23AM6T08 OBJECT ORIENTED ANALYSIS AND DESIGN
(Professional Elective-II)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Become familiar with all phases of OOAD.
2. Master the main features of the UML
3. Master the main concepts of Object Technologies and how to apply the mat work and develop the ability to analyze and solve challenging problem in various domains
4. Learn the Object design Principles and understand how to apply them towards Implementation.

Course Outcomes: At the end of the course, student will be able to

CO1 Understand the necessity of Object Modeling

CO2 Represent classes, responsibilities and states using UML notation.

CO3 Demonstrate knowledge about the conceptual Model of UML

CO4 Model the event driven state of object and transform them into implementation specific layouts.

CO5 Identify, Analyze the subsystems, various components and collaborate them interchangeably

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2	-	-	-	1	1	-	2	2	1
CO2	3	3	3	2	3	-	-	-	2	2	1	2	3	3
CO3	3	2	3	2	3	-	-	-	2	2	1	2	3	3
CO4	3	3	3	3	3	-	-	-	2	2	1	2	3	3
CO5	3	3	3	3	3	-	-	-	3	3	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is nocorrelation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction

The Structure of Complex systems, The Inherent Complexity of Software, Attributes of Complex System, Organized and Disorganized Complexity, Designing Complex Systems.

Case Study: System Architecture: Satellite-Based Navigation

UNIT-II: Introduction to UML

Importance of modeling, principles of modeling, object oriented modeling, conceptual model of the UML, Architecture, and Software Development Life Cycle. Basic Structural Modeling: Classes, Relationships, common Mechanisms, and diagrams.

Case Study: Control System: Traffic Management.

UNIT-III: Class & Object Diagrams

Terms, concepts, modelling techniques for Class & Object Diagrams. Advanced Structural Modeling: Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages.

Case Study: AI: Cryptanalysis.

UNIT-IV: Basic Behavioral Modeling-I

Interactions, Interaction diagrams Use cases, Use case Diagrams, Activity Diagrams.

Case Study: Web Application: Vacation Tracking System

UNIT-V: Advanced Behavioral Modeling

Events and signals, state machines, processes and Threads, time and space, state chart diagrams. Architectural Modeling: Component, Deployment, Component diagrams and Deployment diagrams

Case Study: Weather Forecasting

Text Books

1. Grady BOOCH, Robert A.Maksimchuk, Michael W. ENGLE, BobbiJ. Young, Jim Conallen, Kellia Houston , “Object- Oriented Analysis and Design with Applications”, 3rd edition,2013,PEARSON
2. Grady Booch, James Rumbaugh, Ivar Jacobson : The Unified Modeling Language User Guide, Pearson Education.

Reference Books

1. Meilir Page-Jones: Fundamentals of Object Oriented Design in UML, Pearson Education.
2. Pascal Roques: Modeling Software Systems Using UML2,WILEY Dreamtech India Pvt. Ltd.
3. Atul Kahate: Object Oriented Analysis &Design, The McGraw-Hill Companies.
4. Applying UML and Patterns :An introduction to Object– Oriented Analysis and Design and Unified Process, Craig Larman, Pearson Education.



B.TECH VI SEMESTER

PE-III	L	T	P	C
	3	0	0	3

23AM6T09 COMPUTER VISION
(Professional Elective-III)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. To understand the Fundamental Concepts related to sources, shadows and shading
2. To understand the Geometry of Multiple Views
3. To understand motion analysis.
4. To study some applications of computer vision algorithms

Course Outcomes: At the end of the course, student will be able to

CO1 Understand the fundamental concepts of Measuring Light and Inter reflections

CO2 Apply various filtering techniques

CO3 Analyze and implement image segmentation, feature extraction, and object recognition algorithms

CO4 Apply linear dynamic models such as Kalman filters for object tracking

CO5 Apply geometric principles in real-world case studies like mobile robot localization and medical image registration.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	-	-	-	-	1	-	2	2	1
CO2	3-	3	3	2	3	-	-	-	1	2	-	2	3	3
CO3	3	3	3	3	3	-	-	-	2	2	1	2	3	3
CO4	3	3	3	3	3	-	-	-	2	2	2	2	3	3
CO5	3	3	3	3	3	-	-	-	2	3	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-' 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Cameras

Pinhole Cameras Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Inter reflections: Global Shading Models Color: The Physics of Color, Human Color Perception, Representing Color,

A Model for Image Color, Surface Color from Image Color.

UNIT-II: Linear Filters

Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates, Edge Detection: Noise, Estimating Derivatives, Detecting Edges Texture: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.

UNIT-III: The Geometry of Multiple Views

Two Views Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and Gestalt, Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering,

UNIT-IV: Segmentation by Fitting a Model

The Hough Transform, Fitting Lines, Fitting Curves, Fitting as a Probabilistic Inference Problem, Robustness Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice, Tracking With Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples

UNIT-V: Geometric Camera Models

Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, Case study: Mobile Robot Localization Model- Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtaining Hypotheses by pose Clustering, Obtaining Hypotheses Using Invariants, Verification, Case study: Registration In Medical Imaging Systems, Curved Surfaces and Alignment.

Text Books

1. David A. Forsyth and Jean Ponce: Computer Vision – A Modern Approach, PHI Learning (Indian Edition), 2009.

Reference Books

1. E. R. Davies: Computer and Machine Vision – Theory, Algorithms and Practicalities, Elsevier (Academic Press), 4th edition, 2013.
2. R. C. Gonzalez and R. E. Woods “Digital Image Processing” Addison Wesley 2008.
3. Richard Szeliski “Computer Vision: Algorithms and Applications” Springer-Verlag London Limited 2011

B.TECH VI SEMESTER

	L	T	P	C
PE-III	3	0	0	3

23AM6T10 CLOUD COMPUTING
(Professional Elective-III)**Prerequisites:****Course Objectives:** The main objectives of introducing this course are

1. To explain the evolving utility computing model called cloud computing.
2. To introduce the various levels of services offered by cloud.
3. To discuss the fundamentals of cloud enabling technologies such as distributed computing, service-oriented architecture and virtualization
4. To emphasize the security and other challenges in cloud computing.
5. To introduce the advanced concepts such as containers, serverless computing and cloudcentric Internet of Things.

Course Outcomes: At the end of the course, student will be able to

- CO1** Analyze fundamental cloud computing concepts, models (reference, service, deployment), characteristics, benefits, and leading providers.
- CO2** Summarize enabling cloud technologies: parallel/distributed computing, IPC, SOA, web services, and virtualization.
- CO3** Familiarize with virtualization/container characteristics, taxonomy, technology examples, and their role in cloud computing, including orchestration and public cloud.
- CO4** Explain cloud computing challenges: economics, interoperability, scalability, fault tolerance, energy efficiency, and security (shared responsibility model).
- CO5** Identify advanced cloud concepts: serverless, FaaS, IoT integration, edge/fog computing, DevOps, infrastructure-as-code, and emerging trends like quantum cloud computing

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2	-	-	-	1	1	-	2	2	1
CO2	3	3	2	2	3	-	-	-	1	1	-	2	3	3
CO3	3	3	3	2	3	-	-	-	2	2	1	2	3	3
CO4	3	3	3	3	3	2	2	2	2	2	1	3	3	3
CO5	3	2	3	3	3	1	1	1	2	2	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS**UNIT-I: Introduction to Cloud Computing Fundamentals**

Cloud computing at a glance, defining a cloud, cloud computing reference model, types of

services (IaaS, PaaS, SaaS), cloud deployment models (public, private, hybrid), utility computing, cloud computing characteristics and benefits, cloud service providers (Amazon Web Services, Microsoft Azure, Google AppEngine).

UNIT-II: Cloud Enabling Technologies

Ubiquitous Internet, parallel and distributed computing, elements of parallel computing, hardware architectures for parallel computing (SISD, SIMD, MISD, MIMD), elements of distributed computing, Inter-process communication, technologies for distributed computing, remote procedure calls (RPC), service-oriented architecture (SOA), Web services, virtualization.

UNIT-III: Virtualization and Containers

Characteristics of virtualized environments, taxonomy of virtualization techniques, virtualization and cloud Computing, pros and cons of virtualization, technology examples (XEN, VMware), building blocks of containers, container platforms (LXC, Docker), container orchestration, Docker Swarm and Kubernetes, public cloud VM (e.g. Amazon EC2) and container (e.g. Amazon Elastic Container Service) offerings.

UNIT-IV: Cloud computing challenges

Economics of the cloud, cloud interoperability and standards, scalability and fault tolerance, energy efficiency in clouds, federated clouds, cloud computing security, fundamentals of computer security, cloud security architecture, cloud shared responsibility model, security in cloud deployment models.

UNIT-V: Advanced concepts in cloud computing

Serverless computing, Function-as-a-Service, serverless computing architecture, public cloud (e.g. AWS Lambda) and open-source (e.g. Open FaaS) serverless platforms, Internet of Things (IoT), applications, cloud-centric IoT and layers, edge and fog computing, DevOps, infrastructure-as-code, quantum cloud computing.

Text Books

1. Mastering Cloud Computing, 2nd edition, Rajkumar Buyya, Christian Vecchiola, Thamarai Selvi, Shivananda Poojara, Satish N. Srirama, McGraw Hill, 2024.
2. Distributed and Cloud Computing, Kai Hwang, Geoffery C. Fox, Jack J. Dongarra, Elsevier, 2012.

Reference Books

1. Cloud Computing, Theory and Practice, Dan C Marinescu, 2nd edition, MK Elsevier, 2018.
2. Essentials of cloud Computing, K. Chandrasekhran, CRC press, 2014.
3. Online documentation and tutorials from cloud service providers (e.g., AWS, Azure, GCP)



B.TECH VI SEMESTER

PE-III	L	T	P	C
	3	0	0	3

23AM6T11 DEVOPS
(Professional Elective-III)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Describe the agile relationship between development and IT operations
2. Understand the skill sets and high-functioning teams involved in DevOps and related methods to reach a continuous delivery capability.
3. Implement automated system update and DevOps lifecycle.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand the core concepts of DevOps, its lifecycle, principles, and the importance of automation in the SDLC
- CO2** Effectively utilize Git for source code management, version control, and collaboration, and perform unit testing with code coverage analysis using SonarQube.
- CO3** Implement Continuous Integration (CI) using Jenkins, automate build processes, and manage Jenkins environments.
- CO4** Establish Continuous Delivery (CD) pipelines, containerize applications with Docker, and utilize testing tools like Selenium
- CO5** Perform configuration management using Ansible and orchestrate containerized applications with Kubernetes (OpenShift).

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2	-	-	-	1	1	1	2	2	2
CO2	3	3	2	2	3	-	-	-	2	2	1	2	3	2
CO3	3	3	3	3	3	-	-	-	2	2	2	3	3	3
CO4	3	3	3	2	3	-	-	-	2	2	2	3	3	3
CO5	3	3	3	3	3	-	-	-	2	2	3	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is nocorrelation, Put - 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction to DevOps

Introduction to SDLC, Agile Model. Introduction to Devops. DevOps Features, DevOps Architecture, DevOps Lifecycle, Understanding Workflow and principles, Introduction to DevOps tools, Build Automation, Delivery Automation, Understanding Code Quality,

Automation of CI/ CD. Release management, Scrum, Kanban, delivery pipeline, bottlenecks, examples

UNIT-II: Source Code Management (GIT)

The need for source code control, The history of source code management, Roles and code, source code management system and migrations. What is Version Control and GIT, GIT Installation, GIT features, GIT workflow, working with remote repository, GIT commands, GIT branching, GIT staging and collaboration. UNIT TESTING - CODE COVERAGE: Junit, NUnit & Code Coverage with Sonar Qube, SonarQube - Code Quality Analysis.

UNIT-III: Build Automation - Continuous Integration (CI)

Build Automation, What is CI Why CI is Required, CI tools, Introduction to Jenkins (With Architecture), Jenkins workflow, Jenkins master slave architecture, Jenkins Pipelines, PIPELINE BASICS - Jenkins Master, Node, Agent, and Executor Freestyle Projects & Pipelines, Jenkins for Continuous Integration, Create and Manage Builds, User Management in Jenkins Schedule Builds, Launch Builds on Slave Nodes

UNIT-IV: Continuous Delivery (CD):

Importance of Continuous Delivery, CONTINUOUS DEPLOYMENT CD Flow, Containerization with Docker: Introduction to Docker, Docker installation, Docker commands, Images & Containers, DockerFile, Running containers, Working with containers and publish to Docker Hub. Testing Tools: Introduction to Selenium and its features, JavaScript testing.

UNIT-V: Configuration Management – ANSIBLE

Introduction to Ansible, Ansible tasks, Roles, Jinjatemplating, Vaults, Deployments using Ansible. CONTAINERIZATION USING KUBERNETES (OPENSIFT): Introduction to Kubernetes Namespace & Resources, CI/CD - On OCP, BC, DC & ConfigMaps, Deploying Apps on Openshift Container Pods. Introduction to Puppet master and Chef.

Text Books

1. Joyner, Joseph., DevOps for Beginners: DevOps Software Development Method Guide for Software Developers and It Professionals, 1st Edition MihailsKonoplow, 2015.
2. Alisson Machado de Menezes., Hands-on DevOps with Linux, 1st Edition, BPB Publications, India, 2021.

Reference Books

1. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley; ISBN-10
2. Gene Kim Je Humble, Patrick Debois, John Willis. The DevOps Handbook, 1st Edition, IT Revolution Press, 2016.
3. Verona, Joakim Practical DevOps, 1st Edition, Packt Publishing, 2016.



4. Joakim Verona. Practical Devops, Ingram short title; 2nd edition (2018). ISBN10: 1788392574
5. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications. ISBN: 9788126579952



B.TECH VI SEMESTER

PE-III	L	T	P	C
	3	0	0	3

23AM6T12 SOFT COMPUTING
(Professional Elective-III)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. To introduce the concepts in Soft Computing such as Artificial Neural Networks, Fuzzy logic-based systems, genetic algorithm-based systems and their hybrids.

Course Outcomes: At the end of the course, student will be able to

- CO1** Learn soft computing techniques and their applications.
- CO2** Analyze various neural network architectures.
- CO3** Define the fuzzy systems.
- CO4** Understand the genetic algorithm concepts and their applications.
- CO5** Identify and select a suitable Soft Computing technology to solve the problem; construct a solution and implement a Soft Computing solution.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	-	-	-	-	-	-	2	3	-
CO2	3	3	3	2	2	-	-	-	1	1	-	1	-	2
CO3	3	2	2	2	2	-	-	-	-	-	-	1	3	-
CO4	3	2	2	2	2	-	-	-	-	-	-	2	3	-
CO5	3	3	3	3	3	-	-	-	1	1	1	3	-	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1-Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I:

Introduction to Soft Computing, Artificial neural networks, biological neurons, Basic models of artificial neural networks, Connections, Learning, Activation Functions, McCulloch and Pitts Neuron, Hebb network.

UNIT-II:

Perceptron networks, Learning rule, Training and testing algorithm, Adaptive Linear Neuron, Back propagation Network, Architecture, Training algorithm.

UNIT-III:

Fuzzy logic, fuzzy sets, properties, operations on fuzzy sets, fuzzy relations, operations on fuzzy relations, Fuzzy membership functions, fuzzification, Methods of membership, value assignments, intuition, inference, rank ordering, Lambda –Cuts for fuzzy sets , Defuzzification methods

UNIT-IV:

Truth values and Tables in Fuzzy Logic, Fuzzy propositions, Formation of fuzzy rules, Decomposition of rules, Aggregation of rules, Fuzzy Inference Systems, Mamdani and Sugeno types, Neuro-fuzzy hybrid systems, characteristics, classification

UNIT-V:

Introduction to genetic algorithm, operators in genetic algorithm, coding, selection, crossover, mutation, stopping condition for genetic algorithm flow, Genetic-neuro hybrid systems, Genetic Fuzzy rule based system

Text Books

1. S. N. Sivanandam and S. N. Deepa, Principles of soft computing–John Wiley & Sons,2007.
2. Timothy J. Ross, Fuzzy Logic with engineering applications, John Wiley & Sons, 2016.

Reference Books

1. N.K. Sinha and M.M. Gupta, Soft Computing & Intelligent Systems: Theory & Applications-Academic Press /Elsevier. 2009.
2. Simon Haykin, Neural Network-A Comprehensive Foundation-Prentice Hall International, Inc.1998
3. R. Eberhart and Y. Shi, Computational Intelligence: Concepts to Implementation, Morgan Kaufman/Elsevier, 2007.
3. Bart Kosko, Neural Network and Fuzzy Systems-Prentice Hall, Inc., Englewood Cliffs, 1992
4. Goldberg D.E, Genetic Algorithms in Search, Optimization, and Machine Learning Addison Wesley, 1989.



B.TECH VI SEMESTER	PC	L	T	P	C
		0	0	3	1.5

23AM6L15 SOFTWARE ENGINEERING LAB

Course Objectives:

1. To provide hands-on experience with different aspects of Software Engineering including requirements identification, implementation, testing, and so on.
2. To draw DFD, behavioural and structural design using UML diagrams.

Course Outcomes:

- CO1** Prepare SRS document, design document, test cases and software configuration management and risk management related document.
- CO2** Develop function oriented and object-oriented software design using tools like rational rose
- CO3** Design and develop Test Cases for a system
- CO4** Understand the significance of action analytics
- CO5** Track the progress of a project using various tools.

COURSE ARTICULATION MATRIX(Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	-	-	-	1	1	1	1	3	2
CO2	2	2	3	2	3	-	-	-	1	1	1	1	3	3
CO3	2	2	2	3	3	-	-	-	1	2	1	1	3	2
CO4	2	2	2	2	3	-	-	-	2	2	3	2	2	3
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, Put **-**. **1**-Slight(Low), **2**-Moderate(Medium), **3** Substantial(High).

The following experiments need to be performed using both Hardware and simulation Software.

The experiments need to be simulated using software and the same need to be verified using the hardware.

LIST OF EXPERIMENTS

1. Create the problem statement for a specific system of relevance.
2. Perform requirement analysis and develop Software Requirement Specification Sheet (SRS) for suggested system.
3. To carry out the function-oriented diagram: Data Flow Diagram (DFD) and Structured chart.

-
4. To draw UML diagrams.
 5. To illustrate the test cases, test case preparation and perform Manual Tests.
 6. Perform Estimation of effort using FP Estimation for chosen system.
 7. To prepare time line chart/Gantt Chart/PERT Chart for selected software project.

S.No	Case Study
1	Credit Card Processing
2	Stock Maintenance System
3	Online course reservation system
4	Recruitment system
5	Passport automation System
6	Online Exam Registration

Note: Students shall prepare a document related to all the above activities for at least three real time Case Studies listed below.



B.TECH VI SEMESTER	PC	L	T	P	C
		0	0	3	1.5

23AM6L16 DATA VISUALIZATION LAB

Course Objectives:

1. To visualize the different datasets using histograms, line charts.
2. To understand the use of bar charts and box plots.
3. To understand Scatter plots, mosaic plots
4. To understand different Map visualizations
5. To learn advanced graphs such as correlogram, heatmap and 3D graphs.

Course Outcomes:

CO1: Visualize the different datasets using histograms, line charts.

CO2: Make use of bar charts and box plots on different datasets

CO3: Apply Scatter plots, mosaic plots in R for different datasets

CO4: Apply different Map visualizations in R

CO5: Create advanced graphs such as correlogram, heatmap and 3D graphs

COURSE ARTICULATION MATRIX(Correlation between COs & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	3	-	-	-	1	2	1	2	2	2
CO2	3	2	2	2	3	-	-	-	1	2	1	2	3	3
CO3	3	3	2	3	3	-	-	-	1	2	1	2	3	3
CO4	3	2	3	2	3	1	1	-	1	2	2	2	3	3
CO5	3	3	3	3	3	-	-	-	1	2	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3.If there is no correlation,Put '-'.1-Slight(Low),2-Moderate(Medium),3 Substantial(High).

The following experiments need to be performed using both Hardware and simulation Software.

The experiments need to be simulated using software and the same need to be verified using the hardware.

LIST OF EXPERIMENTS

- 1.a) Load VADeaths(Death Rates in Virginia)dataset in R and visualize the data using different histograms.
 - b) Load air quality dataset in R and visualize La Guardia Airport's dialy maximum temperature using histogram.
2. Load Air Passengers dataset in R and visualize the data using line chart that shows increase in air passengers over given time period.

3. a) Load iris dataset in R, visualize the data using different Bar Charts and also demonstrate the use of stacked plots.
b) Load air quality dataset in R and visualize ozone concentration in air.
4. a) Load iris dataset in R, visualize the data using different Box plots including group by option and also use color palette to represent species.
b) Load air quality dataset in R and visualize air quality parameters using box plots.
5. Visualize iris dataset using simple scatter, multivariate scatter plot and also visualize scatter plot matrix to visualize multiple variables across each other.
6. Load diamonds dataset in R and visualize the structure in datasets with large data points using hexagon binning and also add color palette then use the
7. Load HairEyeColor dataset in R and plot categorical data using mosaic plot.
8. Load mtcars dataset in R and visualize data using heat map.
9. Install leaflet library in R and perform different map visualizations.
10. Visualize iris dataset using 3d graphs such as scatter3d, cloud, xyplot.
11. Make use of correlogram to visualize data in correlation matrices for iris dataset.
12. Install maps library in R and draw different map visualizations.

Processor: Intel i5

RAM: 8 GB

Storage: 256 GB SSD or 500 GB HDD

Graphics: Integrated graphics

Software Requirements:

Operating System - Windows 10/11 (64-bit)

Programming Languages and Libraries

- Python with:
 - pandas, numpy – for data handling
 - matplotlib, seaborn, plotly, bokeh, altair – for visualizations
 - dash, streamlit, voila – for interactive dashboards
- R (optional but useful) with:
 - ggplot2, shiny, plotly, lattice

Visualization Tools (GUI-based or drag-and-drop)

- **Tableau Public or Tableau Desktop (licensed)**

References:

1. <https://www.analyticsvidhya.com/blog/2015/07/guide-data-visualization-r/>
2. <https://www.geeksforgeeks.org/data-visualization-in-r/>



B.TECH VI SEMESTER

SC

L T P C
0 0 4 2

23AM6S17 SOFT SKILLS
(Skill Enhancement Course)

Course Objectives: The main objectives of introducing this course are

1. To equip the students with the skills to effectively communicate in English
2. To train the students in interview skills, group discussions and presentation skills
3. To motivate the students to develop confidence
4. To enhance the students' interpersonal skills
5. To improve the students' writing skills

Course Outcomes: At the end of the course, student will be able to

- CO1** Apply analytical thinking, listening skills, and effective verbal and non-verbal communication.
- CO2** Develop self-management skills, including anger, stress, and time management, alongside social and business etiquette.
- CO3** Apply fundamental grammar rules, pronunciation, and professional writing skills such as letter writing and note-making.
- CO4** Demonstrate job-oriented skills through group discussions; resume building, and interview preparation.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	-	-	-	-	-	2	2	2	3	3	2	2	-	-
CO2	-	-	-	-	-	2	2	2	3	3	3	2	-	-
CO3	-	-	-	-	-	2	2	2	3	3	2	2	-	-
CO4	-	-	-	-	-	2	2	2	3	3	2	2	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, Put **'-'** **1-** Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I:

Analytical Thinking & Listening Skills: Self-Introduction, Shaping Young Minds - A Talk by

Azim Premji (Listening Activity), Self – Analysis, Developing Positive Attitude, Perception.

Communication Skills: Verbal Communication; Non-Verbal Communication (Body Language)

UNIT-II:

Self-Management Skills: Anger Management, Stress Management, Time Management, Six Thinking Hats, Team Building, Leadership Qualities

Etiquette: Social Etiquette, Business Etiquette, Telephone Etiquette, Dining Etiquette

UNIT-III:

Standard Operation Methods: Basic Grammars, Tenses, Prepositions, Pronunciation, Letter Writing; Note Making, Note Taking, Minutes Preparation, Email & Letter Writing

UNIT-IV:

Job-Oriented Skills: Group Discussion, Mock Group Discussions, Resume Preparation, Interview Skills, Mock Interviews

UNIT-V:

Interpersonal relationships: Introduction, Importance, Types, Uses, Factors affecting interpersonal relationships, Accommodating different styles, Consequences of interpersonal relationships.

Text Books

1. Barun K. Mitra, Personality Development and Soft Skills, Oxford University Press, 2011.
2. S.P. Dhanavel, English and Soft Skills, Orient Blackswan, 2010.

Reference Books

1. R.S. Aggarwal, A Modern Approach to Verbal & Non-Verbal Reasoning, S. Chand & Company Ltd., 2018.
2. Raman, Meenakshi & Sharma, Sangeeta, Technical Communication Principles and Practice, Oxford University Press, 2011.

E-resources:

1. https://swayam-plus.swayam2.ac.in/courses/course-details?id=P_CAMBR_01



B.TECH VI SEMESTER

AC	L	T	P	C
	2	0	0	-

23AM6M18 TECHNICAL PAPER WRITING & IPR

Course Objectives: The main objectives of introducing this course are

1. To introduce the fundamentals of technical report writing, including sentence structure, transitions, and tense usage.
2. To develop skills in drafting, designing, and editing technical reports with a focus on clarity, grammar, and visual elements.
3. To enhance proofreading, summarization, and presentation skills for effective communication of technical content. tools PIG & HIVE in Hadoop echo System.

Course Outcomes: At the end of the course, student will be able to

- CO1** Illustrate the structure of technical reports, explain the use of appropriate language and formatting, and discuss how to write for a specific audience.
- CO2** Explain the structure of technical reports, describe how visuals support content, and identify areas for improvement in clarity, grammar, and layout.
- CO3** Summarize, explain, and describe technical reports, including the basics of proposal writing.
- CO4** Utilize word processing tools for advanced document features and document management.
- CO5** Explain Intellectual Property types, the patenting process, and global IP cooperation.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	2	3	2	2	2	2	2	2	2	2
CO2	-	-	-	-	2	3	2	2	2	2	2	-	-	-
CO3	-	-	-	-	2	3	2	2	2	2	2	2	-	-
CO4	-	-	-	-	2	3	2	2	2	2	2	-	-	-
CO5	-	-	-	-	2	3	2	2	2	2	2	2	-	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, Put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).



SYLLABUS

UNIT-I:

Introduction: An introduction to writing technical reports, technical sentences formation, using transitions to join sentences, Using tenses for technical writing.

Planning and Structuring: Planning the report, identifying reader(s), Voice, Formatting and structuring the report, Sections of a technical report, Minutes of meeting writing.

UNIT-II:

Drafting report and design issues: The use of drafts, Illustrations and graphics.

Final edits: Grammar, spelling, readability and writing in plain English: Writing in plain English, Jargon and final layout issues, Spelling, punctuation and Grammar, Padding, Paragraphs, and Ambiguity.

UNIT-III:

Proofreading and summaries: Proofreading, summaries, Activities on summaries.

Presenting final reports: Printed presentation, Verbal presentation skills, Introduction to proposals and practice.

UNIT-IV:

Adding a Table of Contents, Updating the Table of Contents, Deleting the Table of Contents, Adding an Index, Creating an Outline, Adding Comments, Tracking Changes, Viewing Changes, Additions, and Comments, Accepting and Rejecting Changes, Working with Footnotes and Endnotes, inserting citations and Bibliography, Comparing Documents, Combining Documents, Mark documents final and make them read only., Password protect Microsoft Word documents., Using Macros.

UNIT-V:

Nature of Intellectual Property: Patents, Designs, Trade and Copyright.

Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property.

Text Books

1. Kompal Bansal & Parshit Bansal, “Fundamentals of IPR for Beginner’s”, 1st Ed., BS Publications, 2016.
2. William S. Pfeiffer and Kaye A. Adkins, “Technical Communication: A Practical Approach”, Pearson.
3. Ramappa, T., “Intellectual Property Rights Under WTO”, 2nd Ed., S Chand, 2015.: The



Definitive Guide", Third Edition, O'Reilly, 2012

Reference Books

1. Adrian Wallwork, English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011.
2. Day R, How to Write and Publish a Scientific Paper, Cambridge University Press (2006)

E-resources:

1. <https://www.udemy.com/course/reportwriting/>
2. <https://www.udemy.com/course/professional-business-english-and-technical-report-writing/>
3. <https://www.udemy.com/course/betterbusinesswriting/>



OPEN ELECTIVE –I:

S. No.	Course Code	Course Name	L	T	P	C	Offered by
1	23CE5T08	Green Buildings	3	0	0	3	CE
2	23CE5T09	Construction technology and management	3	0	0	3	CE
3	23EE5T08	Renewable Energy Sources	3	0	0	3	EEE
4	23EE5T09	Concepts of Energy Auditing & Management	3	0	0	3	EEE
5	23ME5T09	Sustainable Energy Technologies	3	0	0	3	ME
6	23ME5T10	Design and Analysis of Experiments	3	0	0	3	ME
7	23ME5T11	Thermal Management of Electronic systems	3	0	0	3	ME
8	23EC5T09	Electronic Devices and Circuits	3	0	0	3	ECE
9	23EC5T10	Signals and Systems	3	0	0	3	ECE
10	23EC5T11	Digital Logic Design	3	0	0	3	ECE
11	23CS5T09	Operating Systems	3	0	0	3	CSE
12	23CS5T10	Computer organization and Architecture	3	0	0	3	CSE
13	23IT5T10	Database Management Systems	3	0	0	3	IT
14	23IT5T11	Computer Graphics	3	0	0	3	IT
15	23MB5T01	Entrepreneurship Development and Venture Creation	3	0	0	3	DMS
16	23MB5T03	Innovation and Technology Management	3	0	0	3	DMS
17	23AD5T09	Introduction to Data Science	3	0	0	3	AIDS
18	23AM5T09	Machine Learning	3	0	0	3	AIML

OPEN ELECTIVE –II:

S. No.	Course Code	Course Name	L	T	P	C	Offered by
1	23CE6T12	Disaster management	3	0	0	3	CE
2	23CE6T13	Sustainability in Engineering practices	3	0	0	3	CE
3	23EE6T10	Fundamentals of Electric Vehicles	3	0	0	3	EEE
4	23EE6T11	Electrical Wiring Estimation and Costing	3	0	0	3	EEE
5	23ME6T12	Introduction to Industrial Robotics	3	0	0	3	ME
6	23ME6T13	Introduction to Additive Manufacturing	3	0	0	3	ME
7	23ME6T14	Vehicle Technology	3	0	0	3	ME
8	23EC6T14	Linear & Digital IC Applications	3	0	0	3	ECE
9	23EC6T15	Principles Of Communications	3	0	0	3	ECE
10	23EC6T16	Microprocessors & Microcontrollers	3	0	0	3	ECE
11	23CS6T15	Database Management Systems	3	0	0	3	CSE
12	23CS6T16	Software Engineering	3	0	0	3	CSE
13	23IT6T14	Fundamentals of Machine Learning	3	0	0	3	IT
14	23IT6T15	Computer Networks	3	0	0	3	IT
15	23MB6T01	Industry Specific Employability Skills	3	0	0	3	DMS
16	23MB6T03	Supply Chain Management	3	0	0	3	DMS
17	23AD6T14	Artificial Intelligence	3	0	0	3	AIDS
18	23AM6T14	Social Media Analytics	3	0	0	3	AIML



B.TECH V SEMESTER

OE-I	L	T	P	C
	3	0	0	3

23CE5T08 GREEN BUILDINGS
(Open Elective-I)

Course Learning Objectives:

The objective of this course is:

1. Introduce the fundamental concepts, benefits, and prerequisites of green buildings.
2. Familiarize students with national green building standards, rating systems, and sustainable design strategies.
3. Develop understanding of energy efficiency techniques and integration of renewable energy in green building design.
4. Explain the principles and systems behind green HVAC and lighting design with emphasis on energy conservation.
5. Promote sustainable material selection and practices to enhance indoor air quality and occupant health.

Course Outcomes:

At the end of this course; the student will be able to

- CO1:** Understand the importance, benefits, and basic requirements of green buildings and identify key sustainable features.
- CO2:** Explain Indian green building movements, IGBC and LEED rating systems, and describe practices that enhance resource efficiency and sustainability in building design.
- CO3:** Apply energy-efficient design principles including demand reduction, system efficiency, and renewable energy integration in green building projects.
- CO4:** Analyze HVAC and lighting system designs with eco-friendly interventions and interpret energy modeling strategies for green buildings.
- CO5:** Recommend sustainable building materials and strategies for waste reduction and indoor environmental quality improvement, including air quality and occupant health.

COURSE ARTICULATION MATRIX (Correlation between COs, Pos & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	2	3	3	2	-	2	-	2	3	2
CO2	3	2	2	-	2	3	3	2	-	2	-	2	3	3
CO3	3	3	3	2	3	2	3	1	-	2	-	2	3	3
CO4	3	3	3	2	3	2	3	2	-	2	-	2	3	3
CO5	3	2	3	2	2	3	3	2	-	2	-	2	3	2

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-' 1- Slight (Low), 2 - Moderate (Medium), 3 - Substantial (High).

SYLLABUS:

UNIT-I

Introduction: What is Green Building, why to go for Green Building, Benefits of Green Buildings, Green Building Materials and Equipment in India, What are key Requisites for Constructing a Green Building, Important Sustainable features for Green Building,

UNIT -II

Green Building Concepts And Practices Indian Green Building Council, Green Building Moment in India, Benefits Experienced in Green Buildings, Launch of Green Building Rating Systems, Residential Sector, Market Transformation; Green Building Opportunities And Benefits: Opportunities of Green Building, Green Building Features, Material and Resources, Water Efficiency, Optimum Energy Efficiency, Typical Energy Saving Approach in Buildings, LEED India Rating System and Energy Efficiency,

UNIT-III

Green Building Design Introduction, Reduction in Energy Demand, Onsite Sources and Sinks, Maximize System Efficiency, Steps to Reduce Energy Demand and Use Onsite Sources and Sinks, Use of Renewable Energy Sources. Ecofriendly captive power generation for factory, Building requirement,

UNIT- IV

Air Conditioning Introduction Godrej Green business center, Design philosophy, Design interventions, Energy modeling, HVAC System design, Chiller selection, pump selection, Selection of cooling towers, Selection of air handing units, Precooling of fresh air, Interior lighting system, Key feature of the building. Eco-friendly captive power generation for factory , Building requirement.

UNIT-V

Material Conservation Handling of non process waste, waste reduction during construction, materials with recycled content, local materials, material reuse, certified wood, Rapidly renewable building materials and furniture; Indoor Environment Quality And Occupational Health: Air conditioning, Indore air quality, Sick building syndrome, Tobacco smoke control, Minimum fresh air requirements avoid use of asbestos in the building, improved fresh air ventilation, Measure of IAQ, Reasons for poor IAQ, Measures to achieve Acceptable IAQ levels,

Text Books:

1. Handbook on Green Practices published by Indian Society of Heating Refrigerating and Air conditioning Engineers, 2009.



2. Green Building Hand Book by Tomfoolery and Samkimings, 2009.

Recommended References:

1. Complete Guide to Green Buildings by Trish riley
2. Standard for the design for High Performance Green Buildings by Kent Peterson, 2009

B.TECH V SEMESTER

OE-I **L** **T** **P** **C**
 3 0 0 3

23CE5T09 CONSTRUCTION TECHNOLOGY AND MANAGEMENT
(Open Elective-I)

Course Objectives:

The objective of this course is:

1. To introduce to the student, the concept of project management including network drawing and monitoring
2. To introduce the various equipment related to construction like earth moving equipment, trucks and handling equipment, aggregate production and construction equipment and machinery
3. to introduce the importance of safety in construction projects

Course Outcomes:

Upon the successful completion of this course, the students will be able to:

- CO1:** Understand the principles of construction project management including planning, scheduling, coordination, and monitoring techniques like bar charts, milestone charts, and Critical Path Method (CPM).
- CO2:** Apply PERT techniques for project evaluation and review, and perform cost-time optimization through crashing and resource allocation; gain basic familiarity with construction management software such as Primavera.
- CO3:** Select and evaluate suitable construction equipment for earthwork, hoisting, and material handling based on economic and operational considerations.
- CO4:** Identify and manage concreting equipment including mixers, batching plants, and mobile units for efficient placement and finishing of concrete.
- CO5:** Demonstrate knowledge of various construction methods including earthwork, piling, formwork, and structural erection; apply principles of quality control, safety engineering, and BIM (Building Information Modelling) in civil engineering projects.

COURSE ARTICULATION MATRIX (Correlation between COs, Pos & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	3	3	2	-	-	1	1	2	2
CO2	3	3	3	2	2	3	3	3	-	-	2	2	3	3
CO3	3	2	3	1	3	3	2	2	-	-	1	1	2	2
CO4	3	2	2	1	3	3	2	2	-	-	1	1	2	2
CO5	3	3	2	3	3	3	3	3	-	-	3	3	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, Put **'-'** - Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS:

UNIT-I

Construction project management and its relevance – qualities of a project manager – project planning – coordination –scheduling - monitoring – bar charts – milestone charts – critical path method

UNIT-II

Project revaluation and review technique–cost analysis–updating–crashing for optimum cost–crashing for optimum resources–allocation of resources introduction to software's for construction management, project management using PRIMAVERA (or) equivalent.

UNIT-III

Construction equipment – economical considerations – earthwork equipment – Trucks and handling equipment – rear dump trucks – capacities of trucks and handling equipment – calculation of truck production – compaction equipment – types of compaction rollersHoistingandearthworkequipment–hoists–cranes–tractors–bulldozers–graders–scrapers–draglinesclamshellbuck

UNIT-IV

Concreting equipment—concrete mixers–Batching plants, mobile using plants like “Ajax” etc. mixing and placing of concrete – consolidating and finishing.

UNIT-V

Construction methods – earthwork – piling – placing of concrete – form work – fabrication and erection – quality control and safety engineering. BIM for Civil Engineers (Building Information Modelling)

TEXTBOOKS:

1. 'Construction Planning, Equipment and Methods' by Peurifoyand Schexnayder, Shapira, Tata McGraw hill.
2. 'Construction Project Management Theory and Practice' by Kumar Neeraj Jha (2011), Pearson.
3. 'Construction Technology' by Subir K.Sarkarand Subhajit Sarasvati, Oxford University press

REFERENCES:

1. 'Construction Project Management-An Integrated Approach' by Peter Fewings, Taylor and Francis
2. 'Construction Management Emerging Trends and Technologies' by Trefor Williams, Cengage learning

B.TECH V SEMESTER

OE-I **L** **T** **P** **C**
 3 **0** **0** **3**

23EE5T08 RENEWABLE ENERGY SOURCES
(Open Elective-I)

Pre-requisite: Basic Electrical Engineering

Course Objectives:

- To study the solar radiation data, equivalent circuit of PV cell and its I-V & P-V characteristics.
- To understand the concept of Wind Energy Conversion & its applications.
- To study the principles of biomass, hydel and geothermal energy.
- To understand the principles of ocean Thermal Energy Conversion, waves and power associated with it.
- To study the various chemical energy sources such as fuel cell and hydrogen energy along with their operation and equivalent circuit.

Course Outcomes:

After the completion of the course the student should be able to:

CO1: Analyze solar radiation data, extra-terrestrial radiation, radiation on earth's surface and solar Energy Storage.

CO2: Illustrate the components of wind energy systems.

CO3: Illustrate the working of biomass, hydel plants and Geothermal plants.

CO4: Demonstrate the principle of Energy production from OTEC, Tidal and Waves.

CO5: Evaluate the concept and working of Fuel cells & MHD power generation.

COURSE ARTICULATION MATRIX (Correlation between COs, Pos & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	1	1	-	-	-	3	3	1
CO2	3	2	2	3	3	-	1	1	-	-	-	2	3	1
CO3	3	2	3	3	2	-	1	1	-	-	-	3	2	1
CO4	3	2	3	2	2	-	1	1	-	-	-	2	3	1
CO5	3	3	2	3	1	-	1	1	-	-	-	2	3	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, Put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS:

UNIT-I

Solar Energy

Introduction - Renewable Sources - prospects, solar radiation at the Earth Surface -

Equivalent circuit of a Photovoltaic (PV) Cell - I-V & P-V Characteristics - Solar Energy Collectors: Flat plate Collectors, concentrating collectors - Solar Energy storage systems and Applications: Solar Pond - Solar water heating - Solar Green house.

UNIT-II

Wind Energy

Introduction - basic Principles of Wind Energy Conversion, the nature of Wind - the power in the wind - Wind Energy Conversion - Site selection considerations - basic components of Wind Energy Conversion Systems (WECS) - Classification - Applications.

UNIT-III

Biomass, Hydel and Geothermal Energy

Biomass: Introduction–Biomass conversion technologies–Photosynthesis. Factors affecting Bio digestion.

Hydro plants: Basic working principle – Classification of hydro systems: Large, small, micro hydel plants and pumped storage plants.

Geothermal Energy: Introduction, Geothermal Sources – Applications - operational and Environmental problems

UNIT-IV

Energy from oceans, Waves & Tides:

Oceans: Introduction - Ocean Thermal Electric Conversion (OTEC) – methods - prospects of OTEC in India.

Waves: Introduction - Energy and Power from the waves - Wave Energy conversion devices.

Tides: Basic principle of Tide Energy -Components of Tidal Energy.

UNIT-V

Chemical Energy Sources:

Fuel Cells: Introduction - Fuel Cell Equivalent Circuit - operation of Fuel cell - types of Fuel Cells – Applications

Hydrogen Energy: Introduction - Methods of Hydrogen production - Storage and Applications

TEXTBOOKS:

1. G.D.Rai, Non-Conventional Energy Sources, Khanna Publications, 2011.
2. John Twidell& Tony Weir, Renewable Energy Sources, Taylor & Francis, 2013.



REFERENCES:

1. P.Sukhatme & J.K.Nayak, Solar Energy-Principles of Thermal Collection and Storage, TMH, 2011
2. John Andrews & Nick Jelly, Energy Science- principles, Technologies and Impacts, Oxford, 2nd edition, 2013
3. ShobaNath Singh, Non- Conventional Energy Resources, Pearson Publications, 2015.

Online Learning Resources:

1. <https://archive.nptel.ac.in/courses/103/103/103103206>
2. <https://archive.nptel.ac.in/courses/103/107/103107157>

B.TECH V SEMESTER

OE-I **L T P C**
3 0 0 3

23EE5T09 CONCEPTS OF ENERGY AUDITING & MANAGEMENT
(Open Elective-I)

Pre-requisite: Basics of Conservation of Electrical Energy

Course Objectives:

- To understand basic concepts of Energy Audit & various Energy conservation schemes.
- To design energy an energy management program.
- To understand concept of Energy Efficient Motors and lighting control efficiencies.
- To estimate/calculate power factor of systems and propose suitable compensation techniques.
- To calculate life cycle costing analysis and return on investment on energy efficient technologies.

Course Outcomes:

After the completion of the course the student should be able to:

CO1: Understand the principles of energy audit along with various Energy related terminologies.

CO2: Asses the role of Energy Manager and Energy Management program.

CO3: Design a energy efficient motors and good lighting system.

CO4: Analyse the methods to improve the power factor and identify the energy Instruments for various real time applications

CO5: Evaluate the computational techniques with regard to economic aspects.

COURSE ARTICULATION MATRIX (Correlation between COs, Pos &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	-	-	-	-	-	-	2	3	2
CO2	3	2	2	2	-	-	-	-	-	-	-	2	3	1
CO3	3	2	3	2	-	-	-	-	-	-	-	2	3	1
CO4	3	2	3	2	-	-	-	-	-	-	-	2	3	2
CO5	3	3	2	1	-	-	-	-	-	-	-	2	3	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, **Put '-'** **1-** Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS:

UNIT-I

Basic Principles of Energy Audit:

Energy audit- definitions - concept - types of Energy audit - energy index - cost index - pie charts - Sankey diagrams and load profiles - Energy conservation schemes- Energy audit of industries- energy saving potential - energy audit of process industry, thermal power station - building energy audit - Conservation of Energy Building Codes (ECBC-2017).

UNIT-II

Energy Management:

Principles of energy management - organizing energy management program - initiating - planning - controlling - promoting - monitoring - reporting. Energy manager - qualities and functions - language - Questionnaire – check list for top management.

UNIT-III

Energy Efficient Motors and Lighting:

Energy efficient motors - factors affecting efficiency - loss distribution - constructional details - characteristics – variable speed - RMS - voltage variation-voltage unbalance-over motoring-motor energy audit. lighting system design and practice - lighting control - lighting energy audit

UNIT-IV

Power Factor Improvement and Energy Instruments:

Power factor – methods of improvement - location of capacitors - Power factor with non-linear loads - effect of harmonics on power factor - power factor motor controllers – Energy Instruments- watt meter - data loggers - thermocouples - pyrometers - lux meters - tongue testers.

UNIT-V

Economic Aspects and their Computation:

Economics Analysis depreciation Methods - time value of money - rate of return - present worth method - replacement analysis - lifecycle costing analysis – Energy efficient motors. Calculation of simple payback method - net present value method- Power factor correction - lighting – Applications of life cycle costing analysis - return on investment.

Text Books:

1. Energy management by W.R.Murphy&G.Mckay Butter worth - Heinemann publications 1982.



2. Energy management hand book by W.CTurner - John wiley and sons - 1982.

Reference Books:

1. Energy efficient electric motors by John.C.Andreas - Marcel Dekker Inc Ltd-2nd edition – 1995
2. Energy management by Paul o' Callaghan - Mc-graw Hill Book company-1st edition -1998
3. Energy management and good lighting practice : fuel efficiency- booklet12-EEO

Online Learning Resources:

1. <https://nptel.ac.in/courses/108106022>
2. <https://archive.nptel.ac.in/courses/108/106/108106022>

B.TECH V SEMESTER

OE-I	L	T	P	C
	3	0	0	3

23ME5T09 SUSTAINBLE ENERGY TECHNOLOGIES
(Open Elective-I)

Course Objectives:

- To introduce various renewable energy resources.
- To introduce solar energy and its radiation, collection, storage and applications.
- To introduce the wind energy power plants and its site assessment.
- To make familiar about Biomass and Geothermal energy and its conversion.
- To explore various other renewable energy sources like wave, tidal and hydrogen energy.

Course Outcomes:

At the end of the course student will be able to:

CO1: List and describe the primary renewable energy resources.

CO2: Summarize harnessing of solar energy and thermal systems.

CO3: Infer the harnessing of Biomass energy and Analyze the Geothermal energy harnessing.

CO4: Infer different types of wind mills and their of wind energy

CO5: Explain the significance of Hydrogen energy and Fuel cell technology.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	3	3	-	-	-	-	3	1	3
CO2	3	1	-	-	-	3	3	-	-	-	-	3	1	3
CO3	3	1	-	-	-	3	3	-	-	-	-	3	1	3
CO4	3	1	-	-	-	3	3	-	-	-	-	3	1	3
CO5	3	1	-	-	-	3	3	-	-	-	-	3	1	3

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put'-1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS:

UNIT-I

Renewable Energy (RE) Sources:

Environmental consequences of fossil fuel use, Importance of renewable sources of energy, Sustainable Design and development, Types of RE sources, Limitations of RE sources, Present Indian and international energy scenario of conventional and RE sources.

UNIT-II

Solar PV and Thermal Systems:

Solar Radiation, Radiation Measurement, Solar Thermal Power Plant, Solar Ponds.- Solar Photovoltaic systems (SPV): Basic Principle of SPV conversion – Types of PV Systems- Types of Solar Cells, Photovoltaic cell concepts.

UNIT-III

Biomass Energy:

Introduction-Bio mass resources –Energy from Bio mass: conversion processes-Biomass Cogeneration-Environmental Benefits.

Geothermal Energy:

Basics, Direct Use, Geothermal Electricity. Mini/micro hydro power: Classification of hydropower schemes, Classification of water turbine, Essential components of hydroelectric system.

UNIT-IV

Wind Energy:

Power in the Wind – Types of Wind Power Plants(WPPs)–Components of WPPs-Working of WPPs- Siting of WPPs-Grid integration issues of WPPs.

UNIT-V

Other Energy Sources:

Tidal Energy: Energy from the tides, Barrage and Non Barrage Tidal power systems.

Wave Energy: Energy from waves, wave power devices Ocean. Thermal Energy Conversion (OTEC)-Hydrogen Production and Storage- **Fuel cell:** Principle of working- various types - construction and applications.

Text Books:

1. Solar Energy – Principles of Thermal Collection and Storage/Sukhatme S.P. and J.K.Nayak/TMH.
2. Non-Conventional Energy Resources- Khan B.H/ Tata McGraw Hill, New Delhi, 2006.

Reference Books:

1. Principles of Solar Engineering - D.Yogi Goswami, Frank Krieth& John F Kreider / Taylor & Francis
2. Non-Conventional Energy - Ashok V Desai /New Age International (P) Ltd.
3. Renewable Energy Technologies -Ramesh & Kumar /Narosa.
4. Non-conventional Energy Source- G.D Roy/Standard Publishers.



-
5. Green Manufacturing Processes and Systems - J. Paulo Davim/Springer 2013.

B.TECH V SEMESTER

OE-I **L T P C**
 3 0 0 3

23ME5T10 DESIGN AND ANALYSIS OF EXPERIMENTS
(Open Elective-I)

Course Objectives:

- To identify the role of experimentations in engineering.
- To study the importance of single factor experimentation.
- To gain the concepts of multifactor and fractional factorial experimentation.
- To understand the usage of response surface methodology.
- To study the concepts of Taguchi methods.

Course Outcomes:

At the end of the course student will be able to:

CO1: Plan an experimentation using the experimental design strategies and principles.

CO2: Estimate model parameters and analyse the model adequacy.

CO3: Design experimentation for a multifactor problem using factorial design and Compute fractional factorial design of experimentation.

CO4: Propose design using a response surface methodology and interpret the results.

CO5: Correlate experimentation with a Taguchi based design for a given case study.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	2	3	-	-	-	-	-	-	3	2	-
CO2	2	1	-	2	3	-	-	-	-	-	-	3	2	-
CO3	3	1	-	2	3	-	-	-	-	-	-	3	2	-
CO4	3	1	-	2	3	-	-	-	-	-	-	3	2	-
CO5	3	1	-	2	3	-	-	-	-	-	-	3	2	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial(High).

SYLLABUS:

UNIT-I

Experimental Design Fundamental:

Importance of experiments, experimental strategies, basic principles of design, terminology, ANOVA, steps in experimentation, sample size, normal probability plot, linear regression model.

UNIT-II

Single Factor Experiments:

Completely randomized design, Randomized block design, Latin square design Statistical analysis, estimation of model parameters, model adequacy checking, pair wise comparison tests.

UNIT-III

Multifactor Experiments:

Two and three factor factorial experiments, 2K factorial experiments, confounding and blocking designs

Fractional Factorial design: Fractional factorial design of experiments, nested designs, split plot Design.

UNIT-IV

Special Experimental Design:

Response Surface Methodology, experiments with random factors, rules for expected mean squares, approximate F-tests.

UNIT-V

Taguchi Methods:

Steps in experimentation, design using Orthogonal Arrays, data analysis, Robust design – control and noise factors, S/N ratios, parameter design, case studies.

Text Books:

1. Design and Analysis of Experiments/ Montgomery DC / 7th Edition / John Wiley & Sons, NY / 2008,
2. Taguchi Techniques for Quality Engineering / Ross PJ / McGraw-Hill Book Company, NY/2nd Edition/2005.

Reference Books:

1. Quality Engineering using Robust Design, Madhav S. Phadke, Prentice Hall, 1989
2. Product and Process Design for Quality, Economy and Reliability, Dukkupati, R V and Pradip K Ray, New Age International, 2010, 1st Edn.

B.TECH V SEMESTER

OE-I L T P C
 3 0 0 3

**23ME5T11 THERMAL MANAGEMENT OF ELECTRONIC SYSTEMS
(Open Elective-I)**

Course Objectives:

- To impart concepts of thermodynamics and applications of first law of thermodynamics.
- To infer the modes of heat transfer, basic laws of heat transfer and one dimensional heat conduction
- To acquire knowledge in free and the heat transfer characteristics for free and forced convection heat transfer
- To illustrate the need of thermal management and cooling technologies.
- To expose to different applications of cooling technologies for heat generating systems.

Course Outcomes:

At the end of the course student will be able to:

CO1: Illustrate the basic concepts of thermodynamics and solve problems on applications of first law of thermodynamics.

CO2: Solve one dimensional heat conduction problems on different geometries.

CO3: Solve problems on free and forced convection heat transfer using empirical correlations.

CO4: Discuss the different thermal management and cooling technologies.

CO5: Apply suitable cooling technology for thermal management systems.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	1	-	-	-	-	1	2	3
CO2	3	3	-	-	-	-	1	-	-	-	-	1	2	3
CO3	3	3	-	-	-	-	2	-	-	-	-	1	2	3
CO4	3	3	-	-	-	-	2	-	-	-	-	1	2	3
CO5	3	3	-	-	-	-	2	-	-	-	-	2	2	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put - '1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS:

UNIT-I

Introduction to Thermodynamics: System, Surroundings, Boundaries, Universe, Types of

Systems, Control volume, Perfect Gas Laws, Equation of State, First law of Thermodynamics applied to cyclic process; Non-Flow Energy Equation, Heat and Work Transfer, Changes in Internal Energy and Enthalpy for various Non-flow processes; First law applied to a flow system, Steady Flow Energy Equation and applications.

UNIT-II

Introduction to modes of heat transfer, Basic laws of heat transfer, General heat conduction equation in Cartesian, Cylindrical and Spherical coordinates (No derivation)- Steady, unsteady and periodic heat transfer – Initial and boundary conditions

Steady one dimensional solution for conduction heat transfer in Cartesian, cylindrical and spherical geometry, overall heat transfer coefficient, electrical analogy, critical insulation thickness, heat transfer through fins.

UNIT-III

Basic equations of heat transfer by convection, Introduction to boundary layers, forced and natural convection, external and internal flows, Dimensionless parameters for forced and free convection heat transfer-Correlations for forced and free convection- Estimating heat transfer rates in laminar and turbulent flow situations using appropriate correlations for free and forced convection.

Geothermal Energy:

Basics, Direct Use, Geothermal Electricity. Mini/micro hydro power: Classification of hydropower schemes, Classification of water turbine, Essential components of hydroelectric system.

UNIT-IV

Introduction to Thermal Management - electronics packaging, contact and spreading resistances, heat sink design; Cooling Technologies - thermal interface and phase change materials, passive and novel air cooling approaches, micro-channels, jet impingement, immersion cooling, heat pipes and vapor chambers, thermo-electrics.

UNIT-V

Thermal Management Applications in avionics, data centers, mobile, internet of things, high-performance computing, automotive, and electric vehicle battery cooling, Fuel cell.

Text Books:

1. Engineering Thermodynamics / PK Nag /6th Edition/ Tata McGraw Hill/2017.
2. Heat Transfer: A Basic Approach / M. N. Ozisik/McGraw-Hill/1985.



-
3. Heat Transfer: Thermal Management of Electronics/Younes Shabany/1st edition/S.Chand &Company Ltd/2009

Reference Books:

1. Thermodynamics: An Engineering Approach/ Yunus Cengel and Boles/7th Edition/Tata McGraw Hill/2011.
2. Heat Transfer /J.P. Holman/10th Edition /Tata McGraw Hill/2011.
3. Heat Transfer/P.K. Nag/2nd Edition/ Tata McGraw Hill/2007.
4. Fundamentals of Engineering Heat and Mass Transfer / R. C. Sachdeva / New Age International/2010.
5. Thermal Management Handbook: For Electronic Assemblies/Jerry E. Sergeant / McGraw-Hill /1998.

B.TECH V SEMESTER

OE-I

L	T	P	C
3	0	0	3

23EC5T09 ELECTRONIC DEVICES AND CIRCUITS
(Open Elective-I)

COURSE OBJECTIVES: The main objectives of this course are given below:

- Understand the basic concepts of semiconductor physics and the behaviour of PN junction diodes.
- Explore the characteristics and applications of special semiconductor devices and rectifier circuits.
- Analyze the characteristics and configurations of bipolar junction transistors (BJTs) and field-effect transistors (FETs).
- Understand transistor biasing methods, stability considerations, and thermal stabilization.
- Develop the ability to model and analyze small signal low-frequency transistor amplifiers.

Course Outcomes:

At the end of the course student will be able to:

- CO1:** Explain the fundamental properties of semiconductors and analyze the characteristics of PN junction diodes.
- CO2:** Illustrate the operation and applications of special semiconductor devices and rectifiers.
- CO3:** Analyze the input/output characteristics and configurations of BJTs and FETs.
- CO4:** Design and evaluate biasing circuits for BJTs and FETs with thermal stability considerations.
- CO5:** Model and analyze small-signal transistor amplifier circuits using h-parameters and FET small signal models.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	-	-	-	-	-	-	-	1	3	2	-
CO2	3	2	2	1	-	-	-	-	-	-	-	1	3	2	-
CO3	3	3	3	2	2	-	-	-	-	-	-	1	3	3	-
CO4	3	3	3	2	2	-	-	-	-	-	-	1	3	3	-
CO5	3	3	3	2	3	-	-	-	-	-	-	1	3	3	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial (High).

SYLLABUS:

UNIT-I REVIEW OF SEMI CONDUCTOR PHYSICS:

Review of Semi Conductor Physics: Hall Effect, continuity equation, law of junction, Fermi Dirac function, Fermi level in intrinsic and extrinsic Semiconductors

Junction Diode Characteristics :Energy band diagram of PN junction Diode, Open circuited p-n junction, Biased p-n junction, p-n junction diode, current components in PN junction Diode, diode equation, V-I Characteristics, temperature dependence on V-I characteristics, Diode resistance, Diode capacitance..

UNIT-II SPECIAL SEMICONDUCTOR DEVICES:

Special Semiconductor Devices: Zener Diode, Breakdown mechanisms, Zener diode applications, LED, Varactor Diode, Photodiode, Tunnel Diode, UJT, PNP Diode, SCR. Construction, operation and V-I characteristics.

Rectifiers and Filters: Basic Rectifier setup, half wave rectifier, full wave rectifier, bridge rectifier, derivations of characteristics of rectifiers, rectifier circuits-operation, input and output waveforms, Filters, Inductor filter(Series inductor), Capacitor filter(Shunt inductor), π -Filter, comparison of various filter circuits in terms of ripple factors.

UNIT-III TRANSISTOR CHARACTERISTICS:

BJT: Junction transistor, transistor current components, transistor equation, transistor configurations, transistor as an amplifier, characteristics of transistor in Common Base, Common Emitter and Common Collector configurations, Ebers-Moll model of a transistor, punch through/ reach through, Photo transistor, typical transistor junction voltage values.

FET: FET types, construction, operation, characteristics μ , g_m , r_d parameters, MOSFET-types, construction, operation, characteristics, comparison between JFET and MOSFET.

UNIT-IV TRANSISTOR BIASING AND THERMAL STABILIZATION:

Need for biasing, operating point, load line analysis, BJT biasing- methods, basic stability, fixed bias, collector to base bias, self bias, Stabilization against variations in V_{BE} , I_c , and β , Stability factors, (S, S', S'') , Bias compensation, Thermal runaway, Thermal stability. FET Biasing- methods and stabilization.

UNIT-V SMALL SIGNAL LOW FREQUENCY TRANSISTOR AMPLIFIER MODELS:

BJT: Two port network, Transistor hybrid model, determination of h-parameters, conversion of h-parameters, generalized analysis of transistor amplifier model using h-parameters, Analysis of CB, CE and CC amplifiers using exact and approximate analysis, Comparison of transistor amplifiers.



FET: Generalized analysis of small signal model, Analysis of CG, CS and CD amplifiers, comparison of FET amplifiers.

Text Books:

1. Electronic Devices and Circuits- J. Millman, C. Halkias, Tata Mc-Graw Hill, Second Edition, 2007.
2. Electronic Devices and Circuits by David A. Bell, Oxford University Press.
3. Electronics devices & circuit theory- Robert L.Boylestad and LouiNashelsky, Pearson/Prentice hall, tenth edition, 2009.

Reference Books:

1. Integrated Electronics-J. Millman, C. Halkias,Tata Mc-Graw Hill, Second Edition,2009.
2. Electronic Devices and Circuits-K. Lal Kishore, BS Publications, Fourth Edition, 2016.

B.TECH V SEMESTER

OE-I	L	T	P	C
	3	0	0	3

23EC5T10 SIGNALS AND SYSTEMS
(Open Elective-I)

COURSE OBJECTIVES: The main objectives of this course are given below:

- Understand the fundamental concepts of signals and systems and their classifications.
- Analyze signals using Fourier series and Fourier transforms.
- Study signal correlation, energy and power spectra, and the sampling theorem.
- Apply Laplace transforms for system analysis in the continuous-time domain.
- Understand Z-transforms and their application to discrete-time system analysis.

Course Outcomes:

At the end of the course student will be able to:

- CO1:** Classify signals and systems and perform basic operations on them.
- CO2:** Apply Fourier analysis techniques to continuous-time signals and systems.
- CO3:** Evaluate signal correlation functions and understand signal sampling and reconstruction processes.
- CO4:** Use Laplace transforms for analyzing continuous-time signals and systems.
- CO5:** Analyze discrete-time signals and systems using Z-transforms.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	-	-	-	-	-	-	-	1	3	2	-
CO2	3	3	2	2	-	-	-	-	-	-	-	1	3	3	-
CO3	3	3	3	2	1	-	-	-	-	-	-	1	3	3	-
CO4	3	3	2	3	2	-	-	-	-	-	-	1	3	3	-
CO5	3	3	3	2	2	-	-	-	-	-	-	1	3	3	-

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS:

UNIT-I INTRODUCTION:

Definition of Signals and Systems, Classification of Signals, Classification of Systems, Operations on signals: time-shifting, time-scaling, amplitude-shifting, amplitude-scaling. Problems on classification and characteristics of Signals and Systems, Complex exponential and sinusoidal signals, Singularity functions and related functions: impulse function, step function signum function and ramp function.

UNIT-II FOURIER SERIES AND FOURIER TRANSFORM:

Fourier series representation of continuous time periodic signals, Dirichlet's conditions, Trigonometric Fourier series and Exponential Fourier series, Relation between Trigonometric and Exponential Fourier series, Complex Fourier spectrum. Deriving Fourier transform from Fourier series, Fourier transform of standard signals, properties of Fourier transforms, Fourier transforms involving impulse function and Signum function. Related problems.

UNIT-III CORRELATION & SAMPLING THEOREM:

CORRELATION: Auto-correlation and cross-correlation of functions, properties of correlation function, Energy density spectrum, Parseval's theorem, Power density spectrum, Relation between Convolution and correlation, Detection of periodic signals in the presence of noise by correlation.

SAMPLING THEOREM: Graphical and analytical proof of Band Limited Signals, impulse sampling, Natural and Flat top Sampling, Reconstruction of signal from its samples, Aliasing, Related problems.

UNIT-IV LAPLACE TRANSFORMS:

Introduction, Concept of region of convergence (ROC) for Laplace transforms, constraints on ROC for various classes of signals, Properties of L.T's, Inverse Laplace transform, Relation between L.T's, and F.T. of a signal. Laplace transform of certain signals using waveform synthesis.

UNIT-V Z-TRANSFORMS:

Concept of Z-Transform of a discrete sequence. Region of convergence in Z- Transform, constraints on ROC for various classes of signals, Inverse Z-transform, properties of Z-transforms, Distinction between Laplace, Fourier and Z transforms.

Text Books:

1. Signals, Systems & Communications-B.P.Lathi,BSPublications,2003.
2. Signals and Systems-A.V. Oppenheim, A.S. Willsky and S.H. Nawab,PHI,2ndEdn, 1997.
3. Signals & Systems-SimonHaykinandVanVeen,Wiley,2ndEdition,2007.

Reference Books:

1. Principles of Linear Systems and Signals–BPLathi, Oxford UniversityPress, 2015.
2. Signals and Systems–TK Rawat, Oxford University press, 2011.

B.TECH V SEMESTER

OE-I **L** **T** **P** **C**
 3 **0** **0** **3**

23EC5T11 DIGITAL LOGIC DESIGN
(Open Elective-I)

COURSE OBJECTIVES: The main objectives of this course are given below:

- Learn the fundamentals of number systems, codes, and error-detecting/correcting methods.
- Understand Boolean algebra and its application in digital logic circuit design.
- Design and implement combinational logic circuits.
- Understand the concepts and applications of sequential circuits.
- Familiarize with Programmable Logic Devices (PLDs) and memory components.

Course Outcomes:

At the end of the course student will be able to:

CO1: Perform number system conversions and design digital systems using various codes and error detection/correction techniques.

CO2: Simplify and realize logic functions using Boolean algebra, Karnaugh Maps, and logic gates.

CO3: Design and implement various combinational logic circuits and apply ICs appropriately.

CO4: Develop and analyze sequential circuits using flip-flops, counters, and shift registers.

CO5: Apply PLDs and memory devices for implementation of digital logic circuits.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	-	-	-	-	-	-	-	1	3	2	-
CO2	3	3	3	1	-	-	-	-	-	-	-	1	3	2	-
CO3	3	3	3	2	2	-	-	-	-	-	-	1	3	3	-
CO4	3	3	3	2	2	-	-	-	-	-	-	1	3	3	-
CO5	3	3	3	2	3	-	-	-	-	-	-	1	3	3	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I REVIEW OF NUMBER SYSTEMS & CODES:

Representation of numbers of different radix, conversation from one radix to another radix, r1's compliments and r's compliments of signed members. Graycode, 4 bit codes; BCD, Excess-3, 2421, 84-2-1 code etc. Error detection & correction codes: parity checking, even parity, odd

parity, Hamming code.

UNIT-II BOOLEAN THEOREMS AND LOGIC OPERATIONS:

Boolean theorems, principle of complementation & duality, De-Morgan theorems. Logic operations ; Basic logic operations -NOT, OR, AND, Universal Logic operations, EX-OR, EX-NOR operations. Standard SOP and POS Forms, NAND-NAND and NOR-NOR realizations, Realization of three level logic circuits. Study the pin diagram and obtain truth table for the following relevant ICs 7400,7402,7404,7408,7432,7486.

MINIMIZATION TECHNIQUES: Minimization and realization of switching functions using Boolean theorems, K-Map (up to 6 variables) and tabular method (Quine-mcCluskey method) with only four variables and single function.

UNIT-III COMBINATIONAL LOGIC CIRCUITS DESIGN:

Design of Half adder, full adder, half subtractor, full subtractor, applications of full adders; 4 bit adder-subtractor circuit, BCD adder circuit, Excess 3 adder circuit and carry look-a-head adder circuit, Design code converts using Karnaugh method and draws the complete circuit diagrams.

COMBINATIONAL LOGIC CIRCUITS DESIGN USING MSI &LSI:

Design of encoder ,decoder, multiplexer and de-multiplexers, Implementation of higher order circuits using lower order circuits .Realization of Boolean functions using decoders and multiplexers. Design of Priority encoder, 4-bit digital comparator and seven segment decoder. . Study the relevant ICs pin diagrams and their functions 7442,7447,7485,74154.

UNIT-IV SEQUENTIAL CIRCUITS:

Classification of sequential circuits (synchronous and asynchronous) , operation of NAND & NOR Latches and flip-flops; truth tables and excitation tables of RS flip-flop, JK flip-flop, T flip-flop, D flip-flop with reset and clear terminals. Conversion from one flip-flop to another flip-flop. Design of 5 ripple counters, design of synchronous counters, Johnson counter, ring counter. Design of registers - Buffer register, control buffer register, shift register, bi directional shift register, universal shift, register.

UNIT-V INTRODUCTION OF PLD's:

PLDs: PROM, PAL, PLA -Basics structures, realization of Boolean functions, Programming table. ROM: Internal structure, Static RAM: Internal structure, Dynamic RAM: Internal structure.

Text Books:

1. Switching and finite automata theory Zvi.KOHAVI, Niraj.K.Jha 3rd Edition, Cambridge University Press,2009.



-
2. Digital Design by M.Morris Mano, Michael D Ciletti, 4th edition publication, 2008 PHI.
 3. Switching theory and logic design by Hill and Peterson, Mc-Graw Hill TMH edition, 2012.

Reference Books:

1. Fundamentals of Logic Design by Charles H. Roth Jr, Jaico Publishers, 2006.
2. Digital electronics by R S Sedha.S.Chand & company limited, 2010.
3. Switching Theory and Logic Design by A. Anand Kumar, PHI Learning pvt ltd, 2016.
4. Digital logic applications and design by John M Yarbough, Cengage learning, 2006.

B.TECH V SEMESTER

OE-I **L** **T** **P** **C**
 3 **0** **0** **3**

23CS5T09 OPERATING SYSTEMS
(Open Elective-I)

COURSE OBJECTIVES: The main objectives of introducing this course are:

- Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection.
- Make use of process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.
- Illustrate different conditions for deadlock and their possible solutions.

Course Outcomes: At the end of the course, student will be able to:

- CO1:** Explain the importance, functions and structures of operating systems.
- CO2:** Analyze and compare the performance of various CPU scheduling algorithms.
- CO3:** Apply deadlock handling techniques to avoid deadlocks.
- CO4:** Compare various Memory Management Schemes and analyze various disk Scheduling Algorithms.
- CO5:** Demonstrate the File System concepts and file protection mechanisms.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	2	2
CO2	2	3	1	2	2	-	-	-	-	-	-	1	2	2
CO3	3	3	2	1	2	-	-	-	-	-	-	-	2	2
CO4	2	2	2	-	-	-	-	-	-	-	-	-	2	2
CO5	2	2	-	-	-	-	-	-	-	-	-	1	2	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I

Operating Systems Overview: Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems

System Structures: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure, Building and Booting an Operating System, Operating system debugging.

UNIT-II

Processes: Process Concept, Process scheduling, Operations on processes, Inter-process communication.

Threads and Concurrency: Multithreading models, Thread libraries, threading issues.

CPU Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, multiple processor scheduling.

UNIT-III

Synchronization Tools: The Critical Section Problem, Peterson's Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization.

Deadlocks: system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock.

UNIT-IV

Memory-Management Strategies: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping.

Virtual Memory Management: Introduction, Demand paging, Copy-on-write, Page replacement, Allocation of frames, Thrashing
Storage Management: Overview of Mass Storage Structure, HDD Scheduling.

UNIT-V

File System: File System Interface: File concept, Access methods, Directory Structure; File system Implementation: File-system structure, File-system Operations, Directory implementation, Allocation method, Free space management; File-System Internals: File-System Mounting, Partitions and Mounting, File Sharing.

Secondary Storage Structure: Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, RAID Structure.

Text Books:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson, 2016.

Reference Books:

1. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018.
2. Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013.

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/106/106106144/>
2. <http://peterindia.net/OperatingSystems.html>

B.TECH V SEMESTER**OE-I**
L T P C
3 0 0 3**23CS5T10 COMPUTER ORGANIZATION AND ARCHITECTURE**
(Open Elective-I)**COURSE OBJECTIVES: The main objectives of introducing this course are:**

- To impart basic concepts of computer architecture and organization.
- To explain key skills of constructing cost-effective computer systems.
- To familiarize the basic CPU organization.
- To help students in understanding various memory devices.
- To facilitate students in learning IO communication for deadlock and their possible solutions.

Course Outcomes: At the end of the course, student will be able to:**CO1:** Identify various components of computer and their interconnection.**CO2:** Identify basic components and design of the CPU: the ALU and control unit.**CO3:** Compare and select various Memory devices as per requirement.**CO4:** Compare various types of IO mapping techniques.**CO5:** Illustrate the performance issues of cache memory and virtual memory.**COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):**

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	1	2	-	-	-	-	-	-	-	2	2	2
CO3	2	2	2	1	-	-	-	-	-	-	-	-	-	2
CO4	2	2	2	-	-	-	-	-	-	-	-	-	-	2
CO5	2	2	-	2	-	-	-	-	-	-	-	1	2	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).**SYLLABUS****UNIT-I****STRUCTURE OF COMPUTERS:** Computer types, Functional units, Basic operational concepts, Von- Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point, Error detection and correction codes.**COMPUTER ARITHMETIC:** Addition and Subtraction, Multiplication and Division algorithms, Floating-point Arithmetic Operations, Decimal arithmetic operations.

UNIT-II

BASIC COMPUTER ORGANIZATION AND DESIGN: Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control, Memory-Reference Instructions, Input-Output and interrupt.

Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC.

UNIT-III

REGISTER TRANSFER AND MICRO-OPERATIONS: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit.

MICRO-PROGRAMMED CONTROL: Control Memory, Address Sequencing, Micro-Program example, Design of Control Unit.

UNIT-IV

MEMORY SYSTEM: Memory Hierarchy, Semiconductor Memories, RAM(Random Access Memory), Read Only Memory (ROM), Types of ROM, Cache Memory, Performance considerations, Virtual memory, Paging, Secondary Storage, RAID.

UNIT-V

INPUT OUTPUT: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA. **MULTIPROCESSORS:** Characteristics of multiprocessors, Interconnection structures, Inter Processor Arbitration, Inter processor Communication and Synchronization, Cache Coherence.

Text Books:

1. M. Moris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India.

Reference Books:

1. Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 5th edition, McGraw Hill, New Delhi, India.
2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey.
3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc.
4. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHil

B.TECH V SEMESTER

OE-I

L	T	P	C
3	0	0	3

**23IT5T10 DATABASE MANAGEMENT SYSTEMS
(Open Elective-I)**

COURSE OBJECTIVES:

- Understand the basic database concepts, applications, schema and various models.
- Familiarize with entity relation models for a database and write queries using SQL.
- Emphasize the importance of normalization, transaction management and concurrency control in databases.

Course Outcomes: After learning the course, the student will be able:

CO1: Understand the concept of database, database models and familiarize with Entity Relationship models.

CO2: Demonstrate the use of constraints, relational algebra operations.

CO3: Apply SQL queries to interact with database and understand the basics of NOSQL.

CO4: Apply normalization in database design to eliminate anomalies.

CO5: Understand the basic concepts of transaction processing and concurrency control.

SYLLABUS

UNIT-I Database System Applications:

A Historical Perspective, File Systems versus a DBMS, the Data Model, Levels of Abstraction in DBMS, Data Independence, Structure of a DBMS.

Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes, Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design with the ER Model.

UNIT-II Introduction to the Relational Model:

Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical data base design, introduction to views, destroying/altering tables and views. Relational Algebra, Tuple relational Calculus, Domain relational calculus.

UNIT-III SQL: QUERIES:

CONSTRAINTS, TRIGGERS: form of basic SQL query, UNION, INTERSECT, and EXCEPT, Nested Queries, aggregation operators, NULL values, complex integrity constraints in SQL, triggers and active data bases.

NOSQL: Definition of NOSQL, History of NOSQL and Different NOSQL products Applications, features of NoSQL, Difference between SQL and NoSQL.

UNIT-IV Schema Refinement (Normalization):

Introduction to Schema Refinement, Functional Dependencies Reasoning about FDs, Normal Forms, Properties of decomposition, Normalization, Schema refinement in database design, other kinds of dependencies.

UNIT-V Transaction Management and Concurrency Control:

Properties of transactions, Transactions and Schedules, Concurrent execution of transactions, Lock-based concurrency control, deadlocks, Performance of locking. Concurrency control: 2PL, Serializability, recoverability, Introduction to lock management, dealing with deadlocks.

Text Books:

1. Raghu Rama Krishnan, Johannes Gehrke, “Data base Management Systems”, 3rd Edition, TATA McGraw Hill.
2. "Professional NOSQL" by Shashan k Tiwari, 2-11, WROX Press.

Reference Books:

1. Peter Rob & Carlos Coronel, “Data base Systems design, Implementation, and Management”, 7th Edition, Pearson Education.
2. Silberschatz, Korth, “Data base System Concepts”, 6th Edition, McGraw Hill Elmasri Navathe, “Fundamentals of Database Systems”, 5th Edition, Pearson Education.
3. C.J.Date, “Introduction to Database Systems”, 7th Edition, Pearson Education.

B.TECH V SEMESTER

OE-I	L	T	P	C
	3	0	0	3

**23IT5T11 COMPUTER GRAPHICS
(Open Elective-I)**

Course Objectives:

- To develop, design and implement two- and three-dimensional graphical structures.
- To enable students to acquire knowledge Multimedia compression and animations.
- To learn Creation, Management and Transmission of Multimedia objects.

Course Outcomes: After learning the course, the student will be able:

- CO1: Illustrate the basics of computer graphics, different graphics systems and applications of computer graphics with various algorithms for line, circle and ellipse drawing objects for 2D transformations.
- CO2: Apply projections and visible surface detection techniques for displaying 3D scenes on 2D screens.
- CO3: Illustrate able to create the general software architecture of programs that use 3D object sets with computer graphics.
- CO4: Know and be able to select among models for lighting/shading: Color, ambient light; distant and light with sources; Phong reflection model; and shading (flat, smooth, Gourand, Phong).
- CO5: Know and be able to discuss hardware system architecture for computer graphics. This Includes, but is not limited to graphics pipeline, frame buffers, and graphic accelerators/co-processors.

SYLLABUS

UNIT-I Introduction to Graphics:

Application area of Computer Graphics, overview of graphics systems, video-display devices, graphics monitors and workstations and input devices. 2D Primitives: Output Primitives-Line, Circle and Ellipse drawing algorithms, Attributes of output primitives, Two dimensional Geometric transformations, Two-dimensional viewing Line, Polygon, Curve and Text clipping algorithms.

UNIT-II 3D Concepts:

Parallel and Perspective projections, three-dimensional object representation Polygons, Curved lines, Splines, Quadric Surfaces, Visualization of data sets, 3D transformations, Viewing, Visible surface identification.

UNIT-III Graphics Programming:

Color Models - RGB, YIQ, CMY, HSV, Animations - General Computer Animation, Raster,



Key frame. Graphics programming using OpenGL-Basic graphics primitives, drawing three dimensional objects, Drawing three dimensional scenes.

UNIT-IV Rendering:

Introduction to shading models, Flat and Smooth shading, adding texture to faces, adding shadow of objects, Building a camera in a program, Creating shaded objects.

UNIT-V Overview of Ray Tracing:

Intersecting rays with other primitives, Adding Surface texture, Reflections and Transparency, Boolean operations on Objects.

Text Books:

1. Donald Hearn, Pauline Baker, Computer Graphics– C Version, second edition, Pearson Education, 2--4.

Reference Books:

1. James D. Foley, Andries Van Dam, Steven K. Feiner, John F. Hughes, Computer Graphics- Principles and Practice, Second Edition in C, Pearson Education, 2—7.



		L	T	P	C
B.TECH V SEMESTER	OE-I	3	0	0	3
23MB5T01	ENTREPRENEURSHIP DEVELOPMENT AND VENTURE CREATION (Open Elective-I)				

Course Objectives

By the end of the program, students will be / able to:

1. Inspired; develop entrepreneurial mind-set and attributes; entrepreneurial skill sets for venture creation and intrapreneurial leadership
2. Apply process of problem-opportunity identification and feasibility assessment through developing a macro perspective of the real market, industries, domains and customers while using design thinking principles to refine and pivot their venture idea.
3. Analyse Customer and Market segmentation, estimate Market size, develop and validate Customer Persona.
4. Initiate Solution design, Prototype for Proof of Concept. Understand MVP development and validation techniques to determine Product-Market fit
5. Craft initial Business and Revenue models, financial planning and pricing strategy for profitability and financial feasibility of a venture. Understand relevance and viability of informal and formal funding with respect to different business models.
6. Understand and develop Go-to-Market strategies with a focus on digital marketing channels.
7. Understand and apply story telling skills in presenting a persuasive and defensible Venture Pitch.

Course Outcomes

At the end of the course, students will be able to

1. Develop an entrepreneurial mindset and appreciate the concepts of entrepreneurship, cultivate essential attributes to become an entrepreneur or Intrapreneur and demonstrate skills such as problem solving, team building, creativity and leadership
 2. Comprehend the process of problem-opportunity identification through design thinking, identify market potential and customers while developing a compelling value proposition solution
 3. Analyse and refine business models to ensure sustainability and profitability
 4. Build Prototype for Proof of Concept and validate MVP of their practice venture idea
-

-
5. Create business plan, conduct financial analysis and feasibility analysis to assess the financial viability of a venture
 6. Prepare and deliver an investible pitch deck of their practice venture to attract stakeholders

SYLLABUS

Unit I: Entrepreneurship Fundamentals & Context

Meaning and concept, attributes and mindset of entrepreneurial and intrapreneurial leadership, role models in each and their role in economic development. An understanding of how to build entrepreneurial mindset, skillsets, attributes and networks while on campus.

Core Teaching Tool: Simulation, Game, Industry Case Studies (Personalized for students – 16 industries to choose from), Venture Activity

Unit II: Problem & Customer Identification

Understanding and analysing the macro-Problem and Industry perspective, technological, socio economic and urbanization trends and their implication on new opportunities. Identifying passion, identifying and defining problem using Design thinking principles. Analysing problem and validating with the potential customer. Iterating problem-customer fit. Understanding customer segmentation, creating and validating customer personas. Competition and Industry trends mapping and assessing initial opportunity.

Core Teaching Tool: Several types of activities including Class, game, Gen AI, ‘Get out of the Building’ and Venture Activity.

Unit III: Solution design, Prototyping & Opportunity Assessment and Sizing

Understanding Customer Jobs-to-be-done and crafting innovative solution design to map to customer’s needs and create a strong value proposition. Developing Problem-solution fit in an iterative manner. Understanding prototyping and MVP. Developing a feasibility prototype with differentiating value, features and benefits. Initial testing for proof-of-concept and iterate on the prototype. Assess relative market position via competition analysis, sizing the market and assess scope and potential scale of the opportunity.

Core Teaching Tool: Venture Activity, no-code Innovation tools, Class activity

Unit IV: Business & Financial Model, Go-to-Market Plan

Introduction to Business model and types, Lean approach, 9 block lean canvas model, riskiest assumptions to Business models. Importance of Build - Measure – Lean approach. Business planning: components of Business plan- Sales plan, People plan and financial plan.

Financial Planning: Types of costs, preparing a financial plan for profitability using financial template, understanding basics of Unit economics and analysing financial performance. Introduction to Marketing and Sales, Selecting the Right Channel, creating digital presence, building customer acquisition strategy. Choosing a form of business organization specific to your venture, identifying sources of funds: Debt & Equity, Map the Start-up Lifecycle to Funding Options.

Core Teaching Tool: Founder Case Studies – Sama and Securely Share; Class activity and discussions; Venture Activities.

Unit V: Scale Outlook and Venture Pitch readiness

Understand and identify potential and aspiration for scale vis a vis your venture idea. Persuasive Storytelling and its key components. Build an Investor ready pitch deck.

Core Teaching Tool: Expert talks; Cases; Class activity and discussions; Venture Activities.

Suggested Reading:

1. Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd, Sabyasachi Sinha (2020). Entrepreneurship, McGrawHill, 11th Edition.
2. Ries, E. (2011). The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses. Crown Business
3. Osterwalder, A., & Pigneur, Y. (2010). Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers. John Wiley & Sons.
4. Simon Sinek (2011) Start with Why, Penguin Books limited
5. Brown Tim (2019) Change by Design Revised & Updated: How Design Thinking Transforms Organizations and Inspires Innovation, Harper Business
6. Namita Thapar (2022) The Dolphin and the Shark: Stories on Entrepreneurship, Penguin Books Limited
7. Saras D. Sarasvathy, (2008) Effectuation: Elements of Entrepreneurial Expertise, Elgar Publishing Ltd



Web Resources

1. Learning resource- Ignite 5.0 Course Wadhwani platform (Includes 200+ components of custom created modular content + 500+ components of the most relevant curated content)

B.TECH V SEMESTER

OE-I **L** **T** **P** **C**
 3 0 0 3

23MB5T03 INNOVATION AND TECHNOLOGY MANAGEMENT
(Open Elective-I)

Course Objectives:

- To introduce the fundamental concepts of innovation and its strategic importance.
- To understand innovation processes, models, and leadership involvement.
- To apply tools and methods for managing technology and R&D.
- To explore practical frameworks such as design thinking for implementing innovation.
- To familiarize students with emerging trends, sustainability, and ethical aspects in innovation.

Course Outcomes (Cos): Upon successful completion of the course, the student will be able to:

CO1: Describe the types of innovation, their significance, and distinguish innovation from invention.

CO2: Classify and explain innovation models and strategies, and their relevance in business.

CO3: Apply concepts of technology forecasting, R&D, and IPR management to organizational contexts.

CO4: Use design thinking tools and innovation project techniques in practical settings.

CO5: Identify and explain emerging innovation trends and sustainability practices in the tech industry.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	-	-	2	2	-	2	-	3	1	1
CO2	2	3	2	3	-	-	-	-	2	3	2	3	2	2
CO3	3	3	3	2	3	-	-	-	-	2	3	3	2	3
CO4	-	2	2	3	3	2	2	3	3	3	2	3	3	2
CO5	2	2	2	3	-	3	3	3	2	3	3	3	2	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I Introduction to Innovation and Technology Management:

Definition and importance of innovation and technology management, Types of innovation: product, process, service, and business model innovations, Historical evolution and the role of innovation in economic development. Innovation vs. invention: key differences and interrelations, Barriers to innovation and strategies to overcome them.

UNIT-II Innovation Process and Strategies:

Stages of the innovation process: idea generation, selection, development, and commercialization. Innovation models: linear, interactive, and open innovation. Strategic approaches to innovation: incremental vs. radical innovation. Role of leadership and organizational culture in fostering innovation. Case studies on successful innovation strategies in various industries.

UNIT-III Technology Management:

Understanding technology life cycles and the diffusion of innovations. Technology forecasting and road mapping, Assessment and acquisition of new technologies. Managing research and development (R&D) activities. Intellectual property rights and their role in technology management.

UNIT-IV Innovation in Practice:

Design thinking and its application in innovation. Tools and techniques for idea generation and problem-solving (e.g., brainstorming, TRIZ). Managing innovation projects: planning, execution, and control. Collaborative innovation: partnerships, alliances, and networks, Measuring innovation performance and outcomes.

UNIT-V Emerging Trends and Challenges:

Digital transformation and its impact on innovation. Sustainable innovation and the triple bottom line approach. Innovation in the context of globalisation and competitive markets. Innovation in the context of globalisation and competitive markets. Ethical considerations in innovation and technology management. Future directions: artificial intelligence, IoT, and other emerging technologies.

Text Books:

1. Technology and Innovation Management; Author: Sanjiva Shankar Dubey, Publisher: PHI Learning.
2. Management of Technology and Innovation: Competing Through Technological Excellence; Author: P.N. Rastogi; Publisher: SAGE Publications India

B.TECH V SEMESTER

OE-I

L	T	P	C
3	0	0	3

23AD5T09 INTRODUCTION TO DATA SCIENCE
(Open Elective-I)

Prerequisites:

Course Objectives: The main objectives of introducing this course are:

- Knowledge and expertise to become a data scientist.
- Essential concepts of statistics and machine learning that are vital for data science.
- Significance of exploratory data analysis (EDA) in data science.
- Critically evaluate data visualizations presented on the dashboards.
- Suitability and limitations of tools and techniques related to data science process.

Course Outcomes: At the end of the course, student will be able to:

CO1: Apply principles of NumPy and Pandas to the analysis of data.

CO2: Make use of various file formats in loading and storage of data.

CO3: Identify and apply the need and importance of pre-processing techniques.

CO4: Show the results and present them in a pictorial format.

CO5: Facilitate new solutions for visualization of datasets by using different plotting techniques.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	-	-	-	-	1	-	2	3	2
CO2	3	2	2	2	3	-	-	-	-	-	-	2	2	3
CO3	3	3	2	2	2	-	-	-	-	-	-	2	3	2
CO4	2	2	2	2	3	-	-	-	-	2	-	2	1	1
CO5	2	2	3	2	3	-	-	-	1	2	1	1	1	1

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I Introduction to Data science & Data Science process:

Introduction to Data science, benefits and uses, facets of data, data science process in brief, big data ecosystem and data science.

Data Science process: Overview, defining goals and creating project charter, retrieving data, cleansing, integrating and transforming data, exploratory analysis, model building, presenting findings and building applications on top of them.

UNIT-II Applications & Handling large data:

Applications of machine learning in Data science, role of ML in DS, Python tools like sklearn, modelling process for feature engineering, model selection, validation and prediction, types of ML, semi-supervised learning.

Handling large data: problems and general techniques for handling large data, programming tips for dealing large data, case studies on DS projects for predicting malicious URLs, for building recommender systems.

UNIT-III NoSQL movement for handling Big data:

Distributing data storage and processing with Hadoop framework, case study on risk assessment for loan sanctioning, ACID principle of relational databases, CAP theorem, base principle of NoSQL databases, types of NoSQL databases, case study on disease diagnosis and profiling.

UNIT-IV Tools and Applications of Data Science:

Introducing Neo4j for dealing with graph databases, graph query language Cypher, Applications graph databases, Python libraries like nltk and SQLite for handling Text mining and analytics, case study on classifying Reddit posts.

UNIT-V Data Visualization:

Data Visualization options, Crossfilter, the JavaScript MapReduce library, Creating an interactive dashboard with dc.js, Dashboard development tools.

Text Books:

1. Davy Cielen, Arno D.B.Meysman, and Mohamed Ali, “Introducing to Data Science using Python tools”, Manning Publications Co, Dreamtech press, 2016.
2. Prateek Gupta, “Data Science with Jupyter” BPB publishers, 2019 for basics

Reference Books:

1. Joel Grus, “Data Science from Scratch”, OReilly, 2019.
2. Doing Data Science: Straight Talk from the Frontline, 1 st Edition, Cathy O’Neil and Rachel Schutt, O’Reilly, 2013.

B.TECH V SEMESTER

OE-I	L	T	P	C
	3	0	0	3

**23AM5T09 MACHINE LEARNING
(Open Elective-I)**

Prerequisites:

COURSE OBJECTIVES: The main objectives of introducing this course are:

- To understand the need for machine learning for various problem solving.
- To study the various supervised, semi-supervised and unsupervised learning algorithms in machine learning.
- To understand the latest trends in machine learning.
- To design appropriate machine learning algorithms for problem solving.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Basic understanding of Machine Learning algorithms.

CO2: Apply various ANN algorithms with machine learning techniques.

CO3: Apply the Bayesian concepts to machine learning.

CO4: Implementation of SVM.

CO5: Exposure to types of reinforced learning.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	1	-	2	2	2
CO2	3	3	3	2	3	-	-	-	-	1	-	2	2	2
CO3	3	2	2	3	2	-	-	-	-	1	-	2	2	2
CO4	3	3	3	3	3	-	-	-	-	1	-	2	2	1
CO5	2	2	2	3	2	-	-	-	-	1	-	3	1	2

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I Introduction to Machine Learning:

Evolution of Machine Learning, Paradigms for ML, Well posed Learning Problems, Perspectives and Issues, Designing a Learning system, Concept Learning, FIND-S Algorithm, Version Spaces and Candidate Eliminations, Inductive bias, Decision Tree learning, Decision Tree Representation – ID3 Algorithm, Heuristic Space search.

UNIT-II Neural Networks and Genetic Algorithms:

Neural Network Representation, Appropriate Problems for ANN, Perceptron Learning, Perceptron Models Types, Multi-layered perceptron model, Multilayer Networks and Back Propagation Algorithm, Advanced Topics in ANN, Genetic Algorithm in machine Learning,

Genetic Programming, Models of Evaluation and Learning

UNIT-III Bayesian and Computational Learning:

Bayes Theorem Representation, Brute force Concept Learning , Maximum Likelihood-Least Squared error Hypothesis, Maximum Likelihood hypothesis-Predicting Probabilities, Minimum Description Length Principle, Bayes Optimal Classifier, Naive Bayes Classifier (NBC), Bayesian Belief Network, EM Algorithm.

UNIT-IV Instant Based Learning:

Introduction of Instant based learning, K- Nearest Neighbor Learning-Algorithm , Regression- Linear Regression, Locally weighted regression, Logistic Regression, Random Forests for Classification and Regression, Radial Basis Functions, Case Based Learning-CADET System, Support Vector Machines (SVM).

UNIT-V Reinforcement Learning:

Introduction – Terms, Key features, Approaches, Elements, The Bellman Equation, Types of Reinforcement Learning, Reinforcement Learning Algorithms, The Learning Task, Q – Learning, non-deterministic rewards and actions, temporal difference learning, Algorithms in temporal difference learning, generalizing from examples, relationship to dynamic programming.

Text Books:

- Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.

Reference Books:

- EthemAlpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
- Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009.

B.TECH VI SEMESTER

OE-II **L** **T** **P** **C**
 3 **0** **0** **3**

23CE6T12 DISASTER MANAGEMENT
(Open Elective-II)

Course Learning Objectives: The objective of this course is:

- Develop an understanding of why and how the modern disaster manager is involved with pre- disaster and post-disaster activities.
- Develop an awareness of the chronological phases of natural disaster response and refugee relief operations. Understand how the phases of each are parallel and how they differ.
- Understand the ‘relief system’ and the ‘disaster victim’.
- Describe the three planning strategies useful in mitigation.
- Identify the regulatory controls used in hazard management.
- Describe public awareness and economic incentive possibilities.
- Understand the tools of post-disaster management.

COURSE OUTCOMES:

Upon the successful completion of this course, the students will be able to:

CO1: Affirm the usefulness of integrating management principles in disaster mitigation work.

CO2: Distinguish between the different approaches needed to manage pre- during and post-disaster periods.

CO3: Explain the process of risk management.

CO4: Relate to risk transfer.

CO5: Explain the process of multi-sectional issues, education and community preparedness.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	2	1	3	3	2	2	2	2	2	2	2
CO2	2	3	2	2	2	3	3	2	2	2	2	2	2	2
CO3	2	3	2	3	2	3	3	2	2	2	2	2	2	2
CO4	2	2	2	2	2	3	3	2	2	2	2	2	2	2
CO5	2	2	2	2	1	3	3	3	3	3	2	3	2	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I Natural Hazards and Disaster Management:

Introduction of DM – Inter disciplinary nature of the subject– Disaster Management cycle – Five priorities for action. Case study methods of the following: Vegetal Cover floods, droughts – Earthquakes – landslides – global warming, cyclones & Tsunamis – Post Tsunami hazards along the Indian coast.

UNIT-II Man Made Disaster and Their Management Along With Case Study Methods Of The Following:

Fire hazards – transport hazard dynamics – solid waste management – post disaster – bio terrorism -threat in mega cities, rail and aircraft accidents, ground water, industries - Emerging infectious diseases and Aids and their management

UNIT-III Risk and Vulnerability:

Building codes and land use planning – Social Vulnerability – Environmental vulnerability – Macro-economic management and sustainable development, Climate change risk rendition – Financial management of disaster – related losses.

UNIT-IV Role of Technology in Disaster Managements:

Disaster management for infra structures, taxonomy of infra structure – treatment plants and process facilities-electrical substations- roads and bridges- mitigation programme for earth quakes – flowchart, geospatial information in agriculture drought assessment - Multimedia Technology in disaster risk management and training - Transformable Indigenous Knowledge in disaster reduction – Role of RS & GIS.

UNIT-V Multi-sectional Issues, Education and Community Preparedness:

Impact of disaster on poverty and deprivation - Climate change adaptation and human health - Exposure, health hazards and environmental risk-Forest management and disaster risk reduction -The Red cross and red crescent movement - Corporate sector and disaster risk reduction- Education in disaster risk reduction- Essentials of school disaster education - Community capacity and disaster resilience-Community based disaster recovery - Community based disaster management and social capital-Designing resilience- building community capacity for action.

TEXT BOOKS:

1. An Introduction of Disaster Management- Natural Disasters & Vulnerable Hazards– S. Vaidyanathan: CBS Publishers & Distributors Pvt.Ltd.
2. Natural Hazards & Disaster Management, Vulnerability and Mitigation by RB Singh- Rawat Publications



-
3. 'Disaster Science & Management' by Tushar Bhattacharya, Tata McGraw Hill Education Pvt. Ltd., New Delhi.
 4. 'Disaster Management – Future Challenges and Opportunities' by Jagbir Singh (2007), I K International Publishing House Pvt. Ltd.

REFERENCE BOOKS:

1. 'Disaster Management' edited by H K Gupta (2003), Universities press.
2. 'Disaster Management – Global Challenges and Local Solutions' by Rajib shah & R R Krishnamurthy (2009), Universities press. R. Nishith, Singh AK,
3. "Disaster Management in India: Perspectives, Issues and strategies" New Royal Book Company."

B.TECH VI SEMESTER

	L	T	P	C
OE-II	3	0	0	3

23CE6T13 SUSTAINABILITY IN ENGINEERING PRACTICES
(Open Elective-II)

Course Learning Objectives: The objective of this course is:

- Understand the principles of sustainable development.
- Gain knowledge of environmental ethics, laws, and legislation.
- Identify and analyze local and global environmental issues.
- Learn sustainability assessment tools.
- Explore sustainable practices.
- Recognize the importance of renewable energy sources.

COURSE OUTCOMES: At the end of the course, the student will be able to

- CO1:** Explain sustainable development and different environmental agreements and protocols.
- CO2:** Discuss real time activities causing environmental issues and different methods to use renewable energy resources.
- CO3:** Explain local and global environmental issues.
- CO4:** Differentiate between carbon emissions for regular and sustainable cities and explain different practices to move industries towards sustainability.
- CO5:** Discuss different renewable energy resources and explain methods to implement green technology.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1	1	2	3	3	2	1	2	1	2	2	2
CO2	2	2	2	2	2	3	3	2	2	2	1	2	2	2
CO3	2	2	2	1	1	3	3	2	1	2	1	2	2	2
CO4	2	2	2	2	2	3	3	2	2	2	2	2	2	2
CO5	2	2	3	2	3	3	3	2	2	2	2	2	2	2

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **0**. **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I

Introduction to Sustainable Engineering- Sustainable development, concepts of sustainable development: three pillar model, egg of sustainability model, Atkisson's pyramid model, prism model, principles of sustainable development, sustainable engineering, threats for sustainability.

Environmental Ethics and Legislations – Environmental ethics and education, multilateral environmental agreements and protocols, enforcement of environmental laws in India – The Water Act, The Air Act, The Environment Act..

UNIT-II

Local Environmental Issues- Solid waste, impact of solid waste on natural resources, zero waste concept and three R concept, waste to energy technology: thermo-chemical conversion, biochemical conversion.

Global Environmental Issues- Resource degradation: deterioration of water resources, land degradation, air pollution, climate change and global warming, ozone layer depletion, carbon footprint, carbon trading

UNIT-III

Tools for Sustainability - Environmental management System (EMS), concept of ISO14000, life cycle assessment (LCA): basic components, advantages, disadvantages, case study. Environmental impact assessment (EIA), environmental auditing, bio mimicking, case studies.

UNIT-IV

Sustainable Habitat - Concept of green building, green building materials, green building certification and rating: green rating for integrated habitat assessment (GRIHA), leadership in energy and environmental design (LEED) rating, energy efficient buildings, sustainable cities, sustainable transport, sustainable pavements, case studies in sustainability engineering: Green building, sustainable city, sustainable transport system.

Sustainable Industrialization and Urbanization – Sustainable urbanization, industrialization, material selection, pollution prevention, industrial ecology, industrial symbiosis, poverty reduction.

UNIT-V

Renewable energy resources- Conventional and non- conventional forms of energy, solar energy, fuel cells, wind energy, small hydro plants, biogas systems, biofuels, energy from ocean, geothermal energy, conservation of energy.

Green technology and Green Business: Sustainable business, green technology, green energy, green construction, green transportation, green chemistry, green computing.

TEXT BOOKS:

1. R.L. Rag and Lekshmi Dinach and ran Remesh. Introduction to Sustainable Engineering. 2nd Edition, PHI Learning Pvt. Ltd., 2016.

REFERENCE BOOKS:

1. 'D.T. Allen and D.R. Shonnard. Sustainability Engineering: Concepts, Design and Case Studies, 1st Edition, Prentice Hall, 2011.
2. 'A.S. Bradley, A.O.Adebayo, P.Maria. Engineering applications in sustainable design and development, 1st Edition, Cengage learning, 2016.

B.TECH VI SEMESTER

OE-II **L** **T** **P** **C**
 3 **0** **0** **3**

23EE6T10 FUNDAMENTALS OF ELECTRIC VEHICLES
(Open Elective-II)

Pre-requisite: Basic knowledge in Physics, Chemistry and Basics of Electrical and Electronics.

Course Objectives:

- To familiarize the students with the need and advantages of electric and hybrid electric vehicles.
- To understand various power converters used in electric vehicles.
- To be familiar all the different types of motors suitable for electric vehicles.
- To know various architecture of hybrid electric vehicles.
- To have knowledge on latest developments in batteries and other storage systems.

COURSE OUTCOMES: After the completion of the course the student should be able to:

CO1: Illustrate the use and advantages of different types of electric vehicles.

CO2: Use suitable power converters for EV application.

CO3: Select suitable electric motor for EV power train.

CO4: Design HEV configuration for a specific application.

CO5: Analyse various storage systems and battery management system for EVs.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	1	-	3	-	-	-	3	3	3	1
CO2	3	2	2	2	2	-	3	-	-	-	2	3	3	1
CO3	3	2	3	2	2	-	3	-	-	-	2	2	3	1
CO4	3	2	3	2	2	-	3	-	-	-	2	2	3	1
CO5	3	3	2	2	2	-	3	-	-	-	-	2	3	1

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I Introduction:

Fundamentals of vehicles – Vehicle model – Calculation road load and tractive force – Components of conventional vehicles – Drawbacks of conventional vehicles – Need for electric vehicles– Advantages and applications of Electric Vehicles – History of Electric Vehicles – EV Market in India and outside India –Types of Electric Vehicles.

UNIT-II Components of Electric Vehicles:

Main components of Electric Vehicles – Electric Traction Motor and Controller – Power Converters – Rectifiers used in EVs – Bidirectional DC–DC Converters – Voltage Source Inverters – PWM inverters used in EVs- Electronic control units(ECU)for EV.

UNIT-III Motors for Electric Vehicles:

Characteristics of traction drive – requirements of electric machines for EVs – Comparison of Different motors for Electric and Hybrid Vehicles – Induction Motors – Synchronous Motors – Permanent Magnetic Synchronous Motors –Synchronous reluctance motors- Brushless DC Motors – Switched Reluctance Motors (Construction details and working only).

UNIT-IV Hybrid Electric Vehicles:

Evolution of Hybrid Electric Vehicles – Advantages and Applications of Hybrid Electric Vehicles – Architecture of HEVs – Series and Parallel HEVs – Complex HEVs – Range extended HEVs – Examples – Merits and Demerits.

UNIT-V Energy Sources for Electric Vehicles:

Batteries– Types of Batteries – Lithium-ion – Nickel-metal hydride – Lead-acid – Comparison of Batteries – Battery Charging – Fast Charging –smart charging-Battery Management System – Ultra capacitors – Flywheels – Compressed air energy storage (CAES)– Fuel Cell – it's working-Flow batteries (Basic outline).

TEXT BOOKS:

1. Iqbal Hussein - Electric and Hybrid Vehicles: Design Fundamentals - CRC Press- 2021.
2. Tom Denton, Hayley Pells - Electric and hybrid vehicles, Third Edition, 2024

REFERENCE BOOKS:

1. Kumar - L. Ashok - and S. Albert Alexander. Power Converters for Electric Vehicles.
CRC Press - 2020.
2. Chau - Kwok Tong. Electric vehicle machines and drives: design - analysis and application. John Wiley & Sons – 2015.
3. Berg - Helena. Batteries for electric vehicles: materials and electrochemistry. Cambridge university press – 2015

Online Learning Resources:

1. MOOC at <https://www.edx.org/learn/electric-cars>
2. <https://archive.nptel.ac.in/courses/108/106/108106170>

B.TECH VI SEMESTER

OE-II **L** **T** **P** **C**
 3 **0** **0** **3**

23EE6T11 ELECTRICAL WIRING ESTIMATION & COSTING
(Open Elective-II)

Pre-requisite: Electrical Circuits, Basics of Power Systems and Electrical Machines.

Course Objectives:

- Introduce the electrical symbols and simple electrical circuits.
- Able to learn the design of electrical installations.
- Able to learn the design of electrical installation for different types of buildings and small industries.
- Learn the basic components of electrical substations.
- Familiarize with the motor control circuits.

COURSE OUTCOMES: After the completion of the course the student should be able to:

CO1: Demonstrate the various electrical apparatus and their interconnections.

CO2: Examine various components of electrical installations.

CO3: Estimate the cost for installation of wiring for different types of building and small industries.

CO4: Illustrate the components of electrical substations.

CO5: Design suitable control circuit for starting of three phase induction motor and synchronous motor.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	1	-	2	-	-	-	1	2	3	1
CO2	3	2	2	1	1	-	1	-	-	-	-	2	3	2
CO3	3	2	3	2	2	-	1	-	-	-	1	2	3	2
CO4	3	2	3	1	2	-	-	-	-	-	1	2	3	2
CO5	3	3	2	3	2	-	-	-	-	-	-	1	3	2

Note: Enter Correlation Levels **1**or**2**or**3**. If there is no correlation, put **-**. **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I Electrical Symbols and Simple Electrical Circuits:

Identification of electrical symbols - Electrical wiring Diagrams - Methods of representation of wiring diagrams - introduction to simple light and fan circuits - system of connection of appliances and accessories.

UNIT-II Design Considerations of Electrical Installations:

Electric supply system - Three-phase four wire distribution system - protection of electric installation against overload - short circuit and earth fault - earthing - neutral and earth wire - types of loads - systems of wiring - permissible of voltage drops and sizes of wires - estimating and costing of electrical installations.

UNIT-III Electrical Installation for Different Types of Buildings and Small Industries:

Electrical installations for electrical buildings - estimating and costing of material - simple examples on electrical installation for residential buildings - electrical installations for commercial buildings - electrical installation for small industries-case study.

UNIT-IV Substations:

Introduction - types of substations - outdoor substations-pole mounted type - indoor substations-floor mounted type - simple examples on quantity estimation-case study.

UNIT-V Motor control circuits:

Introduction to AC motors - starting of three phase squirrel cage induction motors - starting of wound rotor motors - starting of synchronous motors - contactor control circuit components - basic control circuits - motor protection – Schematic and wiring diagrams for motor control circuits.

TEXT BOOKS:

1. Electrical Design and Estimation Costing - K. B. Raina and S.K.Bhattacharya – New Age International Publishers - 2007.

REFERENCE BOOKS:

1. Electrical wiring estimating and costing – S.L.Uppal and G.C.Garg – Khanna publishers - 6th edition - 1987.
2. A course in electrical installation estimating and costing – J.B.Gupta –Kataria SK & Sons - 2013

Online Learning Resources:

1. https://onlinecourses.swayam2.ac.in/nou25_ec07/preview

B.TECH VI SEMESTER

OE-II

L	T	P	C
3	0	0	3

23EE6T12 INTRODUCTION TO INDUSTRIAL ROBOTICS
(Open Elective-II)

Course Objectives:

- Define robot, work volume and anatomy and identify the components of a robot.
- To derive the rotation matrix thereby form the transformation of a manipulator.
- To derive differential relationship thereby calculate the Jacobian.
- Identify various actuators and sensors along with their working principles.
- To define the terminology and to know techniques of trajectory.

COURSE OUTCOMES: At the end of the course, student will be able to:

- CO1:** Classify the robots, identify configurations of industrial robots and select the appropriate end effectors.
- CO2:** Apply spatial transformation to obtain forward and inverse kinematics equations for manipulators.
- CO3:** Calculate the Jacobian for 2DOF and 3DOF manipulator.
- CO4:** Classify the actuators and sensors also select them based on the application.
- CO5:** Generate the trajectory of robots for a specific task.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	3	-	-	-	-	-	-	2	3	3
CO2	3	3	3	2	2	-	-	-	-	-	-	2	3	3
CO3	3	3	3	3	3	-	-	-	-	-	-	2	3	3
CO4	3	2	2	2	3	-	-	-	-	-	-	2	3	3
CO5	3	3	3	2	3	-	-	-	-	-	2	2	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I

Introduction: Automation and Robotics, History of Robotics, Definition of a Robot, Three law of robotics, Classification of Robots: By JIRA, By RIA, By AFR, By coordinate system and control systems; Components of the Industrial Robotics System, Robot Anatomy, Work Volume, recision of Movement.

End effectors: Grippers-types, operation, mechanism, force analysis, tools as end effectors,

consideration in gripper selection and design.

UNIT-II

Motion Analysis: Degrees of Freedom, Types of joints, Position and orientation representation, frames, Basic rotation matrix, Composite rotation matrix, Homogenous transformation matrix, rotation matrix with Euler angle presentation, rotation matrix about an arbitrary axis,

Forward and Inverse Kinematic Equations- Position and orientation. Denavit-Hartenberg Representation of Forward Kinematic Equations of Robots and The Inverse Kinematic Solution of Robots for 2DOF and 3DOF robot manipulator.

UNIT-III

Differential Motions and Velocities: Differential Relationships, Jacobian, Differential Motions of a Frame, Interpretation of the Differential Change, Calculation of the Jacobian, Inverse Jacobian.

UNIT-IV

Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors, comparison of Actuators.

Feedback Components: position sensors – potentiometers, resolvers, encoders – Velocity sensors, Tactile and Range sensors, Force and Torque sensors. Machine vision- Introduction, applications of machine vision.

UNIT-V

Trajectory planning: Path Vs Trajectory, Joint Space Trajectory Planning, Third order Polynomial Trajectory planning, fifth order Polynomial Trajectory planning.

TEXT BOOKS:

1. Industrial Robotics/Groover MP/Pearson Edu.
2. Robotics and Control /Mittal R K &Nagrathi J /TMH.

REFERENCE BOOKS:

1. Robotics/Fu KS/ McGraw Hill.
2. Robotic Engineering /Richard D. Klafter, Prentice Hall.
3. Robot Analysis and Control/ H. Asada and J.J.E. Slotine/BSP Books Pvt.Ltd.
4. Introduction to Robotics/John J Craig/PearsonEdu.

B.TECH VI SEMESTER

OE-I	L	T	P	C
	3	0	0	3

23EE6T13 INTRODUCTION TO ADDITIVE MANUFACTURING
(Open Elective-II)

Course Objectives:

- To know the overall history and classification of additive manufacturing technology.
- To follow the concepts of liquid based additive manufacturing system.
- To differentiate the working of solid based additive manufacturing system and to understand the principle of powder based additive manufacturing system.
- To visualize the concepts of CAD and Reverse Engineering.
- To relate the concepts of additive manufacturing to medical applications.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Describe the differences and the application of range of additive manufacturing Processes.

CO2: Illustrate the Liquid based Additive manufacturing system along with its applications.

CO3: Relate the process of Solid based Additive Manufacturing system along with its applications.

CO4: Appreciate the use of reverse engineering with concepts of CAD modelling.

CO5: Infer various medical and bio-additive manufacturing examples with case studies.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	-	-	-	-	-	-	2	3	3
CO2	3	2	2	2	3	-	-	-	-	-	-	2	3	3
CO3	3	2	2	2	3	-	-	-	-	-	-	2	3	3
CO4	3	2	3	2	3	-	-	-	-	-	1	2	3	3
CO5	3	2	2	2	2	2	2	-	-	-	-	2	3	2

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I Introduction:

Prototyping fundamentals, Historical development, Advantages of AMT, commonly used terms, process chain, 3D modelling, Data Conversion, and transmission, Checking and preparing, Building, Post processing, RP data formats, Classification of AMT process, Applications to various fields.

UNIT-II Liquid Based Additive Manufacturing Systems:

Classification – Liquid based system – Stereolithography Apparatus (SLA) – Principle, process, advantages and applications – solid based system.

UNIT-III Solid Based Additive Manufacturing Systems:

Fused Deposition Modelling Principle, process, advantages and applications – Laminated Object Manufacturing

Powder Based Additive Manufacturing Systems Selective Laser Sintering – Principles of SLS process – process, advantages and applications, three-dimensional printing principle, process, advantages and applications, Laser Engineered Net Shaping (LENS), Electron Beam Melting.

UNIT-IV Pre-Processing in Additive Manufacturing:

Preparation of 3D-CAD model, Reverse engineering, Reconstruction of 3D-CAD model using reverse engineering, Part orientation and support generation, STL Conversion, STL error diagnostics, Slicing and Generation of codes for tool path, Surface preparation of materials.

Post-Processing in Additive Manufacturing Support material removal, surface texture improvement, accuracy improvement, aesthetic improvement, preparation for use as a pattern, property enhancements using non-thermal and thermal techniques, Brief information on characterization techniques used in additive manufacturing.

UNIT-V Medical and Bio-Additive Manufacturing:

Customized implants and prosthesis: Design and production, Bio- Additive Manufacturing. Computer Aided Tissue Engineering (CATE) Case Studies.

TEXT BOOKS:

1. Rapid prototyping: Principles and Applications /Chua C.K., Leong K.F. and LIM C.S/World Scientific publication.

REFERENCE BOOKS:

1. Rapid Manufacturing / D.T. Pham and S.S. Dimov/Springer.
2. Wohlers Report 2000 /Terry T Wohlers/Wohlers Associates
3. Rapid Prototyping & Manufacturing / Paul F.Jacobs/ASME Press
4. Rapid Prototyping / Chua and Liou

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

23EE6T14 VEHICLE TECHNOLOGY
(Open Elective-II)

Course Objectives:

- To study the advanced engine technologies.
- To learn various advanced combustion technologies and its benefits.
- To learn the methods of using low carbon fuels and its significance.
- To learn and understand the hybrid and electric vehicle configurations.
- To study the application of fuel cell technology in auto motives.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Discuss the latest trends in engine technology.

CO2: Discuss the need of advanced combustion technologies and its impact on reducing Carbon foot-print on the environment.

CO3: Analyzing the basic characteristics of low carbon fuels, its impact over conventional fuels and in achieving sustainable development goals.

CO4: Discuss the working and energy flow in various hybrid and electric configurations.

CO5: Analyzing the need for fuel cell technology in automotive applications.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	1	-	-	-	-	-	-	1	2	2
CO2	3	-	-	-	-	3	3	-	-	-	-	2	3	3
CO3	3	-	-	-	1	3	3	-	-	-	-	2	3	3
CO4	3	-	-	-	1	3	3	-	-	-	-	2	3	3
CO5	3	-	-	-	1	3	3	-	-	-	2	2	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **0**. **1** Slight (Low), **2**- Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I

Gasoline Direct Injection, Common Rail Direct Injection, Variable Compression Ratio Turbocharged Engines, Electric Turbochargers, VVT, Intelligent Cylinder De-activation, After Treatment Technologies, Electric EGR, Current EMS architecture.

UNIT-II

Spark Ignition combustion, Compression Ignition Combustion, Conventional Dual Fuel Combustion, Low Temperature Combustion Concepts– Controlled Auto Ignition,

Homogeneous Charge Compression Ignition, Premixed Charge Compression Ignition, Partially Premixed Compression Ignition, Reactivity Controlled Compression Ignition, Gasoline Direct Injection Compression Ignition.

UNIT-III

Alcohol Fuels, Ammonia Fuel and Combustion, Methane Technology, Dimethyl Ether, Hydrogen Fuel Technology, Challenges, and way forward.

UNIT-IV

Conventional Hybrids (Conventional ICE + Battery), Modern Hybrids (RCCI/GDCI Engine + Battery), Pure Electric Vehicle Technology – Challenges and Way forward.

UNIT-V

Fuel cells for automotive applications - Technology advances in fuel cell vehicle systems - Onboard hydrogen storage - Liquid hydrogen and compressed hydrogen - Metal hydrides, Fuel cell control system - Alkaline fuel cell - Road map to market.

TEXT BOOKS:

1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004.
2. Rakesh Kumar Maurya, Characteristics and Control of Low Temperature Combustion Engines. ISBN 978-3-319-68507-6, SPRINGER.

REFERENCE BOOKS:

1. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
2. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.
3. Rand D.A.J, Woods, R & Dell RM Batteries for Electric vehicles, John Wiley & Sons, 1998.

B.TECH VI SEMESTER	OE-II	L	T	P	C
		3	0	0	3

23EC6T14 LINEAR & DIGITAL IC APPLICATIONS
(Open Elective-II)

Prerequisites: Electronic Devices & circuits.

Course Objectives: The main objectives of introducing this course are

- Understanding the basic building blocks, characteristics (ideal and practical), and various Configurations of Op-Amps.
- Understanding different types of active filters and their design.
- Understanding the principles and design of D/A converters and A/D converters.
- To understand and implement the working of basic digital circuits & memories.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Analyze and design various configurations of operational amplifiers, and applications such as instrumentation amplifiers, voltage regulators and comparators.

CO2: Design and implement active filters and waveform generators using op-amps, IC-555, and IC-565, and evaluate their performance for signal processing applications.

CO3: Compare different data conversion techniques (DAC and ADC) and implement digital-to-analog and analog-to-digital conversion circuits in real-time applications.

CO4: Apply combinational logic ICs such as multiplexers, de-multiplexers, encoders, decoders, and arithmetic circuits to solve complex digital design problems.

CO5: Develop sequential circuits using flip-flops, counters, and shift registers, and analyze their use in digital memory systems, including ROM, RAM, and their variants.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	-	-	1	-	-	1	3	2
CO2	3	3	3	2	2	-	-	-	1	-	-	1	3	2
CO3	3	3	3	2	3	-	-	-	1	-	-	1	3	3
CO4	3	2	3	2	2	-	-	-	1	-	-	1	3	3
CO5	3	2	3	2	2	-	-	-	1	-	-	1	3	3

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I OPERATIONAL AMPLIFIER:

Ideal and Practical Op-Amp, Op-Amp Characteristics, DC and AC Characteristics, features of 741 Op-Amp, Modes of Operation-Inverting, Non-Inverting, Differential, Instrumentation Amplifier, AC Amplifier, Differentiators and Integrators, Comparators, Schmitt Trigger,

Introduction to Voltage Regulators, Features of 723 Regulator, Three Terminal Voltage Regulators.

UNIT-II OP-AMP, IC-555 & IC565 APPLICATIONS:

Introduction to Active Filters, Characteristics of Bandpass, Band reject and All Pass Filters, Analysis of 1st order LPF & HPF Butterworth Filters, Waveform Generators – Triangular, Sawtooth, Square Wave, IC555 Timer-Functional Diagram, Monostable and Astable Operations, Applications, IC565 PLL-Block Schematic, principle and Applications.

UNIT-III DATA CONVERTERS:

Introduction, Basic DAC techniques, Different types of DACs-Weighted resistor DAC, R-2R ladder DAC, Inverted R-2R DAC, Different Types of ADCs – Parallel Comparator Type ADC, Counter Type ADC, Successive Approximation ADC and Dual Slope ADC, DAC and ADC Specifications.

UNIT-IV COMBINATIONAL LOGIC ICs:

Specifications and Applications of TTL-74XX & CMOS 40XX Series ICs - Code Converters, Decoders, LED & LCD Decoders with Drivers, Encoders, Priority Encoders, Multiplexers, Demultiplexers, Priority Generators/Checkers, Parallel Binary Adder/Subtractor, Magnitude Comparators.

UNIT-V SEQUENTIAL LOGIC IC'S AND MEMORIES:

Familiarity with commonly available 74XX & CMOS40XX Series ICs - All Types of Flip-flops, Synchronous Counters, Decade Counters, Shift Registers.

Memories - ROM Architecture, Types of ROMs & Applications, RAM Architecture, Static & Dynamic RAMs

TEXT BOOKS:

1. Ramakanth A.Gayakwad-Op-Amps & LinearICs, PHI, 2003.
2. Floydand Jain- Digital Fundamentals, 8thEd., Pearson Education, 2005.

REFERENCE BOOKS:

1. D.Roy Chowdhury–Linear Integrated Circuits, New Age International (p) Ltd, 2ndEd., 2003.
2. John. F.Wakerly–Digital Design Principles and Practices, 3rdEd., Pearson, 2009.
3. Salivahana – Linear Integrated Circuits and Applications, TMH, 2008.
4. William D.Stanley – Operational Amplifiers with Linear Integrated Circuits, 4thEd., Pearson Education India, 2009.

B.TECH VI SEMESTER

OE-II **L** **T** **P** **C**
 3 **0** **0** **3**

23EC6T15 PRINCIPLES OF COMMUNICATIONS
(Open Elective-II)

COURSE OBJECTIVES: The main objectives of this course are given below

- Understand the fundamental concepts and mathematical tools used in communication systems.
- Analyze various analog modulation techniques such as AM, DSB-SC, SSB, and VSB.
- Comprehend the principles of angle modulation techniques like FM and PM.
- Study sampling and pulse modulation techniques and understand quantization.
- Explore the basics of probability, random variables, and noise in communication systems.

COURSE OUTCOMES: At the end of the course, student will be able to:

- CO1:** Explain the fundamental concepts of Fourier analysis, correlation, and their applications in communication systems.
- CO2:** Analyze amplitude modulation techniques and their spectral and power characteristics.
- CO3:** Describe angle modulation methods and evaluate bandwidth requirements and demodulation schemes.
- CO4:** Apply sampling theorem and pulse modulation techniques, including quantization and PCM systems.
- CO5:** Demonstrate understanding of probability theory, noise analysis, and their role in communication system performance.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	-	-	-	-	-	-	-	1	3	2
CO2	3	2	3	2	-	-	-	-	-	-	-	1	3	2
CO3	3	3	3	2	1	-	-	-	-	-	-	1	3	3
CO4	3	3	3	2	2	-	-	-	-	-	-	1	3	3
CO5	3	3	3	2	2	-	-	-	-	-	-	1	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, put '-'. 1-Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I BASIC TOOLS AND AMPLITUDE MODULATION:

Basic tools for communication, Fourier Series/Transform, Properties, Autocorrelation, Energy Spectral Density, Parseval's Relation, Amplitude Modulation (AM), Spectrum of AM, Envelope Detection, Power Efficiency, Modulation Index.

UNIT-II DOUBLE SIDEBAND AND SINGLE SIDEBAND MODULATION:

Double Sideband Suppressed Carrier (DSB-SC) Modulation, Demodulation, Costas Receiver, Single Sideband Modulation (SSB), Hilbert Transform, Complex Pre-envelope/Envelope, Demodulation of SSB, Vestigial Sideband Modulation (VSB).

UNIT-III ANGLE MODULATION TECHNIQUES:

Angle Modulation, Frequency Modulation (FM), Phase Modulation (PM), Modulation Index, Instantaneous Frequency, Spectrum of FM Signals, Carsons Rule for FM Bandwidth, Narrowband FM Generation, Wideband FM Generation via Indirect Method, FM Demodulation.

UNIT-IV SAMPLING AND PULSE MODULATION TECHNIQUES:

Introduction to Sampling, Spectrum of Sampled Signal, Aliasing, Nyquist Criterion, Signal Reconstruction from Sampled Signal, Pulse Amplitude Modulation, Quantization, Uniform Quantizers – Midrise and Midtread, Quantization noise, , Non uniform Quantizers, Delta Modulation, Differential Pulse Code Modulation (DPCM).

UNIT-V PROBABILITY AND NOISE IN COMMUNICATION:

Basics of Probability, Conditional Probability, MAP Principle, Random Variables, Probability Density Functions, Applications in Wireless Channels, Basics of Random Processes, Gaussian Random Process, Noise.

TEXT BOOKS:

1. Simon Haykin, Communications Systems, 4th Edition. John Wiley and Sons, Inc.
2. Fundamentals of Wireless Communication by David Tse.

REFERENCE BOOKS:

1. Principles of Communication Systems – H. Taub and D. Schilling, TMH, 2003.

B.TECH VI SEMESTER

OE-II **L** **T** **P** **C**
 3 **0** **0** **3**

23EC6T16 MICROPROCESSOR AND MICROCONTROLLERS
(Open Elective-II)

Prerequisites: Digital Electronics

COURSE OBJECTIVES:

- To understand microprocessor architectures and the 8086 architecture in detail.
- To develop basic programming skills using 8086 assembly language.
- To learn interfacing techniques for memory and I/O devices with 8086.
- To study 8051 microcontroller architecture and its interfacing applications.
- To understand ARM Cortex-M3 architecture and develop embedded applications.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Understand the architecture, memory organization, and operation of the 8086 microprocessor.

CO2: Develop and write assembly language programs using the 8086 instruction set.

CO3: Applying 8086 processor to interface with necessary peripherals.

CO4: Understand the architecture of the 8051 microcontroller and its interfacing techniques.

CO5: Explain the basic concepts, architecture, and features of ARM Cortex-M processors.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	-	-	-	-	-	-	-	-	-	2	3
CO2	2	2	2	-	-	-	-	-	-	-	-	-	2	3
CO3	3	3	2	-	-	-	-	-	-	-	-	-	2	3
CO4	2	3	2	-	-	-	-	-	-	-	-	-	2	3
CO5	3	3	2	1	-	-	-	-	-	-	-	1	2	3

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2- Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I INTRODUCTION:

Microprocessor based system, Origin of microprocessors, Harvard and Von Neumann architectures with examples, Microprocessor Unit versus Microcontroller Unit.

8086 Architecture: internal architecture of 8086 microprocessor, register organization, physical memory organization, general bus operation.

UNIT-II 8086 PROGRAMMING:

Instruction set, addressing modes, assembler directives, programming with assembler, writing

simple programs with an assembler, stack and stack structure, interrupts and interrupt service routines, interrupt cycle of 8086.

UNIT-III 8086 INTERFACING:

Semiconductor memories interfacing (RAM, ROM), Intel 8255 programmable peripheral interface, Interfacing switches and LEDs, Interfacing seven segment displays, Intel 8251 USART architecture and interfacing, stepper motor, A/D and D/A converters.

UNIT-IV INTEL 8051 MICROCONTROLLER AND INTERFACING:

Introduction to microcontrollers, internal architecture of 8051 microcontroller, I/O ports and memory organization, MCS51 addressing modes and instruction set, assembly language programming, simple programs, counters/timers, serial data input/output, interrupts. Interfacing to 8051: A/D and D/A Convertors, keyboard, LCD Interfacing.

UNIT-V ARM ARCHITECTURES AND PROCESSORS:

Introduction to CISC and RISC architectures, ARM Architecture, ARM Processors Families, Introduction to 16/32 bit processors, ARM7 architecture and organization, Thumb instructions, ARM Cortex-M3 Processor Functional Description.

TEXT BOOKS:

1. Advanced microprocessors and peripherals by K. M. Bhurchandi, A. K. Ray 3e.
2. The 8051 Microcontrollers and Embedded systems Using Assembly and C, Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D.McKinlay; Pearson 2-Edition, 2011.
3. Microprocessors and Microcontrollers by N. Senthil Kumar, M. Saravanan and S. Jeevanathan Oxford higher education

REFERENCE BOOKS:

1. Embedded Systems Fundamentals with Arm Cortex-M based Microcontrollers: A Practical Approach in English, by Dr. Alexander G. Dean, Published by Arm Education Media, 2017.
2. Cortex-M3 Technical Reference Manual

B.TECH VI SEMESTER**OE-II**

L	T	P	C
3	0	0	3

23CS6T15 DATABASE MANAGEMENT SYSTEMS
(Open Elective-II)

COURSE OBJECTIVES: The main objectives of introducing this course are

- Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra.
- Introduce the concepts of basic SQL as a universal Database language.
- Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization.
- Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques Illustrate different conditions for deadlock and their possible solutions.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Explain the concept of database, database models and familiarize with Entity relationship models.

CO2: Demonstrate the use of constraints, relational algebra operations.

CO3: Apply SQL queries to interact with database and understand the basics of NOSQL.

CO4: Apply normalization in database design to eliminate anomalies.

CO5: Illustrate the basic concepts of transaction processing and concurrency control.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSO):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	-	-	-	-	-	-	-	-	2	2
CO2	2	3	2	2	-	-	-	-	-	-	-	-	2	2
CO3	2	2	3	2	3	-	-	-	-	-	-	2	3	3
CO4	2	2	3	2	2	-	-	-	-	-	-	-	2	2
CO5	2	2	2	1	2	-	-	-	-	-	-	-	2	2

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put'-'1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS**UNIT-I**

Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT-II

Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT-III

SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT-IV

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form (BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT-V

Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

TEXT BOOKS:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4).
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5).

REFERENCE BOOKS:

1. Introduction to Database Systems, 8th edition, C J Date, Pearson.
2. Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe,



Pearson.

3. Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

ONLINE LEARNING RESOURCES:

1. <https://nptel.ac.in/courses/106/105/106105175/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

23CS6T16 SOFTWARE ENGINEERING
(Open Elective-II)

Course Objectives: The main objectives of introducing this course are

- Software life cycle models, Software requirements and SRS document.
- Project Planning, quality control and ensuring good quality software.
- Software Testing strategies, use of CASE tools, Implementation issues.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Summarize the software life cycle models.

CO2: Analyze the software requirements and generate SRS document along with project estimation.

CO3: Demonstrate the approach for Agile project management.

CO4: Explain various approaches to achieving software quality and ensuring reliability.

CO5: Explain the Software maintenance and reuse issues.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	-	-	-	-	2	2	3	-	3	3
CO2	2	2	3	2	3	-	-	2	2	3	3	-	3	3
CO3	2	2	2	1	-	2	-	-	2	2	3	-	3	3
CO4	2	3	3	3	3	-	-	-	-	-	3	2	3	3
CO5	2	2	3	2	3	-	-	2	-	-	3	2	3	3

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I

Introduction: Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering.

Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.

UNIT-II

Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead's software science, risk management.

Requirements Analysis And Specification: Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.

UNIT-III

Software Design: Overview of the design process, How to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling. approaches to software design.

Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models, Tool Set for the Agile Process (Text Book 2)

Function-Oriented Software Design: Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review.

User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.

UNIT-IV

Coding And Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing, Testing object-oriented programs, Smoke testing, and Some general issues associated with testing.

Software Reliability And Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000.SEI Capability maturity model. Few other important quality standards and Six Sigma.

UNIT-V

Computer-Aided Software Engineering (Case): CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment.

Software Maintenance: Characteristics of software maintenance, Software reverse engineering, and Software maintenance process models and Estimation of maintenance cost.

Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse at organization level.

TEXT BOOKS:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.



-
2. Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, McGraw Hill International Edition.

REFERENCE BOOKS:

1. Software Engineering, Ian Sommerville, 10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

ONLINE LEARNING RESOURCES:

1. <https://nptel.ac.in/courses/106/105/106105182/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003

B.TECH VI SEMESTER	OE-II	L	T	P	C
		3	0	0	3
23IT6T14	FUNDAMENTALS OF MACHINE LEARNING (Open Elective-II)				

Course Objectives: This course will enable students to

- To introduce the basic concepts and techniques of Machine Learning.
- To develop the skills in using recent machine learning software for solving practical problems.
- To be familiar with a set of well-known supervised, semi-supervised and unsupervised learning algorithms.

Course Outcomes: After studying this course, the students will be able to:

CO1: Choose the learning techniques and investigate concept learning.

CO2: Identify the characteristics of decision tree and solve problems associated with.

CO3: Apply effectively neural networks for appropriate applications.

CO4: Apply Bayesian techniques and derive effectively learning rules.

CO5: Evaluate hypothesis and investigate instant based learning and reinforced Learning.

SYLLABUS

UNIT-I

Introduction: Well-posed learning problems, Designing a Learning system, Perspective and Issues in Machine Learning.

Concept Learning: Concept learning task, Concept learning as search, Find-S algorithm, Version space, Candidate Elimination algorithm, Inductive Bias.

UNIT-II

Decision Tree Learning: Decision tree representation, Appropriate problems for Decision tree learning, Basic decision tree learning algorithm, hypothesis space search in decision tree learning, Inductive bias in decision tree learning, Issues in decision tree learning.

UNIT-III

Artificial Neural Networks: Introduction, Neural Network representation, Appropriate problems, Perceptions, Back propagation algorithm.

Bayesian Learning: Introduction, Bayes theorem, Bayes theorem and concept learning, Naive Bayes classifier, Bayesian belief networks.

UNIT-IV



Learning Sets of Rules: Introduction, sequential covering algorithms, learning rule sets: summary, learning First-Order rules, learning sets of First-Order rules: FOIL, Induction as inverted deduction, inverting resolution.

UNIT-V

Instance Based Learning: Introduction, k-nearest neighbor learning, locally weighted regression, radial basis function, case-based reasoning,

Reinforcement Learning: Introduction, Learning Task, Q Learning.

TEXT BOOKS:

1. Tom M. Mitchell, Machine Learning, India Edition 2-13, McGraw Hill Education.

REFERENCE BOOKS:

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman, h The Elements of Statistical Learning, 2nd edition, springer series in statistics.
2. Ethem Alpaydın, Introduction to machine learning, second edition, MIT press.

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

**23IT6T15 COMPUTER NETWORKS
(Open Elective-II)**

Pre-requisite: Fundamental Operating System Concepts, Introduction to C, Programming and Data Structures

COURSE OBJECTIVES:

- Familiarize the student with the basic taxonomy and terminology of the computer networking area.
- Introduce the students to basic principles of networking using goals like protocol layering and top-down approach.
- Build an understanding of the basics of internetworking and routing used in the computer networks.
- To provide guidelines in developing network applications.

COURSE OUTCOMES: At the end of the course, students will be able to:

CO1: Independently enumerate the layers of the OSI model and TCP/IP.

CO2: Identify the different types of network topologies and protocols.

CO3: Compare and contrast methods to identify Errors and correct them.

CO4: Differentiate between various network routing algorithms.

CO5: Understand WWW and HTTP Architectures.

SYLLABUS

UNIT-I Introduction:

OSI overview, TCP/IP and other networks models, Examples of Networks: Arpanet, Internet, Network Topologies Wide Area Networks (WAN), Local Area Networks (LAN), Metropolitan Area Networks (MAN).

UNIT-II Physical Layer and overview of PL Switching:

Multiplexing: frequency division multiplexing, wavelength division multiplexing, synchronous time division multiplexing, statistical time division multiplexing, introduction to switching: Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.

UNIT-III Data link layer: Design issues, Framing: fixed size framing, variable size framing, flow control, error control, error detection and correction, CRC, Checksum: idea, one's complement internet checksum, services provided to Network. Elementary Data Link Layer protocols: Simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel. Sliding window protocol: One bit, Go-back N, Selective Repetitive protocol, Stop

and wait protocol.

UNIT-IV Random Access:

ALOHA, MAC addresses, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, Controlled Access: Reservation, Polling, Token Passing,

Channelization: Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA).

Network layer: Shortest Path, Distance Vector Routing Algorithm, Hierarchical routing algorithm.

UNIT-V Application layer (WWW and HTTP):

WWW ARCHITECTURE: Client (Browser), Server, Uniform Resource Locator, Resource Record, HTTP: HTTP Transaction, HTTP Operational Model and Client/Server Communication, HTTP Request Message Format, HTTP Response Message Format.

TEXT BOOKS:

1. Data Communications and Networks – Behrouz A. Forouzan. Third Edition TMH. (Units 1,2,4,5).
2. Computer Networks - Andrew S Tanenbaum, 4th Edition. Pearson Education (Units 1, 3, 4)

REFERENCE BOOKS:

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education.
2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson.

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

23MB6T01 INDUSTRY SPECIFIC EMPLOYABILITY SKILLS
(Open Elective-II)

COURSE OBJECTIVES:

- To build strong fundamentals in number systems, simplification, and numerical logic.
- To develop problem-solving skills in financial aptitude and logical arrangements.
- To apply ratio, mixture, partnership, and age-related concepts to practical scenarios.
- To solve time-based quantitative problems and evaluate logical syllogisms and assumptions.
- To improve spatial reasoning, inference drawing, and critical analysis through verbal and non-verbal reasoning.

COURSE OUTCOMES (COS):

Upon successful completion of the course, the student will be able to:

CO1: Solve numerical problems involving number properties, averages, series, and basic reasoning patterns.

CO2: Apply concepts of profit & loss, interest, direction, ranking, and puzzles to real-time aptitude tests.

CO3: Solve problems based on ratios, partnerships, mixtures, and evaluate logical conditions in decision-making.

CO4: Analyze time-work-distance problems and evaluate statements, assumptions, and syllogisms.

CO5: Apply geometry and mensuration to aptitude problems and critically analyze statements and arguments.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	3	-	-	-	-	-	2	-	2	-	-
CO2	3	3	-	3	-	-	-	-	-	2	-	2	-	-
CO3	3	3	-	3	-	-	-	-	-	2	-	2	-	-
CO4	3	3	-	3	-	-	-	-	-	3	-	2	-	-
CO5	3	3	-	3	-	-	-	-	-	3	-	2	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I

Divisibility and remainder rules of numbers, Unit digit, square root, cube root and simplification of numbers, HCF and LCM of numbers, Averages and Percentages, Alphabetical and miscellaneous series, Coding and decoding and Blood Relations.

UNIT-II

Profit & loss, Simple interest and Compound interest, Direction, Order and Ranking, Sitting arrangement and Puzzle.

UNIT-III

Ratio & proportions, Partnership, Alligation and mixtures and Ages. Data sufficiency Inequalities and Decision making.

UNIT-IV

Time and work, Pipes & cisterns and Time and distance. Syllogism, Statement and course of action and Statement and Assumption.

UNIT-V

Boats and streams, Areas, Volume and surface areas, Statement and argument, Cause and effect and Drawing inference.

Text Books:

1. "Objective Arithmetic" by R.S. Agarwal, S.Chand Publications.
2. Verbal and non-verbal Reasoning, R.S. Agarwal, S.Chand Publications.

Reference Books:

3. Quantitative Aptitude by Dinesh Khattar, Pearson Education.
4. Quantitative Aptitude by Abhjit Guha.
5. Fast Track objective Arithmetic, Rajesh Verma, Arihant publications.

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

23MB6T03 SUPPLY CHAIN MANAGEMENT
(Open Elective-II)

COURSE OBJECTIVES:

- To understand basic concepts and strategic decisions in supply chain management.
- To learn inventory management techniques and transportation models.
- To Explore IT applications in supply chain operations.
- To Analyze reverse supply chain processes and environmental impacts.
- To Develop skills for integration and collaboration in supply chains.

COURSE OUTCOMES (COS):

Upon successful completion of the course, the student will be able to:

CO1: Explain the fundamental concepts and strategic decision areas of supply chain management.

CO2: Apply inventory control techniques and transportation models to supply chain scenarios.

CO3: Analyze IT tools and systems used in managing supply chains.

CO4: Evaluate reverse supply chain processes and their environmental impact.

CO5: Design collaborative and integrated supply chain strategies.

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	3	-	-	-	-	2	2	-	2	-	-
CO2	3	2	3	3	-	-	-	-	2	2	-	2	-	-
CO3	3	3	2	2	3	-	-	-	2	3	-	3	-	2
CO4	3	3	2	3	-	-	2	2	2	2	-	2	-	-
CO5	2	2	3	3	-	-	3	2	3	3	-	3	-	-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **-**. **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I Introduction to Supply Chain Management:

Introduction, Generic Types of supply chain, Various Definitions and Implications, Major Drivers of Supply chain. Strategic Decisions–in Supply Chain Management. Business Strategy, Core Competencies in Supply Chain, Strategic SC Decisions, Customer Relationship Management Strategy, Supplier Relationship Management Strategy.

UNIT-II Inventory Management in Supply Chain:

Introduction, Types of Inventory, Supply/ Demand Uncertainties, Inventory costs, Selective

Inventory Control, Vendor Manage Inventory system, Inventory Performance, Measure Transportation Selection, Trade-off, Models for Transportation and Distribution, Overview of Indian Infrastructure for Transportation.

UNIT-III Information Technology in Supply Chain:

Introduction, Types of IT Solutions like Electronic Data Inter change (EDI), Intranet/ Extranet, Data Mining/ Data Warehousing and Data Marts, E-Commerce, E-Procurement, Bar Coding Technology, Information System in Supply Chain, Computer Based Information Systems, Computer Models and Perceptions about ERP, ERP & SCM....

UNIT-IV Reverse Supply Chain:

Introduction, Reverse Supply Chain v/s Forward Supply Chain, Types of Reverse Flows, Issues in Management of Reverse Supply Chain, Reverse Supply Chain for Food items, Reverse Logistic and Environment Impact.

UNIT-V Integration & Collaborative Supply Chain:

Introduction, Evolution of collaborative SCM, Efficient Customer response, Collaboration at various levels, Imperatives for Successful Integrative Supply Chains.

Text Books & Reference Books:

1. Supply Chain Management Theories & Practices, R. P. Mohanty, S. G. Deshmukh, Dream tech Press, 19-A, Anari Road, Daryaganj, New Delhi
2. Supply Chain Management Strategy, Planning & Operation by Sunil Chopra, Peter Meindl.
3. Total Supply Chain Management by Ron Basu, J. Nevan Wright.
4. Supply Chain Management, Chopra, Pearson.
5. Logistics Engineering and Management, Blanchard, pearson.

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

23AD6T14 ARTIFICIAL INTELLIGENCE
(Open Elective-II)

COURSE OBJECTIVES: The main objectives of introducing this course are

- The student should be made to study the concepts of Artificial Intelligence.
- The student should be made to learn the methods of solving problems using Artificial Intelligence.
- The student should be made to introduce the concepts of Expert Systems.
- To understand the applications of AI, namely game playing, theorem proving, and machine learning.
- To learn different knowledge representation techniques.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Explain the history and foundations of Artificial Intelligence.

CO2: Apply the basic principles of AI in problem solving.

CO3: Choose the appropriate representation of Knowledge.

CO4: Explain the perspectives in Machine Learning.

CO5: Discuss the concepts in Machine Learning.

COURSE ARTICULATION MATRIX (Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	1	-	-	-	-	2	-	1	3	2
CO2	3	3	2	2	2	-	-	-	-	2	-	1	3	3
CO3	3	2	3	2	2	-	-	-	-	2	-	1	3	3
CO4	3	2	2	2	3	-	-	-	-	2	-	2	3	3
CO5	3	2	2	2	3	-	-	-	-	2	-	2	3	3

Note: Enter Correlation Levels **1**or**2**or**3**. If there is no correlation, put **'-'** **1** Slight (Low), **2**-Moderate (Medium), **3**-Substantial (High).

SYLLABUS

UNIT-I Introduction:

Foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents.

UNIT-II Searching:

Types of Search Strategies, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms,

Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning.

UNIT-III Representation of Knowledge:

Knowledge representation issues, predicate logic logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems.

UNIT-IV Logic concepts:

First order logic. Inference in first order logic, propositional vs first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Reinforcement Learning.

UNIT-V Expert Systems:

Architecture of expert systems, Roles of expert systems Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

TEXT BOOKS:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, Second Edition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, Mc Graw Hill.

REFERENCE BOOKS:

1. David Poole, Alan Mackworth, Randy Goebel, ”Computational Intelligence: a logical approach”, Oxford University Press.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problem solving”, Fourth Edition, Pearson Education.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.

B.TECH VI SEMESTER

OE-II	L	T	P	C
	3	0	0	3

23AM6T14 SOCIAL MEDIA ANALYTICS
(Open Elective-II)

Prerequisites:

COURSE OBJECTIVES: The main objectives of introducing this course are

- Knowledge on social media and its analytics Course.
- To introduce the fundamentals of social media platforms, data types, and user behaviour.
- To provide insights into techniques for collecting, processing, and analyzing data.
- To familiarize students with key analytical methods.

COURSE OUTCOMES: At the end of the course, student will be able to:

CO1: Understanding characteristics and types of social media.

CO2: Knowledge on layers of social media analytics.

CO3: Apply text analysis tools on social media data.

CO4: Understand the significance of action analytics.

CO5: Detect viral topics on social media (YouTube).

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSO's):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	1	1	2	-	-	1	2	1	2	2	2
CO2	3	2	2	2	3	1	-	-	1	2	1	2	3	3
CO3	3	3	2	3	3	-	-	-	1	2	1	2	3	3
CO4	3	2	2	2	3	1	-	-	1	2	2	3	3	3
CO5	3	3	3	3	3	3	-	-	2	2	2	3	3	3

Note: Enter Correlation Levels 1or2or3. If there is no correlation, put '-'. 1 Slight (Low), 2-Moderate (Medium), 3-Substantial (High).

SYLLABUS

UNIT-I Introduction to Social Media World Wide Web:

Web 1.0, Web 2.0, Web3.0, Social Media, Core Characteristics Of Social Media, Types Of Social Media, Social Networking Sites, Using Facebook For Business Purposes, Content Communities.

UNIT-II Social Media Analytics:

Overview Purpose Of Social Media Analytics, Social Media Vs. Traditional Business Analytics, Seven Layers Of Social Media Analytics, Types Of Social Media Analytics, Social Media Analytics Cycle, Challenges To Social Media Analytics, Social Media Analytics Tools. Case Study: The Underground Campaign That Scored Big.

UNIT-III Social Media Text Analytics:

Types Of Social Media Text, Purpose Of Text Analytics, Steps In Text Analytics, Social Media Text Analysis Tools.

Case Study: Tapping Into Online Customer Opinions.

UNIT-IV Social Media Actions Analytics:

Introduction To Actions Analytics, Common Social Media Actions, Actions Analytics Tools.

Case Study: Cover-More Group.

UNIT-V Social Media Hyperlink Analytics:

Types Of Hyperlinks, Hyperlink Analytics, Types Of Hyperlink Analytics, Hyperlink Analytics Tools. Case Study: Hyperlinks And Viral YouTube Videos.

TEXT BOOKS:

1. Seven Layers Of Social Media Analytics Mining Business Insights From Social Media Text, Actions, Networks, Hyperlinks, Apps, SearchEngine, And Location Data By Gohar F. Khan Isbn: 1507823207, Isbn-13: 9781507823200.

REFERENCE BOOKS:

1. Social Media Analytics: Techniques And Insights For Extracting Business Value Out Of Social Media By Matthew Ganis, AvinashKohirkar, Pearson Education.
2. Social Media Analytics: Effective Tools for Building, Interpreting, and Using Metrics, Marshall Sponder, MGH.
3. Big Data and Analytics, Seema Acharya, Subhasinin Chellappan, Wiley Publications.
4. Big Data, Black Booktm, Dream tech Press,2015 Edition.

B.TECH MINOR		MN	L	T	P	C
			3	0	0	3
23AMMN01	INTRODUCTION TO MACHINE LEARNING (Minor Engineering Course)					

Prerequisites: Probability and Statistics, Linear Algebra

Course Objectives:

- To understand the need for machine learning for various problem solving
- To study the various supervised, semi-supervised and unsupervised learning algorithms in machine learning
- To understand the latest trends in machine learning
- To design appropriate machine learning algorithms for problem solving

Course Outcomes: At the end of the course, student will be able to

- CO1** Basic understanding of Machine Learning algorithms
- CO2** Apply various ANN algorithms with machine learning techniques.
- CO3** Apply the Bayesian concepts to machine learning.
- CO4** Implementation of SVM.
- CO5** Exposure to types of reinforced learning

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	1	2	-	-	-	1	1	-	2	3	2
CO2	2	2	3	1	2	-	-	-	1	1	-	2	3	3
CO3	2	2	2	1	2	-	-	-	1	1	-	2	3	3
CO4	2	2	2	1	2	-	-	-	1	1	-	2	3	2
CO5	2	1	2	1	2	-	-	-	1	1	-	2	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1-Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction to Machine Learning :

Evolution of Machine Learning, Paradigms for ML, Well posed Learning Problems, Perspectives and Issues, Designing a Learning system, Concept Learning, FIND-S Algorithm, Version Spaces and Candidate Eliminations, Inductive bias, Decision Tree learning, Decision Tree Representation – ID3 Algorithm, Heuristic Space search

UNIT-II: Neural Networks and Genetic Algorithms:

Neural Network Representation, Appropriate Problems for ANN, Perceptron Learning, Perceptron Models Types, Multi-layered perceptron model, Multilayer Networks and Back Propagation Algorithms, Advanced Topics in ANN, Genetic Algorithm in machine Learning

,Genetic Programming, Models of Evaluation and Learning.

UNIT-III: Bayesian and Computational Learning

Bayes Theorem Representation, Brute force Concept Learning , Maximum Likelihood-Least Squared error Hypothesis, Maximum Likelihood hypothesis-Predicting Probabilities, Minimum Description Length Principle, Bayes Optimal Classifier, Naïve Bayes Classifier (NBC), Bayesian Belief Network, EM Algorithm.

UNIT-IV: Instant Based Learning

Introduction of Instant based learning, K- Nearest Neighbor Learning-Algorithm ,RegressionLinear Regression, Locally weighted regression, Logistic Regression, Random Forests for Classification and Regression, Radial Basis Functions, Case Based Learning-CADET System, Support Vector Machines (SVM).

UNIT-V: Reinforcement Learning:

Introduction – Terms, Key features, Approaches, Elements, The Bellman Equation, Types of Reinforcement Learning, Reinforcement Learning Algorithms, The Learning Task, Q – Learning, non-deterministic rewards and actions, temporal difference learning, Algorithms in temporal difference learning, generalizing from examples, relationship to dynamic programming.

TEXT BOOKS:

1.Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.

References:

- 1.EthemAlpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
- 2.Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009



B.TECH MINOR	MN	L	T	P	C
		3	0	0	3

23AMMN02 SOCIAL NETWORK ANALYSIS (Minor Engineering Course)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. To understand the concept of semantic web and related applications.
2. To understand functionality of various communities in web social networks.
3. To understand human behaviour and privacy issues in social web and related communities.

Course Outcomes: At the end of the course, student will be able to

- CO1** Understand and Develop semantic web related applications
CO2 Represent knowledge using ontology
CO3 Understand Web Mining
CO4 Predict human behavior in social web and related communities
CO5 Interpret data from Social Networks using visualization techniques

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	–	–	–	1	2	–	1	3	2
CO2	3	2	3	2	3	–	–	–	–	2	–	1	3	2
CO3	3	3	3	3	3	–	–	–	–	2	–	2	3	3
CO4	2	3	3	3	3	2	2	2	2	2	–	2	3	3
CO5	2	2	2	3	3	–	–	–	1	2	–	2	3	

Note: Enter Correlation Levels 1 or 2 or 3. If there is nocorrelation, Put ‘-’ 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: INTRODUCTION

Introduction to Semantic Web: Limitations of current Web – Development of Semantic Web – Emergence of the Social Web – Social Network analysis: Development of Social Network Analysis – Key concepts and measures in network analysis – Electronic sources for network analysis: Electronic discussion networks, Blogs and online communities – Web-based networks – Applications of Social Network Analysis.

UNIT-II: MODELLING, AGGREGATING AND KNOWLEDGE REPRESENTATION

Ontology and their role in the Semantic Web: Ontology-based knowledge Representation – Ontology languages for the Semantic Web: Resource Description Framework – Web Ontology Language – Modelling and aggregating social network data: State-of-the-art in network data

representation – Ontological representation of social individuals – Ontological representation of social relationships – Aggregating and reasoning with social network data – Advanced representations.

UNIT-III: EXTRACTION AND MINING COMMUNITIES IN WEB SOCIAL NETWORKS

Extracting evolution of Web Community from a Series of Web Archive – Detecting communities in social networks – Definition of community – Evaluating communities – Methods for community detection and mining – Applications of community mining algorithms – Tools for detecting communities social network infrastructures and communities – Decentralized online social networks – Multi-Relational characterization of dynamic social network communities.

UNIT-IV: PREDICTING HUMAN BEHAVIOR AND PRIVACY ISSUES

Understanding and predicting human behavior for social communities – User data management – Inference and Distribution – Enabling new human experiences – Reality mining – Context – Awareness – Privacy in online social networks – Trust in online environment – Trust models based on subjective logic – Trust network analysis – Trust transitivity analysis – Combining trust and reputation – Trust derivation based on trust comparisons – Attack spectrum and counter measures.

UNIT-V: VISUALIZATION AND APPLICATIONS OF SOCIAL NETWORKS

Graph theory – Centrality – Clustering – Node-Edge Diagrams – Matrix representation – Visualizing online social networks, visualizing social networks with matrix-based representations – Matrix and Node-Link Diagrams – Hybrid representations – Applications – Cover networks – Community welfare – Collaboration networks – Co-Citation networks.

Text Books

1. Peter Mika, —Social Networks and the Semantic Web, First Edition, Springer 2007.
2. Borko Furht, —Handbook of Social Network Technologies and Applications, 1st Edition, Springer, 2010.

Reference Books

1. Guandong Xu, Yanchun Zhang and Lin Li, -Web Mining and Social Networking – Techniques and applications, First Edition, Springer, 2011.
2. Dion Goh and Schubert Foo, -Social information Retrieval Systems: Emerging Technologies and Applications for Searching the Web Effectively, IGI Global Snippet, 2008.



B.TECH MINOR	MN	L	T	P	C
		3	0	0	3

**23AMMN03 DATA VISUALIZATION
(Minor Engineering Course)**

Prerequisites: Computer Graphics, Image Processing

Course Objectives: The main objectives of introducing this course are

1. familiarize students with the basic and advanced techniques of information visualization and scientific visualization
2. learn key techniques of the visualization process
3. a detailed view of visual perception, the visualized data and the actual visualization, interaction and distorting techniques

Course Outcomes: At the end of the course, student will be able to

- CO1 Explain** the fundamental concepts of visualization, including its history, the visualization process, and principles of visual perception such as Gestalt laws to effectively represent data.
- CO2 Apply** various visual mapping and design techniques to create meaningful visual representations and develop visualization applications based on the visualization reference model.
- CO3 Classify** different types of visualization systems and techniques, including interaction methods, and identify common pitfalls such as misleading visualizations.
- CO4 Demonstrate** visualization methods for complex data structures such as groups, trees, graphs, clusters, and networks, including metaphorical visualization approaches.
- CO5 Evaluate** advanced visualization techniques for volumetric data, vector fields, geographic information systems (GIS), and collaborative visualizations, incorporating recent trends and perception techniques.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	-	-	-	-	-	2	-	3	3	-
CO2	2	3	2	2	3	-	-	-	1	2	1	3	3	2
CO3	2	3	-	3	2	-	-	-	-	2	-	3	3	2
CO4	2	3	2	2	3	-	-	-	1	2	1	3	3	3
CO5	2	2	3	3	3	1	2	1	1	2	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is nocorrelation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I:

What Is Visualization?, History of Visualization, Relationship between Visualization and Other Fields. The Visualization Process, Introduction of visual perception, visual representation of data, Gestalt principles, and information overloads.



UNIT-II:

Creating visual representations, visualization reference model, visual mapping, visual analytics, Design of visualization applications

UNIT-III:

Classification of visualization systems, Interaction and visualization techniques misleading, Visualization of one, two and multi-dimensional data, text and text documents.

UNIT-IV:

Visualization of groups, trees, graphs, clusters, networks, software, Metaphorical visualization

UNIT-V:

Visualization of volumetric data, vector fields, processes and simulations, Visualization of maps, GIS systems, collaborative visualizations, Evaluating visualizations

Recent trends in various perception techniques, various visualization techniques, data structures used in data visualization.

Text Books

1. WARD, GRINSTEIN, KEIM. Interactive Data Visualization: Foundations, Techniques, and Applications. Natick : A K Peters, Ltd.
2. E. Tufte, The Visual Display of Quantitative Information, Graphics Press.

Reference Books

1. https://kdd.cs.ksu.edu/Courses/CIS536/Lectures/Slides/Lecture-34-Main_6up.pdf

B.TECH MINOR

MN	L	T	P	C
	3	0	0	3

**23AMMN04 EXPERT SYSTEMS
(MINOR ENGINEERING COURSE)****Prerequisites:****Course Objectives:** The main objectives of introducing this course are

1. **To introduce the fundamental concepts and architecture** of Expert Systems, including their relationship to Artificial Intelligence and Knowledge-Based Systems.
2. **To explore various knowledge representation techniques** such as rules, frames, and semantic networks, and understand their role in reasoning and decision-making.
3. **To study different types of inference mechanisms** (deduction, induction, abduction) and how they are applied in expert systems.
4. **To provide hands-on exposure to expert system development**, including methodologies for knowledge acquisition, system design, and evaluation.
5. **To familiarize students with existing expert system tools and shells**, such as CLIPS, Jess, and EMYCIN, and guide them in selecting appropriate tools for real-world applications.

Course Outcomes: At the end of the course, student will be able to

- CO1 Describe the structure and function of expert systems** and explain their role in decision support across various domains.
- CO2 Apply knowledge representation techniques** to model expert knowledge using rules, logic, frames, and networks.
- CO3 Implement inference techniques** such as forward and backward chaining to simulate expert reasoning in problem-solving scenarios.
- CO4 Use expert system shells and development tools** (e.g., CLIPS or Jess) to design, build, and test a simple expert system.
- CO5 Critically evaluate expert systems** in terms of architecture, performance, limitations, and applicability to real-world problems.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	1	-	-	-	2	-	1	3	-
CO2	3	3	2	-	2	-	-	-	-	-	-	1	3	2
CO3	3	3	2	2	2	-	-	-	-	-	-	1	3	2
CO4	2	2	3	3	3	-	-	-	1	2	2	2	3	3
CO5	2	3	2	2	2	2	1	1	1	2	1	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put - '1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction to Expert Systems

Introduction and Definitions, Nature and Characteristics of Expert Systems, The Nature of Expertise, Differences between Expert Systems, AI, and Knowledge-Based Systems (KBS), Applications and Domains of Use (Medical, Finance, Engineering, etc.), Advantages and Limitations of Expert Systems, Classic Expert Systems: MYCIN, DENDRAL, XCON, PROSPECTOR: Criteria for Selecting an Application Area.

UNIT-II: Theoretical Foundations and Inference Mechanisms

What an Expert System Is and How It Works, Architecture of Expert Systems; Knowledge Base, Inference Engine, User Interface. Basic Inference Techniques: Deduction, Induction, Abduction, Reasoning Methods: Forward Chaining, Backward Chaining, Conflict Resolution, SLD, Resolution refraction methods using predicate and proposition logic.

UNIT-III: Knowledge Representation and Manipulation

Knowledge Representation Approaches: Rule-Based Representation (Production Rules), Logic-Based Representation (Predicate Logic, Resolution), Frames (Inheritance, Exceptions), Semantic Networks and Partitioned Semantic Nets, Taxonomies and Meronomies, Rule Interpreters and Pattern Matching, Query Handling and Explanation Capabilities, Knowledge Organization Techniques.

UNIT-IV: Expert System Architectures and Case Studies

Basic Components of Expert Systems, Detailed Architectures: Rule-Based, Frame-Based, Blackboard Architecture (e.g., HEARSAY), Deep Expert Systems, Co-operating Expert Systems, Explanation Mechanisms, Limitations of First-Generation Expert Systems, Natural Language Interfaces for Expert Systems.

UNIT-V: Building Expert Systems and Expert System Shells

Methodologies for Building Expert Systems: Knowledge Acquisition and Elicitation, Knowledge Formalization, Evaluation and Validation, Development Stages and Challenges: Working with Domain Experts, Common Pitfalls in Development, Knowledge Engineering Tools **Expert System Shells** (Pre-existing Platforms): **CLIPS (C Language Integrated Production System)**, **Jess (Java Expert System Shell)**, **Prolog-based Shells**, **KEE (Knowledge Engineering Environment)**, **EMYCIN (Empty MYCIN)**, Selection of Expert System-Based Tools (ESBT) for Organizations **Case Study**: Building a Mini Expert System Using a Shell (e.g., CLIPS or Jess)

Text Books

1. P Jackson, Introduction to Expert Systems, Addison Wesley, 1990 (2nd Edition)
2. Giarratano, J., & Riley, G. – *Expert Systems: Principles and Programming*, Thomson.



-
3. Durkin, J. – *Expert Systems: Design and Development*, Macmillan.
 4. Jackson, P. – *Introduction to Expert Systems*, Addison-Wesley.

B.TECH MINOR	MN	L	T	P	C
		3	0	0	3
23AMMN05 NATURAL LANGUAGE PROCESSING (Minor Engineering Course)					

Prerequisites:

Course Objectives: This course introduces the fundamental concepts and techniques of natural language processing (NLP).

- Students will gain an in-depth understanding of the computational properties of natural languages and the commonly used algorithms for processing linguistic information.
- The course examines NLP models and algorithms using both the traditional symbolic and the more recent statistical approaches.
- Enable students to be capable to describe the application based on natural language processing and to show the points of syntactic, semantic and pragmatic processing.

Course Outcomes: At the end of the course, student will be able to

- CO1** Demonstrate a given text with basic Language features
- CO2** To design an innovative application using NLP components
- CO3** Explain a rule based system to tackle morphology/syntax of a language
- CO4** To design a tag set to be used for statistical processing for real-time applications
- CO5** To compare and contrast the use of different statistical approaches for different types of NLP applications.

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	1	-	-	-	-	2	-	1	3	-
CO2	2	2	3	-	3	-	-	-	1	-	2	1	3	2
CO3	3	3	-	-	2	-	-	-	-	-	-	2	3	2
CO4	2	3	2	2	3	-	-	-	-	1	1	2	3	3
CO5	3	3	-	3	3	-	-	-	-	2	-	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put -1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: INTRODUCTION

Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.

UNIT-II: WORD LEVEL ANALYSIS

Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word

Classes, Part- of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum.

UNIT-III: SYNTACTIC ANALYSIS

Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures

UNIT-IV: SEMANTICS AND PRAGMATICS

Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.

UNIT-V: DISCOURSE ANALYSIS AND LEXICAL RESOURCES

Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).

Text Books

1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, 2nd Edition, Daniel Jurafsky, James H. Martin - Pearson Publication, 2014.
2. Natural Language Processing with Python, First Edition, Steven Bird, Ewan Klein and Edward Loper, O'Reilly Media, 2009.

Reference Books

1. Language Processing with Java and Ling Pipe Cookbook, 1st Edition, Breck Baldwin, Atlantic Publisher, 2015.
2. Natural Language Processing with Java, 2nd Edition, Richard M Reese, O'Reilly Media, 2015.
3. Handbook of Natural Language Processing, Second, Nitin Indurkha and Fred J. Damerau, Chapman and Hall/CRC Press, 2010. Edition
4. Natural Language Processing and Information Retrieval, 3rd Edition, Tanveer Siddiqui, U.S. Tiwary, Oxford University Press, 2008.



B.TECH MINOR	MN	L	T	P	C
		3	0	0	3
EXPLORATORY DATA ANALYSIS WITH PYTHON					
23AMMN06 (Minor Engineering Course)					

Prerequisites:

Course Objectives: The main objectives of introducing this course are

1. Introduce the fundamentals of Exploratory Data Analysis
2. Cover essential exploratory techniques for understanding multivariate data by summarizing it through statistical methods and graphical methods.
3. Evaluate the Models and select the best model

Course Outcomes: At the end of the course, student will be able to

CO1 Understand the fundamental concepts of Exploratory Data Analysis

CO2 Apply data handling techniques using Python libraries

CO3 Demonstrate the use of various data visualization tools like matplotlib and seaborn to generate insightful visualizations such as bar charts, histograms, scatter plots, and polar charts.

CO4 Analyze datasets using statistical measures such as central tendency, dispersion, correlation, and distributions, and perform univariate, bivariate, multivariate, and time series analysis.

CO5 Develop and evaluate machine learning models by performing hypothesis testing, model training, and applying evaluation metrics such as R² score, MAE, and MSE to select the best performing models.

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	1	-	-	2	3
CO2	3	3	2	2	2	-	-	-	-	1	-	-	2	2
CO3	2	2	3	2	3	-	-	-	1	2	-	-	2	2
CO4	3	2	3	3	2	-	-	-	-	2	-	-	2	2
CO5	3	2	3	3	3	-	-	-	-	2	2	1	2	2

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put - '1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Exploratory Data Analysis Fundamentals:

Understanding data science, The significance of EDA, Steps in EDA, Making sense of data, Numerical data, Categorical data, Measurement scales, Comparing EDA with classical and

Bayesian analysis, Software tools available for EDA, Getting started with EDA.

Sample Experiments:

1. a) Download Dataset from Kaggle using the following link :

<https://www.kaggle.com/datasets/sukhmanibedi/cars4u>

b) Install python libraries required for Exploratory Data Analysis (numpy, pandas, matplotlib, seaborn)

2. Perform Numpy Array basic operations and Explore Numpy Built-in functions.

3. Loading Dataset into pandas dataframe

4. Selecting rows and columns in the dataframe

UNIT-II: Visual Aids for EDA:

Technical requirements, Line chart, Bar charts, Scatter plot using seaborn, Polar chart, Histogram, Choosing the best chart

Case Study: EDA with Personal Email, Technical requirements, Loading the dataset, Data transformation, Data cleansing, Applying descriptive statistics, Data refactoring, Data analysis.

Sample Experiments:

1. Apply different visualization techniques using sample dataset

a. Line Chart b. Bar Chart c. Scatter Plots d. Bubble Plot

2. Generate Scatter Plot using seaborn library for iris dataset

3. Apply following visualization Techniques for a sample dataset

a. Area Plot b. Stacked Plot c. Pie chart d. Table Chart

4. Generate the following charts for a dataset.

a. Polar Chart b. Histogram c. Lollipop chart

5. Case Study: Perform Exploratory Data Analysis with Personal Email Data

UNIT-III: Data Transformation:

Merging database-style data frames, Concatenating along with an axis, Merging on index, Reshaping and pivoting, Transformation techniques, Handling missing data, Mathematical operations with NaN, Filling missing values, Discretization and binning, Outlier detection and filtering, Permutation and random sampling, Benefits of data transformation, Challenges.

Sample Experiments:

1. Perform the following operations

a) Merging Dataframes b) Reshaping with Hierarchical Indexing

c) Data Deduplication d) Replacing Values

2. Apply different Missing Data handling techniques

a) NaN values in mathematical Operations b) Filling in missing data

c) Forward and Backward filling of missing values d) Filling with index values

e) Interpolation of missing values

3. Apply different data transformation techniques

- a) Renaming axis indexes b) Discretization and Binning
- c) Permutation and Random Sampling d) Dummy variables

UNIT-IV: Descriptive Statistics:

Distribution function, Measures of central tendency, Measures of dispersion, Types of kurtosis, Calculating percentiles, Quartiles, Grouping Datasets, Correlation, Understanding univariate, bivariate, multivariate analysis, Time Series Analysis

Sample Experiments:

1. Study the following Distribution Techniques on a sample data
 - a) Uniform Distribution b) Normal Distribution
 - c) Gamma Distribution d) Exponential Distribution
 - e) Poisson Distribution f) Binomial Distribution
2. Perform Data Cleaning on a sample dataset.
3. Compute measure of Central Tendency on a sample dataset
 - a) Mean b) Median c) Mode
4. Explore Measures of Dispersion on a sample dataset
 - a) Variance b) Standard Deviation c) Skewness d) Kurtosis
5. a) Calculating percentiles on sample dataset
 - b) Calculate Inter Quartile Range(IQR) and Visualize using Box Plots
6. Perform the following analysis on automobile dataset.
 - a) Bivariate analysis b) Multivariate analysis
7. Perform Time Series Analysis on Open Power systems dataset

UNIT-V: Model Development and Evaluation:

Unified machine learning workflow, Data preprocessing, Data preparation, Training sets and corpus creation, Model creation and training, Model evaluation, Best model selection and evaluation, Model deployment
Case Study: EDA on Wine Quality Data Analysis

Sample Experiments:

1. Perform hypothesis testing using stats models library
 - a) Z-Test b) T-Test
2. Develop model and Perform Model Evaluation using different metrics such as prediction score, R2 Score, MAE Score, MSE Score.
3. Case Study: Perform Exploratory Data Analysis with Wine Quality Dataset



Text Books

1. Suresh Kumar Mukhiya, Usman Ahmed, Hands-On Exploratory Data Analysis with Python, Packt Publishing, 2020.

Reference Books

1. Ronald K. Pearson, Exploratory Data Analysis Using R, CRC Press, 2020

2. Radhika Datar, Harish Garg, Hands-On Exploratory Data Analysis with R: Become an expert in exploratory data analysis using R packages, 1st Edition, Packt Publishing, 2019



B.TECH HONORS	HN	L	T	P	C
		3	0	0	3

**20AMHN01 OPTIMIZATION TECHNIQUES IN MACHINE LEARNING
(Honors Engineering Course)**

Prerequisites: Basics of Machine Learning and Operations Research.

Course Objectives: The main objectives of introducing this course are

- To familiarize and dealing with changing data, identify and interpret potential unintended effects of data in project.
- Also to operationalize and maintain applied machine learning model.

Course Outcomes: At the end of the course, student will be able to

CO1 To understand and analyze how to deal with changing data.

CO2 To understand and interpret potential machine learning strategy.

CO3 To understand and define procedures to operationalize and maintain the applied machine learning model.

CO4 To understand how to optimize the use of Machine Learning in real-life problems.

CO5 To understand basic concepts of mathematics to formulate an optimization problem.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	–	–	–	1	2	–	1	3	2	-
CO2	3	2	3	2	3	–	–	–	–	2	–	1	3	2	-
CO3	3	3	3	3	3	–	–	–	–	2	–	2	3	3	-
CO4	2	3	3	3	3	2	2	2	2	2	–	2	3	3	-
CO5	2	2	2	3	3	–	–	–	1	2	–	2	3		-

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put - 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: INTRODUCTION

Optimization, Formulation of LPP, Solution of LPP: Simplex method, Basic Calculus for optimization: Limits and multivariate functions, Derivatives and linear approximations: Single variate functions and multivariate functions.

UNIT-II: MACHINE LEARNING STRATEGY

ML readiness, Risk mitigation, Experimental mindset, build/buy/partner, setting up a team, Understanding and communicating change.

UNIT-III: RESPONSIBLE MACHINE LEARNING

AI for good and all, Positive feedback loops and negative feedback loops, Metric design and

observing behaviours, Secondary effects of optimization, Regulatory concerns.

UNIT-IV: MACHINE LEARNING IN PRODUCTION AND PLANNING

Integrating info systems, users break things, time and space complexity in production, when to retain the model? Logging ML model versioning, Knowledge transfer, Reporting performance to stakeholders.

UNIT-V: CARE AND FEEDING OF YOUR MACHINE LEARNING MODEL

MLPL Recap, Post deployment challenges, QUAM monitoring and logging, QUAM Testing, QUAM maintenance, QUAM updating, Separating Datastack from Production, Dashboard Essentials and Metrics monitoring.

Text Books

1. Suvrit Sra, Sebastian Nowozin and Stephen J. Wright, "Optimization for Machine Learning", MIT Press, 2011.
2. Mykel J. Kochenderfer and Tim A. Wheeler, "Algorithms for Optimization", 1st Edition, MIT Press, 2019.
3. Jeeva Jose, "Introduction to Machine Learning", Khanna Book Publishing, 1st Edition, 2020.
4. Rajiv Chopra, "Machine Learning", Khanna Book Publishing, 1st Edition, 2021.

Reference Books

1. Cong Fang, Huan Li, and Zhouchen Lin, "Accelerated Optimization for Machine Learning: First-Order Algorithms", Springer, 2020.
2. <https://www.coursera.org/learn/optimize-machine-learning-model-performance>
ONLINE/ NPTEL COURSES: <https://nptel.ac.in/courses/106106245>

B.TECH	HN	L	T	P	C
		3	0	0	3

23AMHN02 MACHINE LEARNING FOR SECURITY
(Honors Engineering Course)

Prerequisites:

Course Objectives: The main objectives of introducing this course are

- To DESCRIBE the fundamental concepts of machine learning for devising security mechanisms.
- To ENUMERATE the techniques for Intrusion Detection and Malware detection and analysis using Machine Learning.
- To learn the machine learning techniques for network traffic analysis

Course Outcomes: At the end of the course, student will be able to

- CO1** have a knowledge of the limitations of the conventional security software in the wake of machine learning based attacks on the security software
- CO2** be able to apply the concepts machine learning based intrusion detection to analyze the IDSs.
- CO3** be able to analyze the malware analysis and mitigation based solutions for the probable threats therein.
- CO4** be able to design the threat models based on machine learning approaches for network analysis.
- CO5** be able to use the concepts of machine learning to prevent security design faults.

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	3	2	-	2	-	-	-	2	3	2
CO2	3	3	3	3	3	-	-	2	-	-	-	2	3	3
CO3	3	3	3	3	3	-	-	2	-	-	-	2	3	3
CO4	3	3	3	3	3	2	-	2	-	-	2	3	3	3
CO5	3	3	3	3	3	2	-	3	-	-	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put - 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS**UNIT-I: INTRODUCTION & REVIEW OF THE MACHINE LEARNING BASICS**

Review of the basic concepts in Linear Algebra, Probability and Statistics. Introduction to the ML techniques. Machine Learning problems viz. Classification, Regression, Clustering, Association rule learning, Structured output, Ranking. The Supervised and Unsupervised

learning algorithms. Linear Regression, Gradient descent for convex functions.

Logistics Regression and Bayesian Classification Support Vector Machines, Decision Tree and Random Forest, Neural Networks, DNNs , Ensemble learning. Principal Components Analysis. Unsupervised learning algorithms: K-means for clustering problems, K-NN (k nearest neighbors). Apriori algorithm for association rule learning problems. Generative vs Discriminative learning. Empirical Risk Minimization, loss functions, VC dimension. Data partitioning (Train/test/Validation), cross-validation, Biases and Variances, Regularization.

UNIT-II: MACHINE LEARNING FOR SECURITY

Introduction to Information Assurance. Review of Cybersecurity Solutions: Proactive Security Solutions, Reactive Security Solutions: Misuse/Signature Detection, Anomaly Detection, Hybrid Detection, Scan Detection. Profiling Modules. Understanding the Fundamental Problems of Machine-Learning Methods in Cybersecurity. Incremental Learning in Cyber infrastructures. Feature Selection/Extraction for Data with Evolving Characteristics. Privacy-Preserving Data Mining. Motivation for ML in security with real-world case studies. Topics of interest in applications of machine learning for security.

UNIT-III: MACHINE LEARNING TECHNIQUES FOR INTRUSION DETECTION

Emerging Challenges in Cyber Security for Intrusion Detection: Unifying the Current Anomaly Detection Systems, Network Traffic Anomaly Detection. Imbalanced Learning Problem and Advanced Evaluation Metrics for IDS. Reliable Evaluation Data Sets or Data Generation Tools. Privacy Issues in Network Anomaly Detection. Machine Learning Techniques: for Anomaly Detection, for Misuse/Signature detection, for Hybrid detection, for Scan detection. Cost-Sensitive Modeling for Intrusion Detection. Data Cleaning and Enriched Representations for Anomaly Detection in System Calls.

UNIT-IV: MACHINE LEARNING TECHNIQUES FOR MALWARE ANALYSIS

Emerging Cyber Threats in malwares: Threats from Malware, Botnets, Cyber Warfare, Mobile Communication. Cyber Crimes. Malware Analysis: Feature generation, Features to classification. Taxonomy of malware analysis approaches based on machine learning. Malware Detection, Similarity Analysis, Category Detection. Feature Extraction. PE Features. Supervised, Unsupervised and Semi-supervised learning algorithms for Malware Detection. Using Deep Learning Approaches: Generative Adversarial Networks.

UNIT-V: NETWORK TRAFFIC ANALYSIS & WEB ABUSE DETECTION

Machine Learning for Profiling Network Traffic: Theory of Network defense (access control, authentication, detecting in-network attackers, data-centric security, honeypots), Predictive model for classifying network attacks.

k-anonymity; l-diversity; differentially private data storage/release; verifiable differential privacy;



privacy preserving inference of social networking data; privacy-preserving recommender system; privacy versus utility. Machine learning techniques for Privacy Preserving Data Mining.

Text Books

1. Clarence Chio, David Freeman, Machine Learning and Security. Protecting Systems with Data and Algorithms, O'Reilly Media Publications.
2. Marcus A. Maloof (Ed.), Machine Learning and Data Mining for Computer Security: Methods and Applications, Springer-Verlag London Limited.
3. Sumeet Dua and Xian Du, Data Mining and Machine Learning in Cybersecurity. CRC Press, Taylor and Francis Group, LLC.

Reference Books

1. Brendan Creane, Bhaskar Krishnamachari, AI and Machine Learning for Network and Cybersecurity, Wiley, 2022.
2. Soma Halder, Sinan Ozdemir, Hands-On Machine Learning for Cybersecurity, Packt Publishing, 2018.



B.TECH HONORS	HN	L	T	P	C
		3	0	0	3

20AMHN03 SENTIMENT ANALYSIS
(Honors Engineering Course)

Prerequisites:

Data mining, Natural Language Processing

Course Objectives

- To learn the fundamentals of Sentiment analysis and Opinion Mining
- To understand the use of Sentiment analysis.
- To understand the role of Sentiment analysis and Opinion Mining
- To apply the Sentiment analysis in applications

Course Outcomes: At the end of the course, student will be able to

CO1 To apply the Sentiment analysis for Document level analysis

CO2 To model the Sentiment analysis

CO3 To implement a shallow processing models to tackle morphology/syntax of a language

CO4 To Examine Syntagmatic and Paradigmatic relations be used for processing the real-time applications

CO5 To apply the algorithms for Sentiment Analysis

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	—	—	—	—	—	—	2	3	3	-
CO2	3	2	3	2	3	—	—	—	—	—	2	2	3	2	-
CO3	3	2	2	2	3	—	—	—	—	—	2	2	3	2	-
CO4	3	3	3	3	3	—	—	—	—	2	2	3	3	3	-
CO5	3	3	3	2	3	—	—	—	—	—	2	3	3	3	-

Note: Enter Correlation Levels 1 or 2 or 3. If there is nocorrelation, Put -'1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction

What is Sentiment analysis, Types of Sentiment Analysis, Why Is Sentiment Analysis Important, Many Facets of Sentiment Analysis, Affective Computing and Sentiment Analysis Sentiment Analysis Applications, Different Levels of Analysis, Sentiment Lexicon and Its Issues, Natural Language Processing Issues, Opinion Spam Detection. The Problem of Sentiment Analysis: Opinion Definition, Sentiment Analysis Tasks, Opinion Summarization, Opinion Summarization, Opinion Summarization.

UNIT-II: Document Sentiment Classification

Sentiment Classification Using Supervised Learning, Sentiment Classification Using Unsupervised Learning, Sentiment Rating Prediction Cross Domain Sentiment Classification, Cross-Language Sentiment Classification. Sentence Subjectivity and Sentiment Classification: Subjectivity Classification, Sentence Sentiment Classification, 4.3 Dealing with Conditional Sentences, Dealing with Sarcastic Sentences, Cross-language Subjectivity and Sentiment Classification, Using Discourse Information for Sentiment Classification.

UNIT-III: Aspect-based Sentiment Analysis

Aspect Sentiment Classification, Basic Rules of Opinions and Compositional Semantics. Aspect Extraction, Identifying Resource Usage Aspect, Simultaneous Opinion Lexicon Expansion and Aspect Extraction, Grouping Aspects into Categories, Entity, Opinion Holder and Time Extraction, Co reference Resolution and Word Sense Disambiguation.

UNIT-IV: Sentiment Lexicon Generation

Dictionary-based Approach, Corpus-based Approach Desirable and Undesirable Facts. Opinion Summarization: Aspect-based Opinion Summarization, improvements to Aspect based Opinion Summarization, Contrastive View Summarization, Traditional Summarization, Analysis of Comparative Opinions.

UNIT-V: Opinion Search and Retrieval

Web Search vs. Opinion Search, Existing Opinion Retrieval Techniques. Opinion Spam Detection: Types of Spam and Spamming, Harmful Fake Reviews, Individual and Group Spamming, Types of Data, Features and Detection, Supervised Spam Detection, Unsupervised spam Detection, Spam Detection based on Atypical Behaviors, Spam Detection Using Review Graph, Group Spam Detection.

Text Books

1. Bing Liu. Sentiment Analysis and Opinion Mining, Morgan & Claypool Publishers, May 2012.
2. Eric Cambria, Dipankar Das, Sivaji Bandyopadhyay, Antonia feraco. A Practical guide to Sentiment Analysis

Reference Books

1. Liu, Bing. Sentiment Analysis: Mining Opinions, Sentiments, And Emotions, Cambridge University Press.
2. Basant Agarwal, Richi Nayak, Namita Mittal, Srikanth Patnaik Deep Learning-Based Approaches for Sentiment Analysis (Algorithms for Intelligent Systems), 1st ed. 2020 Edition



B.TECH SEMESTER	HN	L	T	P	C
		3	0	0	3
20AMHN04 CYBER SECURITY					
(Honors Engineering Course)					

Prerequisites:

Course Objectives: The main objectives of introducing this course are

- Understand the importance of Cyber Security principles, Security architecture, risk management, attacks, incidents, and emerging IT and IS technologies.
- Students will gain insight into the importance of Cyber Security and the integral role of Cyber Security professionals.

Course Outcomes: At the end of the course, student will be able to

CO1 Understand and classify various forms of Cybercrimes

CO2 Interpret the reasons for Cyber offence

CO3 Detect and analyze vulnerabilities in Mobile and Wireless devices

CO4 Analyze tools used to perform cyber crimes

CO5 Understand cyber security Laws

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	—	—	—	1	2	—	1	3	2	-
CO2	3	2	3	2	3	—	—	—	—	2	—	1	3	2	-
CO3	3	3	3	3	3	—	—	—	—	2	—	2	3	3	-
CO4	2	3	3	3	3	2	2	2	2	2	—	2	3	3	-
CO5	2	2	2	3	3	—	—	—	1	2	—	2	3		-

Note: Enter Correlation Levels **1** or **2** or **3**. If there is nocorrelation, Put **—**. **1**- Slight(Low), **2**-Moderate(Medium), **3**-Substantial(High).

SYLLABUS

UNIT-I: Introduction

Cybercrime: Definition and Origins of the Word, Cybercrime and Information Security, Who are Cybercriminals? , Classifications of Cybercrimes, Cybercrime: The Legal Perspectives, Cybercrimes: An Indian Perspective, Cybercrime and the Indian ITA 2000, A Global Perspective on Cybercrimes.

UNIT-II: Cyber offenses

How Criminals Plan Them –Introduction, How Criminals Plan the Attacks, Social Engineering, Cyber stalking, Cyber cafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector Cloud Computing.



UNIT-III: Cybercrime Mobile and Wireless Devices

Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit Card Frauds in Mobile and Wireless Computing Era, security Challenges Posed by Mobile Devices, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for Organizations, Organizational Measures for Handling Mobile.

UNIT-IV: Tools and Methods Used in Cybercrime

Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Key loggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks.

UNIT-V: Cybercrimes and Cyber security

The Indian Context, The Indian IT Act, Challenges to Indian Law and Cybercrime Scenario in India, Consequences of Not Addressing the Weakness in information Technology Act, Digital Signatures and the Indian IT Act, Information Security Planning and Governance, Information Security Policy Standards, Practices, The information Security Blueprint, Security education, Training and awareness program

Text Books

1. Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Nina Godbole, SunitBelapure, Wiley.
2. Principles of Information Security, MichealE.Whitman and Herbert J.Mattord, Cengage Learning.

Reference Books

1. Information Security, Mark Rhodes, Ousley, MGH.



B.TECH HONORS	HN	L	T	P	C
		3	0	0	3
20AMHN05 RESPONSIBLE & ETHICAL AI (Honors Engineering Course)					

Prerequisites:

Course Objectives: The main objectives of introducing this course are

- The objective of the course is to know about the responsibility of artificial intelligence (AI) to make AI more useful for society and humanity.
- The course will also teach principles and practices to perform responsible AI.

Course Outcomes: At the end of the course, student will be able to

CO1 To be able to state aspects of responsible AI such as fairness, accountability, bias, privacy etc.

CO2 To be able to assess the fairness and ethics of AI modules.

CO3 To be able to enforce fairness in models and remove bias in data.

CO4 To be able to preserve the privacy of individuals while learning from them.

CO5 To be able to develop responsible AI modules for given practical problems and estimate the tradeoff with accuracy.

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	2	3	2	3	-	-	-	2	3	2
CO2	3	3	2	2	2	3	2	3	-	-	-	2	3	3
CO3	3	3	3	3	3	2	2	3	-	-	-	2	3	3
CO4	3	2	3	2	3	2	2	3	-	-	-	2	3	2
CO5	3	3	3	3	3	3	2	3	2	2	2	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I:

AI Capabilities Improvement, Imminent risks from AI Models: Toxicity, bias, goal misspecification, adversarial examples etc., Long-term risks from AI Models: Misuse, Misgeneralization, Rogue AGI, Principles of RAI - Transparency; Accountability; Safety, Robustness and Reliability; Privacy and Security; Fairness and non-discrimination; Human Centred Values; Inclusive and Sustainable development, Interpretability – The AI Alignment Problem.

UNIT-II:

Introduction to Deep Learning Techniques, Language/Vision Models, AI Risks for Gen models,

Bias in LLMs – Policing and surveillance - Adversarial Attacks – Vision, NLP – Social media and attention Engineering – autonomous weapon systems and military.

UNIT-III:

ML Poisoning Attacks like Trojans, Implications for current and future AI safety, Explainability, Taxonomy of ML Explainability Methods – Shapley Additive exPlanations (SHAP) – Working example of SHAP – LIME (Local Interpretable Model-Agnostic Explanations) – Working example of LIME - Imminent and Long-term potential for transparency techniques, Mechanistic Interpretability, Representation Engineering, model editing and probing, Critiques of Transparency for AI Safety

UNIT-IV:

ML attacks and counter measures – Attack prevention and Monitoring - Model Drift – concept Drift - Privacy & Fairness in AI, Metrics and Tools for RAI - measuring bias/fairness, audit mechanisms and Compliance.

UNIT-V:

Data and Model Privacy – Differential Privacy – Federated learning – RAI lifecycle – RAI Canvas – Sustainability RAI in domain such as Legal, Health care, Education, Policy issues in RAI.

Text Books

1. Virginia Dignum, “Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way” Springer Nature, 04-Nov-2019; ISBN-10 : 3030303705, ISBN-13 : 978-3030303709.
2. Adnan Masood, Heather Dawe, Ed Price, “Responsible AI in the Enterprise: Practical AI Risk Management for explainable, auditable and safe models with hyperscalers and Azure OpenAI”, Pack Publishing, 2023.

Reference Books

1. Christoph Molnar “Interpretable Machine Learning”. Lulu, 1st edition, March 24, 2019; eBook. ISBN-10 : 0244768528, ISBN-13 : 978-0244768522 [available online]
2. Patrick Hall, James Curtis, Parul Pandey, “Machine learning for High-Risk Applications”, O’Reilly Media, 2023
3. Beena Ammanath, “Trustworthy AI”, Wiley, 2022
4. Sray Agarwal, Shashin Mishra, “Responsible AI” Implementing Ethical and Unbiased algorithms”, Springer, 2021.

B.TECH SEMESTER

HN	L	T	P	C
	3	0	0	3

20AMHN06 DESIGN OF ARTIFICIAL INTELLIGENCE PRODUCTS
(Honors Engineering Course)**Prerequisites:**

Course Objectives: The main objectives of introducing this course are

- Key objectives include understanding the AI product lifecycle.
- Developing and implementing AI models
- Designing and scaling AI products, and addressing ethical considerations in AI development.

Course Outcomes: At the end of the course, student will be able to

CO1 Explore the basics of AI and Machine learning

CO2 Design and Develop AI based solution to healthcare industry

CO3 Structure the design of Gaming with AI

CO4 Apply Artificial Intelligent techniques in Robotics

CO5 Examine the Human Centered AI products

COURSE ARTICULATION MATRIX(Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	3	-	-	-	-	-	-	3	3	2
CO2	3	3	3	3	3	2	2	2	-	2	2	3	3	3
CO3	3	2	3	3	3	-	-	1	2	2	2	2	3	2
CO4	3	3	3	3	3	2	2	2	2	2	2	3	3	3
CO5	2	2	2	2	2	3	3	3	2	2	2	3	2	2

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put -1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS**UNIT-I: Introduction to AI Design**

What's AI - Risks and Benefits of AI – Types of intelligence - Narrow artificial intelligence - General intelligence ,Super intelligence – Training AI with Design - How can AI help in a creative process?- AI Characteristics - AI Design Principles - How does AI become intelligent - Machine Learning, the first step to AI - Types Of Machine Learning - Supervised Machine Learning Algorithms - Unsupervised Machine Learning AlgorithmsSupport Vector Machines - Neural Networks - Naïve Bayesian Classifier - Hidden Markov Models - k-Means Clustering.

UNIT-II: AI in Healthcare

The rise of AI in healthcare applications – Drug discovery and Molecular modeling in AI – Cancer diagnostics and treatment decisions using AI – AI for medical imaging – Outputs of AI in radiology/medical imaging – X-rays and AI in medical imaging (Zebra medical Vision) – Ultrasound and AI in medical imaging (Butterfly iQ) – development of AI in medical devices – Limitations of AI in medical imaging – The future frontiers of AI in medical devices.

UNIT-III: AI in Gaming

Model of Game AI – Algorithms and Data structures used in Gaming - The Complexity Fallacy – The kind of AI in Games – Speed and Memory Constraints – The AI engine – Basic and Kinematic movement algorithms – Steering Behaviors – Combining Steering behaviors – Coordinated movement – Path finding – Hierarchical Path finding – Continuous time path finding – Movement planning – Designing Game AI – AI Based Game Genres

UNIT-IV: AI in Robotics

Foundation for advanced robotics and AI technical requirements – Setting up your robot technical requirements – A concept for a practical robot design process a system engineering-based approach – use cases for cleaning up the playroom – Storyboards.

UNIT-V: Human Centered AI

What is Human Centered AI – HCAI framework – Implications for design – The influence of AI + HCI on Teachers' psychological changes in academic management in colleges – Case study about AI products (Self driving cars, Smart assistant, social media monitoring).

Text Books

1. Stuart J. Russell and Peter Norvig, Artificial Intelligence A Modern Approach, Fourth Edition, Pearson, 2020.
2. Adam Bohr and Kaveh Memarzadeh, Artificial Intelligence in Healthcare, Academic press, 2020
3. Ian Millington, Artificial Intelligence for Games, CRC Press, Third Edition, 2019.
4. Francis X Govers, Artificial Intelligence for Robotics, Packt publishing, 2018.

Reference Books

1. AI Game Programming Wisdom Charles River Media, Volume 1–4, various years.
2. Robin R. Murphy, Introduction to AI Robotics MIT Press, 2nd Edition, 2019.
3. Stuart Russell, Peter Norvig Artificial Intelligence: A Modern Approach Pearson Education, 4th Edition, 2020.

B.TECH SEMESTER**HN**
L T P C
3 0 0 3**20AMHN07 ARTIFICIAL NEURAL NETWORKS**
(Honors Engineering Course)**Prerequisites:****Course Objectives:** The main objectives of introducing this course are

- On completion of this course the students will be able to expose themselves towards intelligence systems and knowledge based systems.
- It also provides knowledge of learning networks.

Course Outcomes: At the end of the course, student will be able to**CO1** Explain the basic concepts of neural networks**CO2** Describe perceptron and back propagation**CO3** Apply various training rules in neural networks**CO4** Explain unsupervised neural networks**CO5** Fine tune the neural networks**COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):**

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	3	-	-	-	-	-	-	3	3	2
CO2	3	3	3	3	3	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	3	-	-	-	-	-	-	3	3	3
CO4	3	2	2	2	3	-	-	-	-	-	-	3	3	2
CO5	3	3	3	3	3	-	-	-	-	-	2	3	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, Put **-**. **1**- Slight(Low), **2**-Moderate(Medium), **3**-Substantial(High).**SYLLABUS****UNIT-I: Introduction to Neural Networks**

History of neural network research- Biological inspiration: Neural computation, Models of computation, Elements of computing models- Network of neurons: structure, Information processing at neurons and synapses,

Information storage, Neurons as self-organizing systems- Artificial Neural Networks: Network of primitive functions, approximation of functions- Neuron Model: Single and multiple input neurons, Transfer functions Network architectures: Single layered and multi layered neurons, Recurrent Networks.

UNIT-II: Perceptron

Introduction to Perceptron- Perceptron Architecture: Single Neuron Perceptron, Multi Neuron Perceptron- Perceptron learning rule: Constructing Learning rules, Unified Learning rule,

Training multi neuron perceptron Complexity of perceptron learning-Computational Limits of Perception-Linearly separable functions- Learning XOR-Feed forward Networks- Back propagation: Chain of rule Calculus, Back-Propagation Computation in Fully-Connected Multi-layer Perceptron.

UNIT-III: Learning and Training

Paradigms of Learning - Using training samples - Gradient Optimization Procedure- Batch Gradient Descent, Stochastic Gradient Descent, Mini-Batch Gradient Descent – Hebbian learning rule – Delta learning rule Convergence and local minima, representational power of feed forward networks, hypothesis search space and inductive bias, generalization, overfitting and stopping criterion-Error functions, Error minimizing procedures-Hebbian learning.

UNIT-IV: Unsupervised Neural Networks

Unsupervised learning in neural networks: Hebbian learning rule, Principle Component Learning, Learning Vector Quantizer-I- Self Organizing Maps: Functionality, Training, Topology Function, Decreasing Learning Rate, Variations of SOMs, Neural gas, Multi-SOM, Multi-neural gas, Growing neural gas- Adaptive Resonance Theory, Orienting subsystems, Learning Laws.

UNIT-V: Tuning the Neural Networks

Pretraining the model: Data selection, Preprocessing, Selection of network architecture- Training the network: Initializing weights, Choice of training algorithm, stopping criteria, Choice Of performance function, Committees of Networks-Post Training Analysis: Fitting, Pattern Recognition, Clustering- Time delay and Recurrent Neural Networks-Case Studies: Smart Sensor system with function approximation- Myocardial Infarction Recognition using pattern recognition-Forest cover problem using Clustering.

Text Books

1. Martin T. Hagan, Neural Network Design, 2nd edition.
2. Tom M. Mitchell, Machine Learning, McGraw-Hill Education (India) Private Limited.
3. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016
4. Andries P., Computational Intelligence: An Introduction, Second Edition, Wiley, 2007
5. David Kriesel, A Brief Introduction to Neural Networks, 2009

Reference Books

1. Simon Haykin *Neural Networks and Learning Machines* Pearson Education, 3rd Edition, 2009.
2. .Laurene V. Fausett, *Fundamentals of Neural Networks: Architectures, Algorithms and Applications* Pearson Education, 1994.



B.TECH HONORS	HN	L	T	P	C
		3	0	0	3
20AMHN08 TEXT MINING AND TIME SERIES ANALYSIS					
(Honors Engineering Course)					

Prerequisites:

Course Objectives: The main objectives of introducing this course are

- Develop the skills needed to do empirical research in fields operating with time series data sets.
- The course aims to provide students with techniques and receipts for estimation and assessment of quality of economic models with time series data.

Course Outcomes: At the end of the course, student will be able to

- CO1** Student will be aware of fundamental concepts of text mining, unsupervised information extraction.
- CO2** Student will be aware of text clustering algorithms like feature selection, distance-based clustering and latent semantic indexing.
- CO3** Student will be aware of Text classification algorithm and text mining techniques.
- CO4** Student should aware of all the characteristics of time series and measures of dependencies.
- CO5** Student will be able to understand the ARIMA Models.

COURSE ARTICULATION MATRIX(Correlation between COs, POs &PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	3	-	-	-	-	-	-	3	3	2
CO2	3	3	2	2	3	-	-	-	-	-	-	3	3	3
CO3	3	3	3	2	3	-	-	-	-	-	-	3	3	3
CO4	3	3	2	3	3	-	-	-	-	-	-	3	3	3
CO5	3	3	3	3	3	-	-	-	-	-	-	3	3	3

Note: Enter Correlation Levels 1 or 2 or 3. If there is no correlation, Put '-'. 1- Slight(Low), 2-Moderate(Medium), 3-Substantial(High).

SYLLABUS

UNIT-I: Introduction

Introduction to Text Mining: Introduction, Algorithms for Text Mining, Information Extraction from Text: Introduction, Named Entity Recognition, Relation Extraction, Unsupervised Information Extraction. Text Summarization Techniques: Extractive Summarization, Topic Representation Approaches, Influence of Context, Indicator Representation and Machine Learning for Summarization.

UNIT-II: Text Clustering Algorithms

Introduction, Feature Selection and Transformation Methods for Text Clustering, Distance-

Based Clustering Algorithms, Word and Phrase-based Clustering, Probabilistic Document Clustering and Topic Modelling. Dimensionality Reduction and Topic Modelling: Latent Semantic Indexing, Topic Models and Dimension Reduction.

UNIT-III: Text Classification Algorithms

Introduction, Feature Selection for Text Classification, Decision Tree Classifiers, Rule-based Classifier, Probabilistic and Naïve Bayes Classifiers, Linear Classifier, Proximity based Classifier, Meta-Algorithms for Text Classification, Probabilistic Models for Text Mining: Mixture models, Stochastic Processes in Bayesian Nonparametric Models, Graphical Models.

UNIT-IV: Characteristics of Time Series

Introduction, Nature of Time Series Data, Time Series Statistical Models, Measures of Dependence: Autocorrelation and Cross-Correlation, Stationary Time Series, Time Series Regression and Exploratory Data Analysis: Classical Regression, Exploratory Data Analysis, Smoothing.

UNIT-V: ARIMA Models

Introduction, Autoregressive Moving Average Models, Difference Equations, Autocorrelation and Partial Autocorrelation, Building ARIMA Models, Multiplicative Seasonal ARIMA Models, Spectral Analysis and Filtering: Cyclical Behaviour and Periodicity, Spectral Density, Periodogram and Discrete Fourier Transform, Nonparametric and Parametric Spectral Estimation, Linear Filters, Dynamic Fourier Analysis and Wavelets.

Text Books

1. Charu C. Aggarwal, Chengxing Zhai, “Mining Text Data”, Kluwer Academic Publishers, Springer, 2012.
2. Robert H. Shumway and David S. Stoffer, “Time Series Analysis and Its Applications with R examples”, Springer, 2016

Reference Books

1. James D. Hamilton, Time Series Analysis, Princeton University Press, 2004.
2. Avishek Pal and PKS Prakash, Practical Time Series Analysis, Birmingham - Mumbai, 2017.